



AI VIET NAM

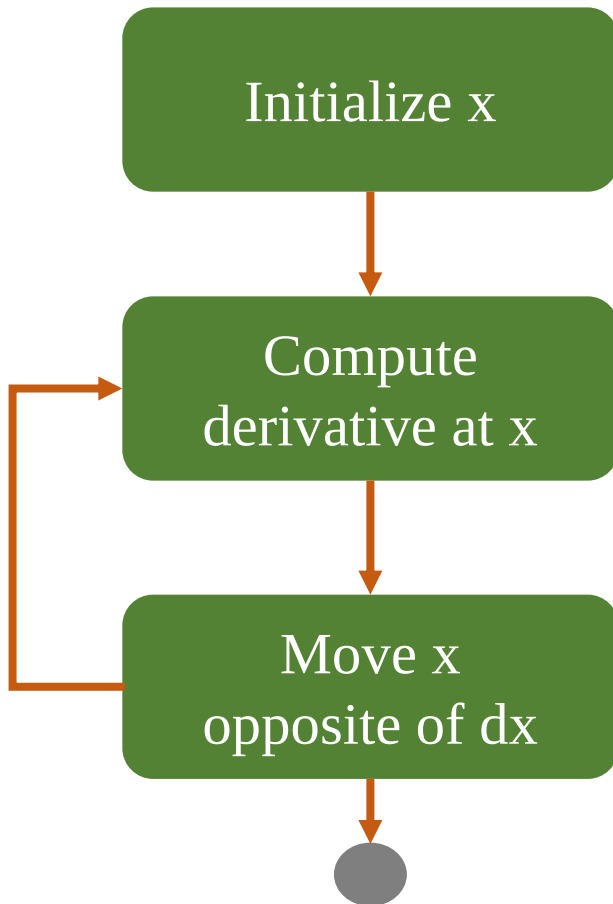
@aivietnam.edu.vn

From Linear Regression to Logistic Regression

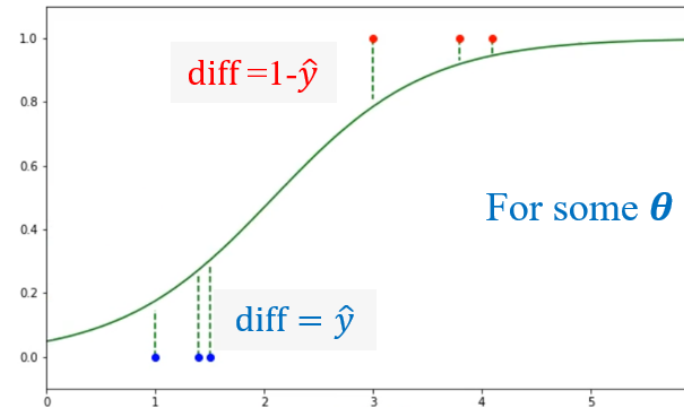
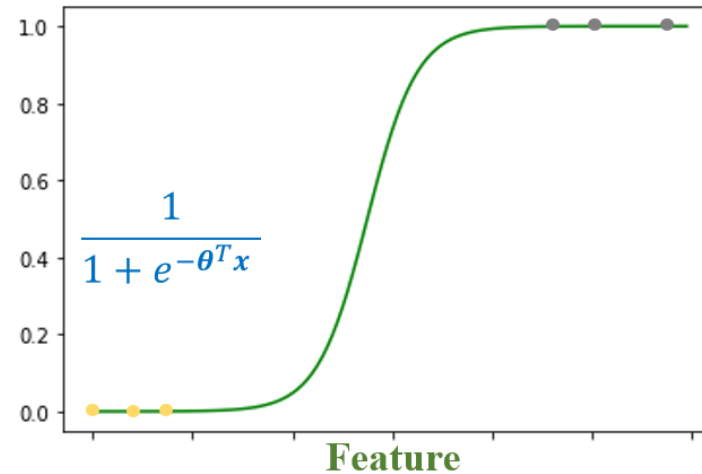
Quang-Vinh Dinh
PhD in Computer Science

Objectives

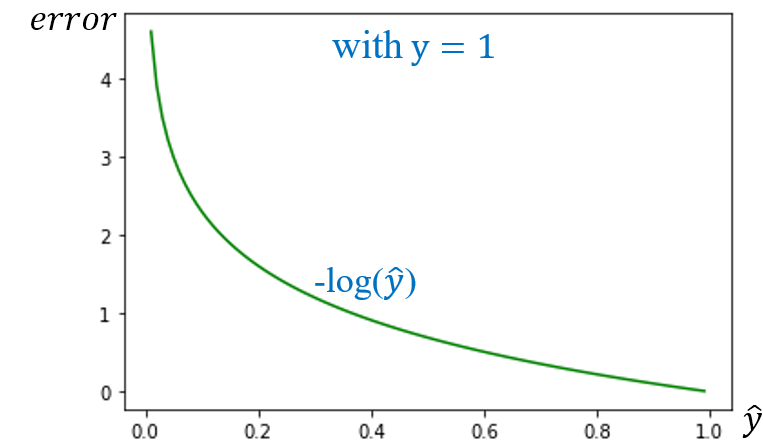
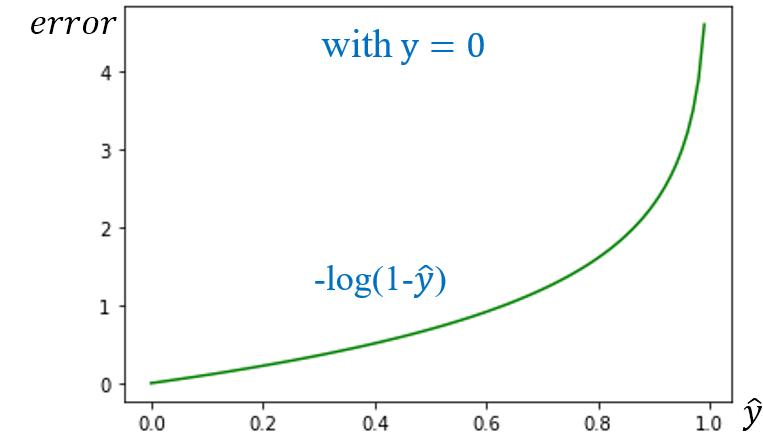
Review



Lo. R. Using MSE



Lo. R. Using BCE



Outline

SECTION 1

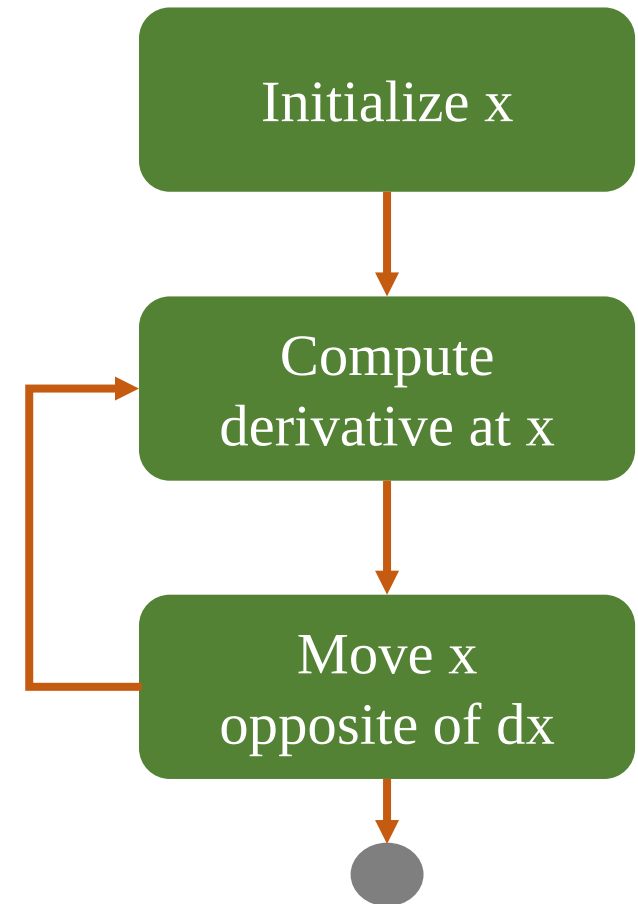
**Review Optimization
and Linear Regression**

SECTION 2

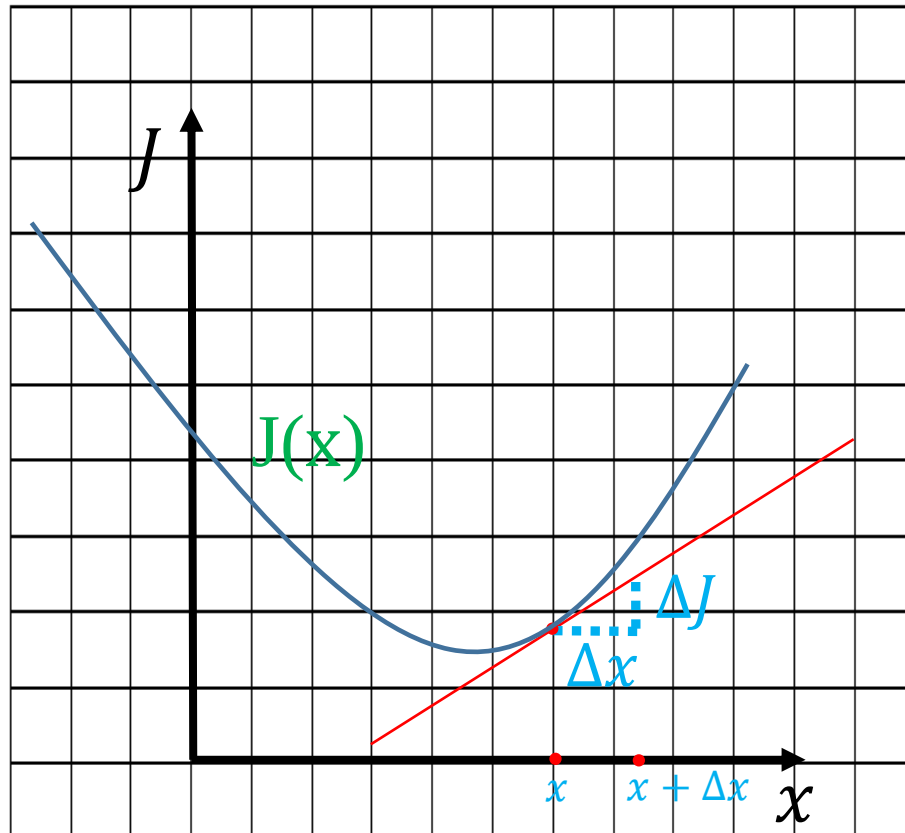
Lo.R. Using MSE

SECTION 3

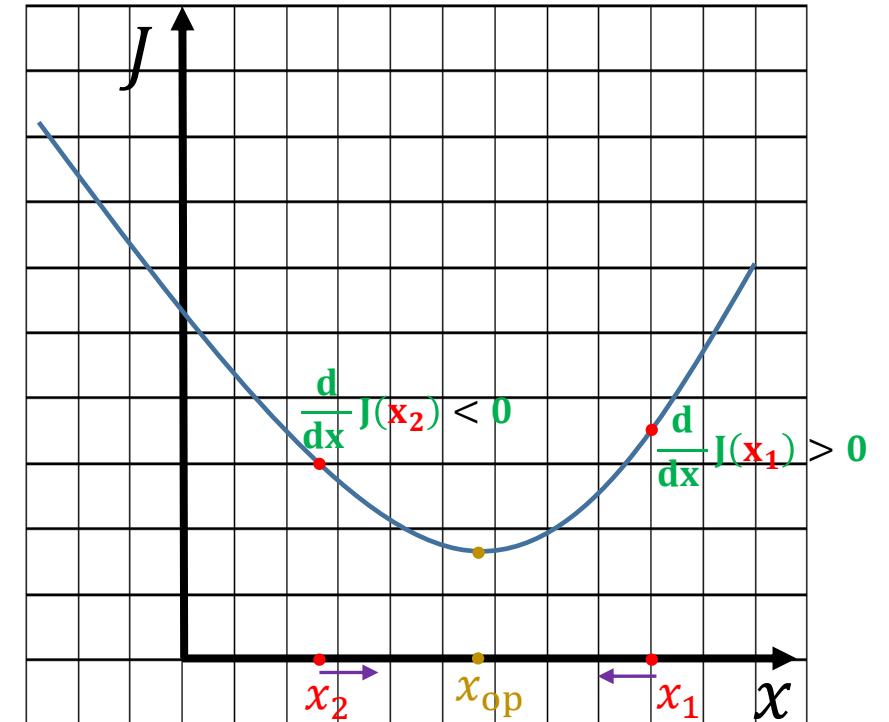
Lo.R. Using BCE



❖ Gradient descent



$$\frac{d}{dx}J(x) = \lim_{\Delta x \rightarrow 0} \frac{J(x + \Delta x) - J(x)}{\Delta x}$$

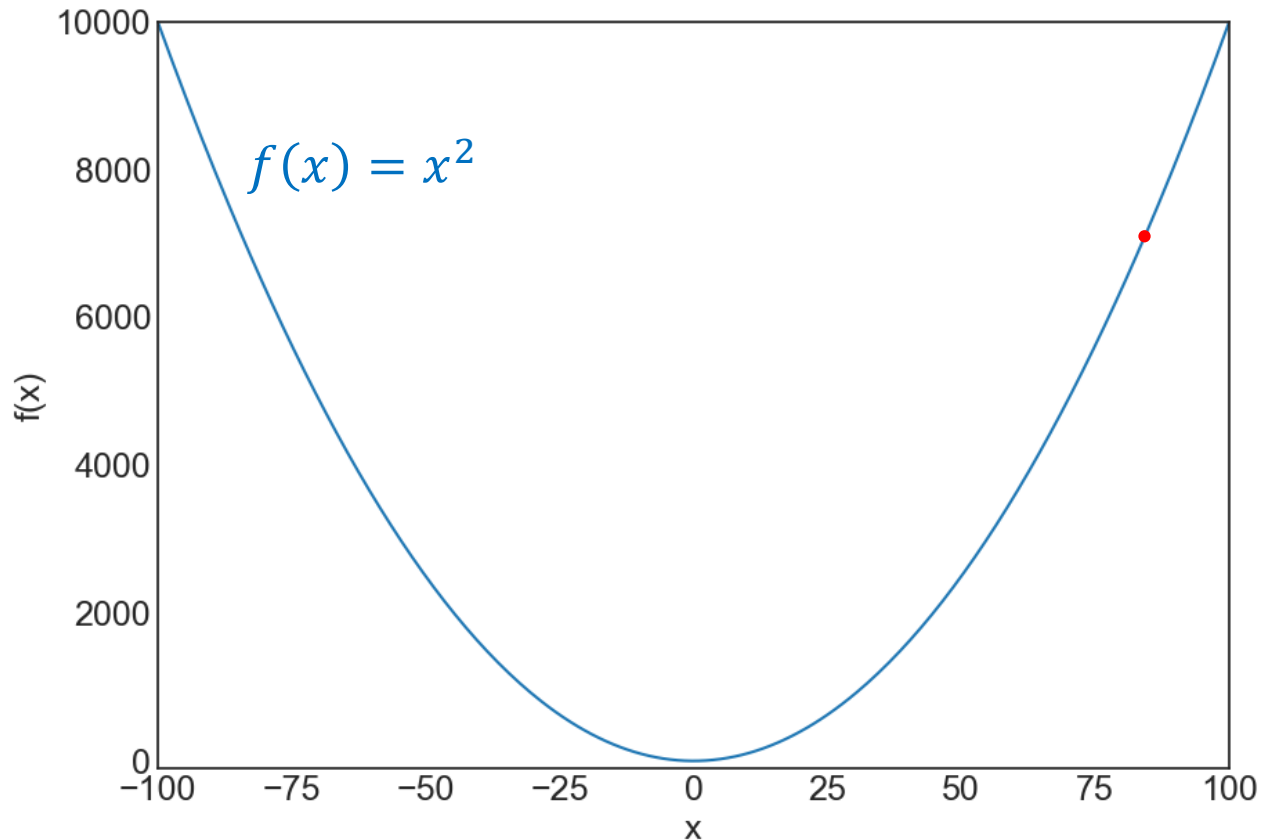


$$x_{new} = x_{old} - \eta \frac{d}{dx}J(x_{old})$$

Derivate at x_{old}

learning rate

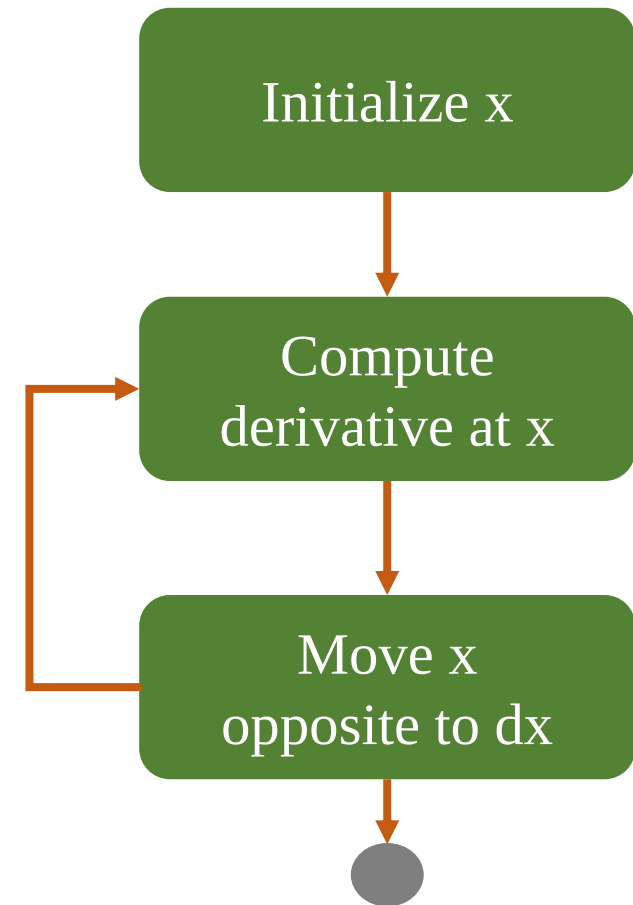
❖ Square function



$$-100 \leq x \leq 100$$

$$x \in \mathbb{N}$$

$$x_t = x_{t-1} - \eta f'(x_{t-1})$$

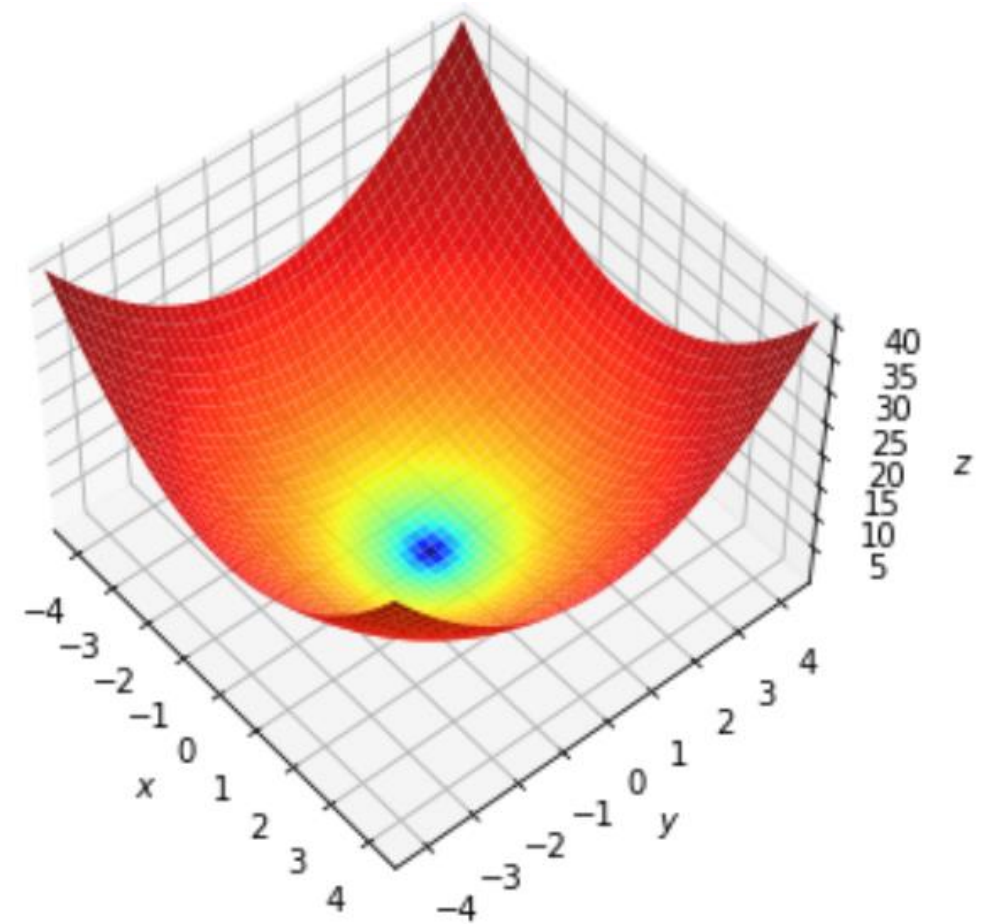
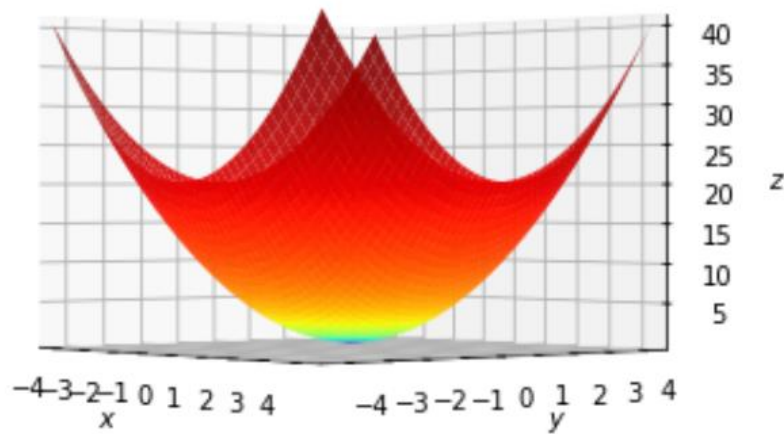


❖ Optimization: 2D function

$$f(x, y) = x^2 + y^2$$

$$-100 \leq x, y \leq 100$$

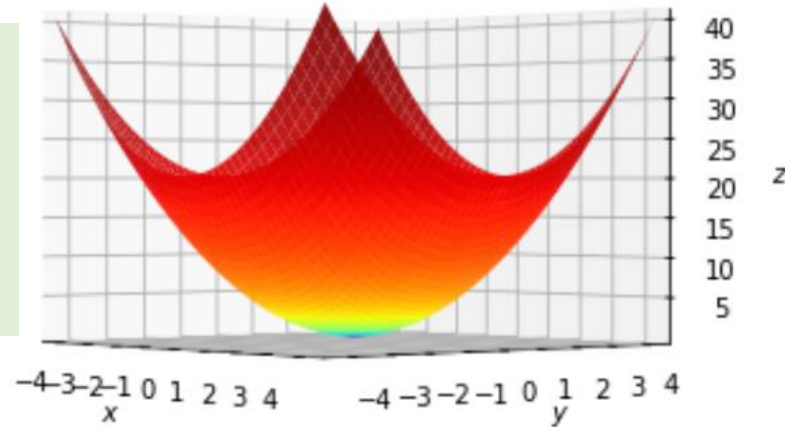
$$x, y \in \mathbb{N}$$



Derivative

❖ Optimization: 2D function

$$f(x, y) = x^2 + y^2$$
$$-100 \leq x, y \leq 100$$
$$x, y \in \mathbb{N}$$



$$x = x - \eta \frac{\partial f(x, y)}{\partial x}$$

$$y = y - \eta \frac{\partial f(x, y)}{\partial y}$$

$$\eta = 1.0$$

$$x_0 = 3.0$$

$$y_0 = 4.0$$

$$\frac{\partial f(x_0, y_0)}{\partial x} = 6.0$$

$$\frac{\partial f(x_0, y_0)}{\partial y} = 8.0$$

$$x_1 = 2.0$$

$$y_1 = 3.0$$

$$\frac{\partial f(x_1, y_1)}{\partial x} = 4.0$$

$$\frac{\partial f(x_1, y_1)}{\partial y} = 6.0$$

$$x_2 = 1.0$$

$$y_2 = 2.0$$

$$\frac{\partial f(x_2, y_2)}{\partial x} = 2.0$$

$$\frac{\partial f(x_2, y_2)}{\partial y} = 4.0$$

$$x_3 = 0.0$$

$$y_3 = 1.0$$

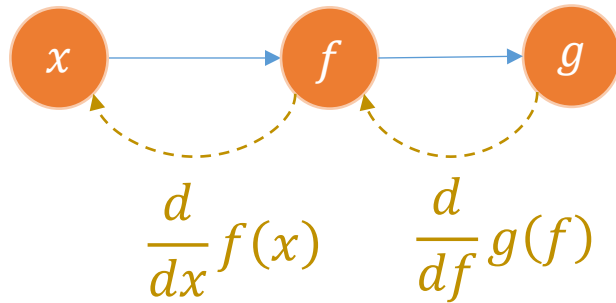
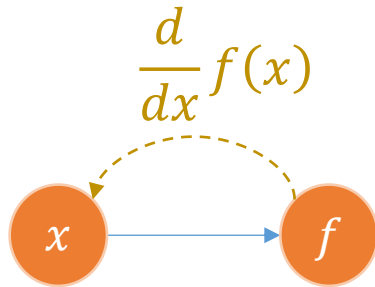
$$\frac{\partial f(x_3, y_3)}{\partial x} = 0.0$$

$$\frac{\partial f(x_3, y_3)}{\partial y} = 0.0$$

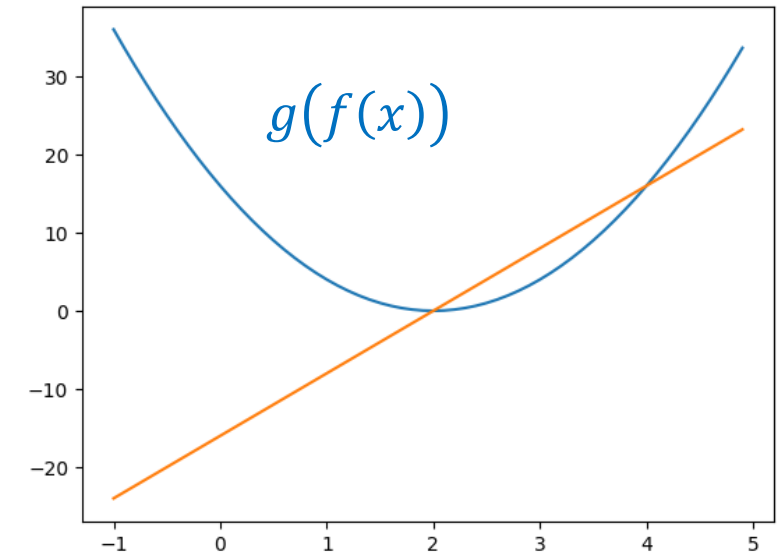
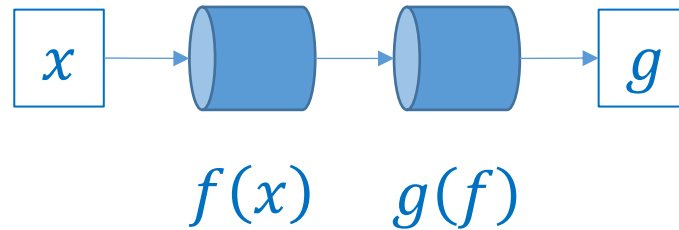
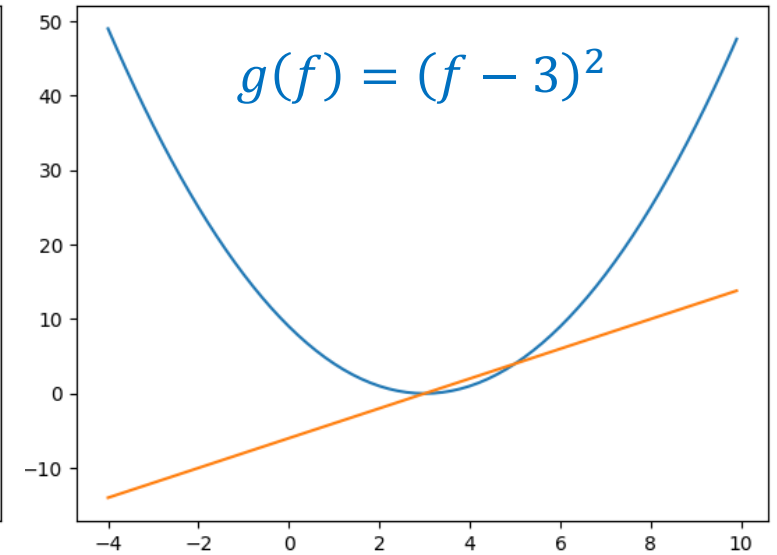
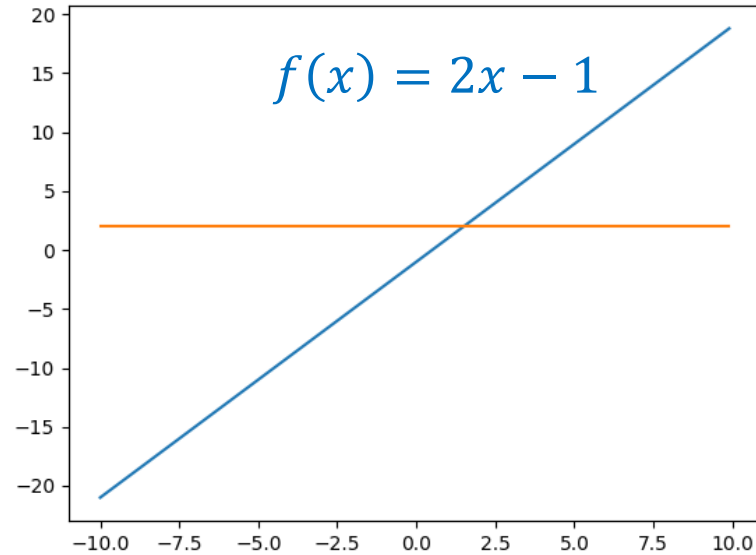
$$x_4 = 0.0$$

$$y_4 = 0.0$$

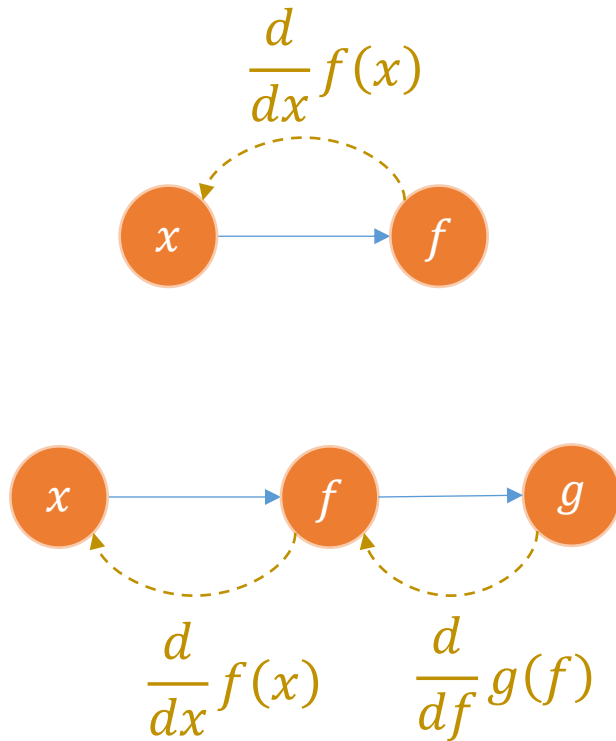
❖ For composite function



$$\frac{d}{dx} g(f(x)) = \left[\frac{d}{df} g(f) \right] * \left[\frac{d}{dx} f(x) \right]$$



❖ For composite function and chain rule



$$\frac{d}{dx} g(f(x)) = \left[\frac{d}{df} g(f) \right] * \left[\frac{d}{dx} f(x) \right]$$



$$f(x) = 2x - 1$$

$$g(f) = (f - 3)^2$$

$$f'(x) = 2$$

$$g'(f) = 2(f - 3)$$



$$\frac{dg}{dx} = \frac{dg}{df} \frac{df}{dx}$$

$$= 2(f - 3)2$$

$$= 4(2x - 1 - 3)$$

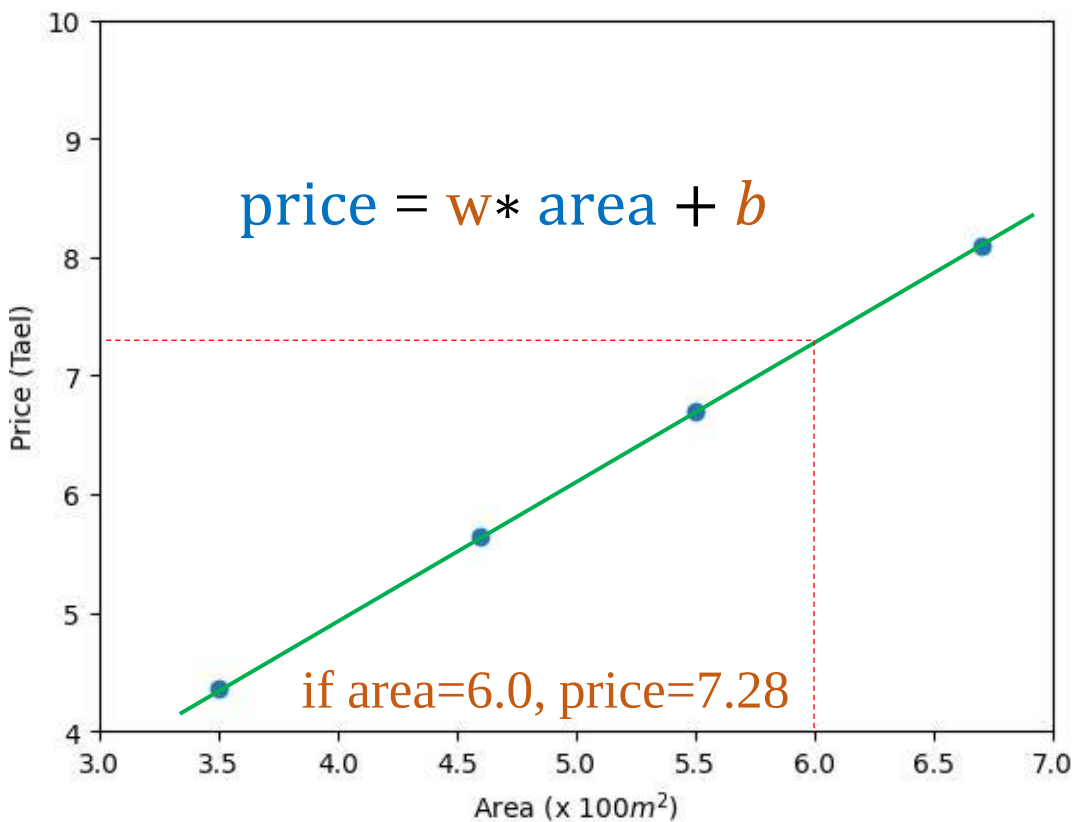
$$= 8x - 16$$

Review Linear Regression

House Price Prediction

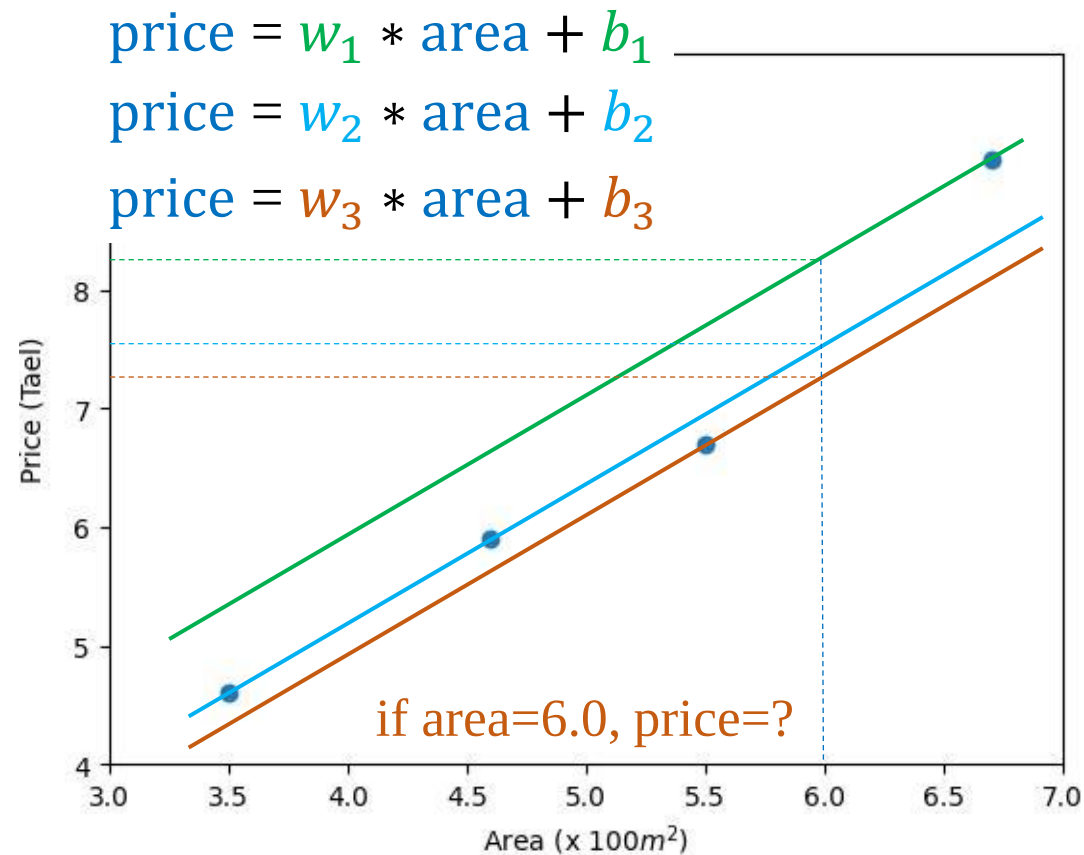
House price data

Feature	Label
area	price
6.7	8.1
4.6	5.6
3.5	4.3
5.5	6.7



Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

House price data



❖ Area-based house price prediction

$$\text{predicted_price} = w * \text{area} + b$$

$$\text{error} = (\text{predicted_price} - \text{real_price})^2$$

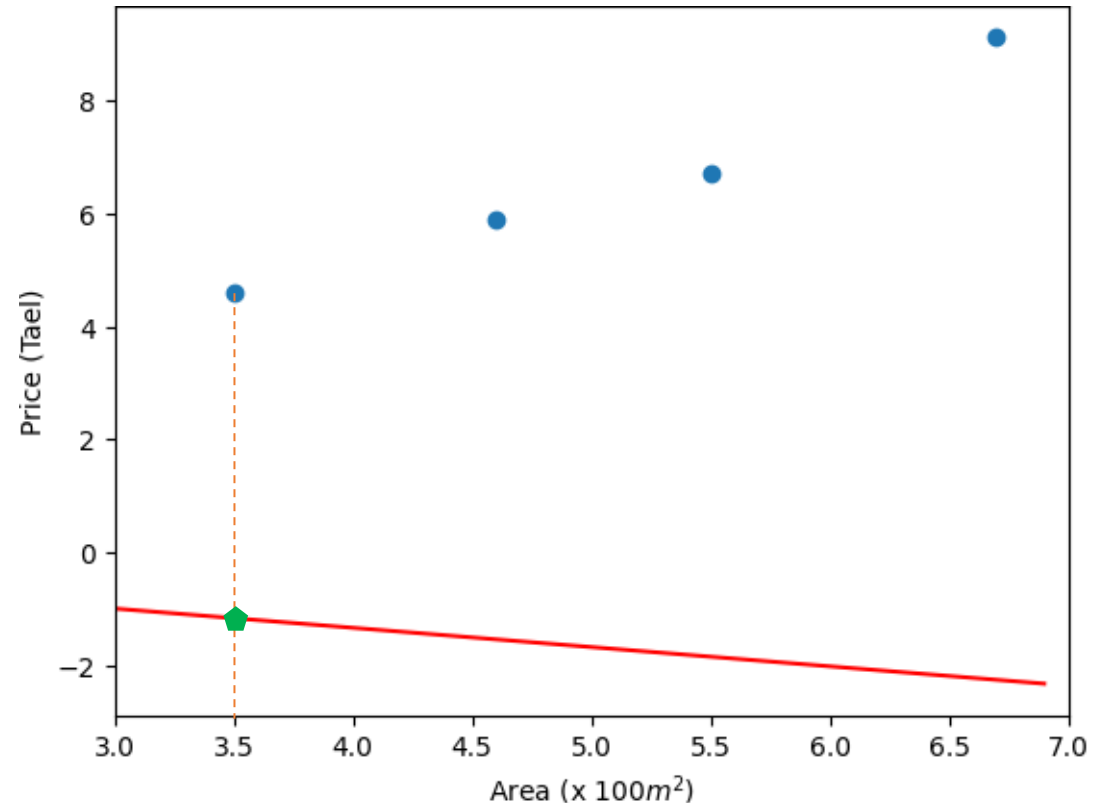
$$\hat{y} = wx + b$$

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

area	price	predicted	error
6.7	9.1	-2.238	128.55
4.6	5.9	-1.524	55.11
3.5	4.6	-1.15	33.06
5.5	6.7	-1.83	72.76

$$w = -0.34$$

$$b = 0.04$$



❖ Area-based house price prediction

$$\text{predicted_price} = w * \text{area} + b$$

$$\text{error} = (\text{predicted_price} - \text{real_price})^2$$

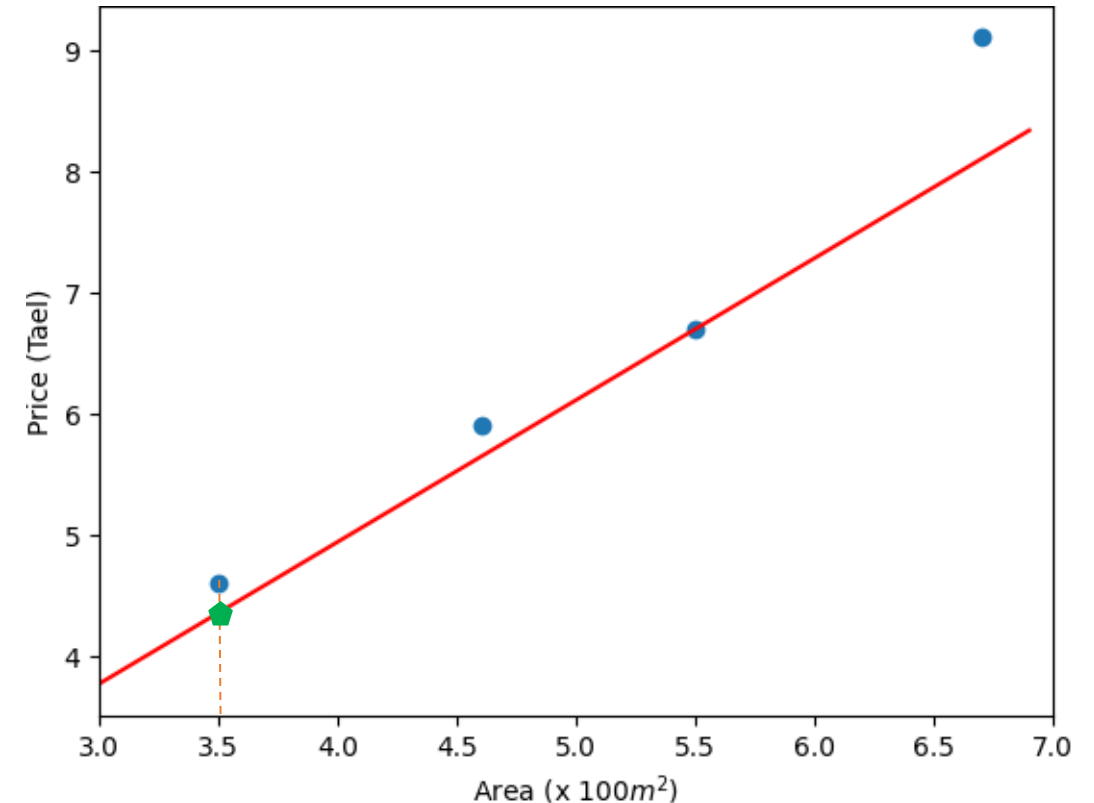
$$\hat{y} = wx + b$$

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

area	price	predicted	error
6.7	9.1	8.099	1.002
4.6	5.9	5.642	0.066
3.5	4.6	4.355	0.06
5.5	6.7	6.695	0.00002

$$w = 1.17$$

$$b = 0.26$$



Linear Regression

10

Linear equation

$$\hat{y} = wx + b$$

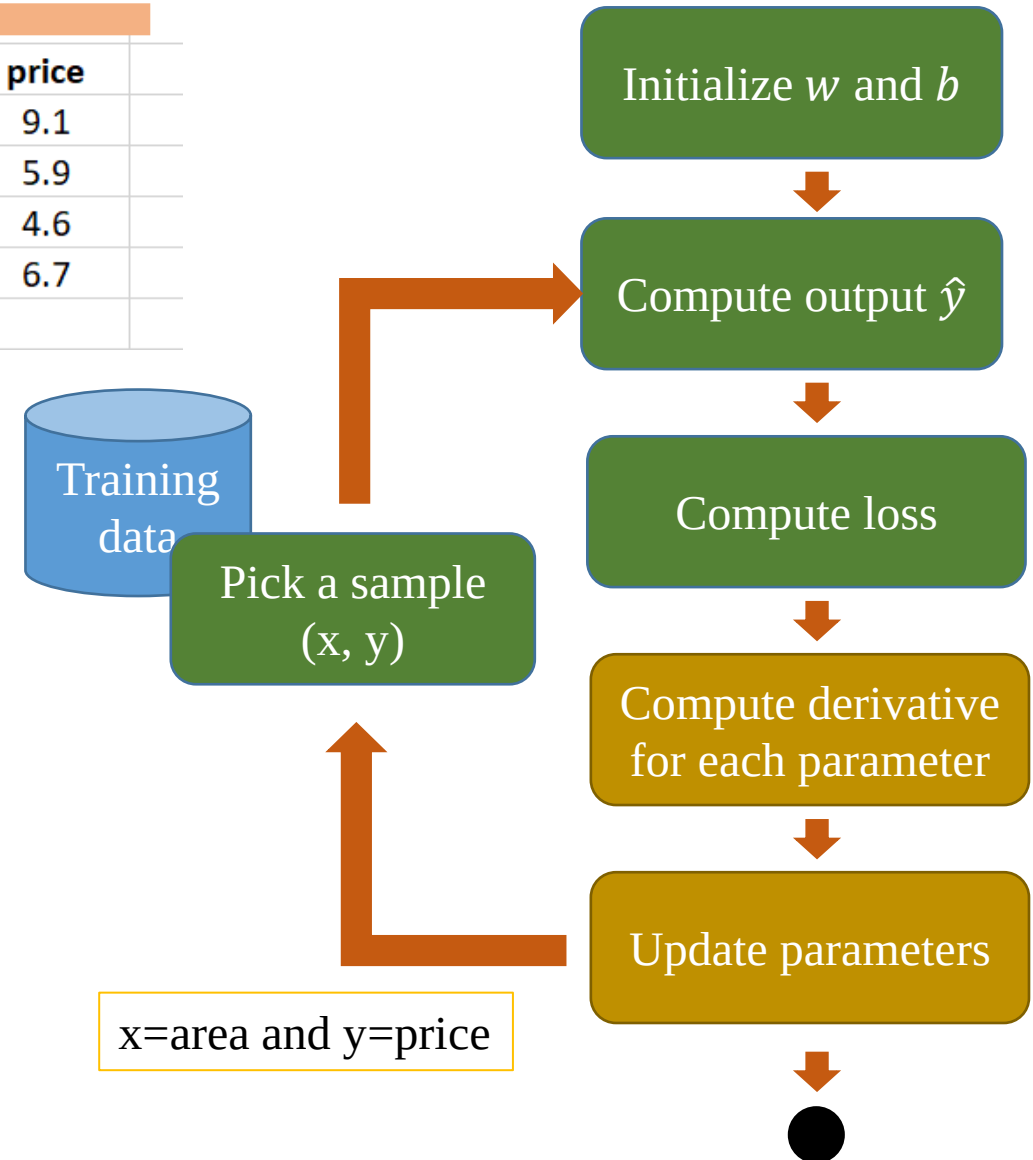
where $\hat{y}$ is a predicted value,
 w and b are parameters
and x is input feature

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Error (loss) computation

Idea: compare predicted values $\hat{y}$ and label values y
Squared loss

$$L(\hat{y}, y) = (\hat{y} - y)^2$$



Linear equation

$$\hat{y} = wx + b$$

where $\hat{y}$ is a predicted value,

w and b are parameters

and x is input feature

Error (loss) computation

Idea: compare predicted values $\hat{y}$ and label values y

Squared loss

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

Find better w and b

Use gradient descent to minimize the loss function

Compute derivate for each parameter

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w} = 2x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b} = 2(\hat{y} - y)$$

Update parameters

$$w = w - \eta \frac{\partial L}{\partial w} \quad b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Outline

SECTION 1

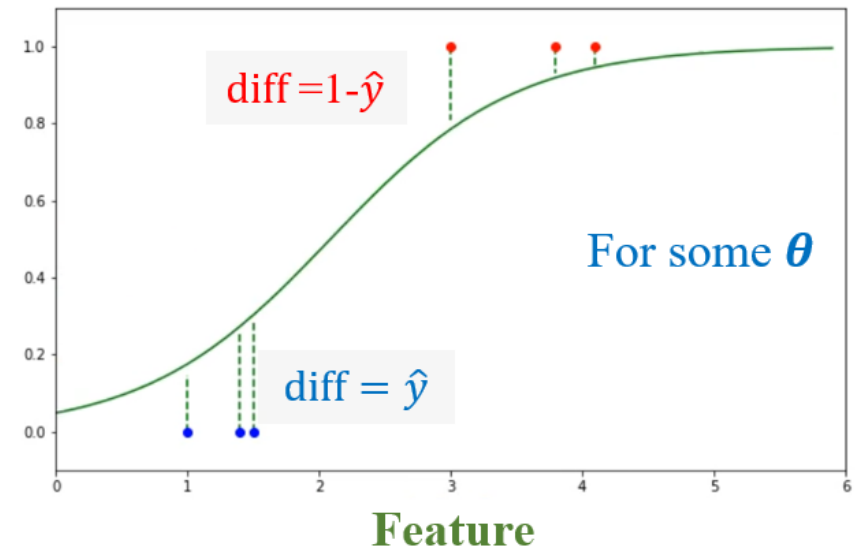
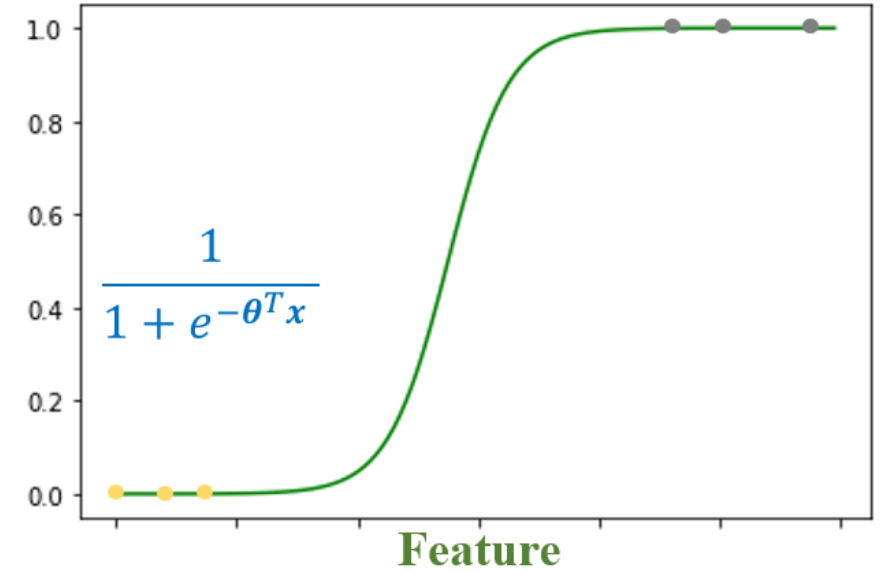
Review

SECTION 2

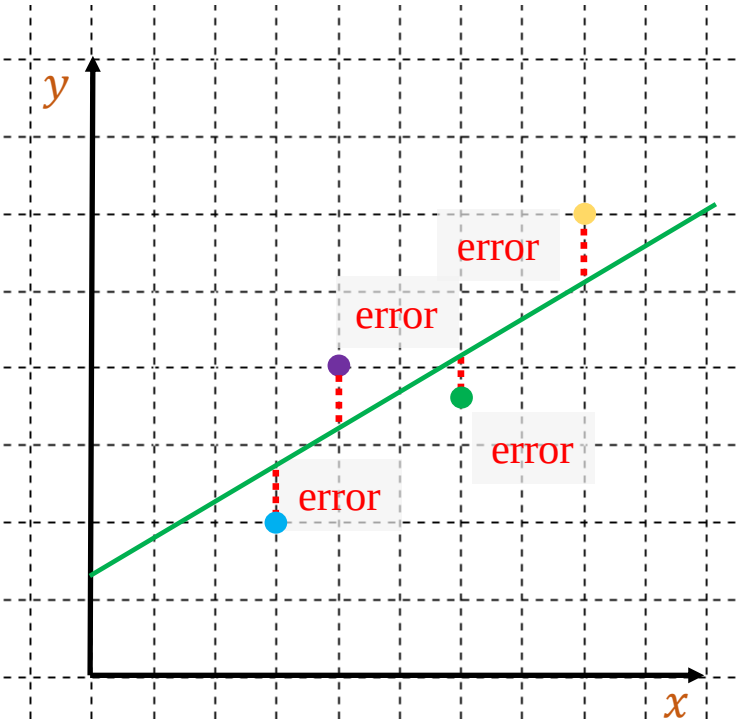
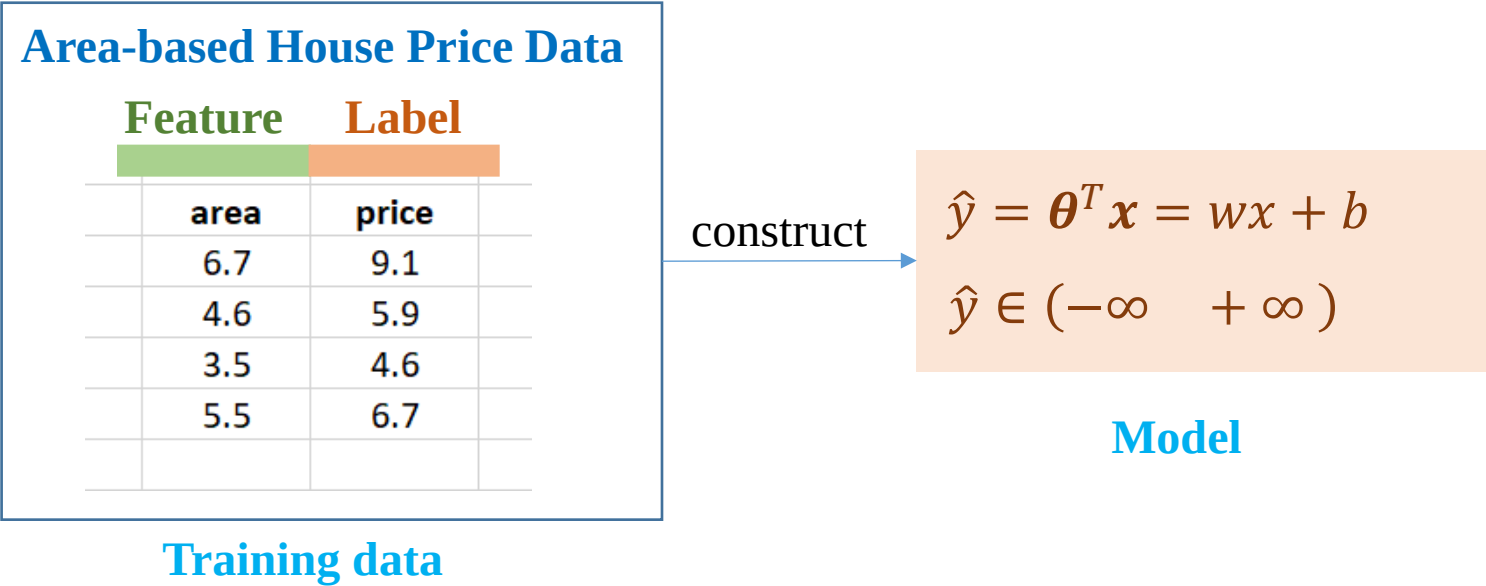
Lo.R. Using MSE

SECTION 3

Lo.R. Using BCE



❖ Using the approach of Linear regression



Find the line $\hat{y} = \theta^T x$ that is best fitting to given data,
then use $\hat{y}$ to predict for new data

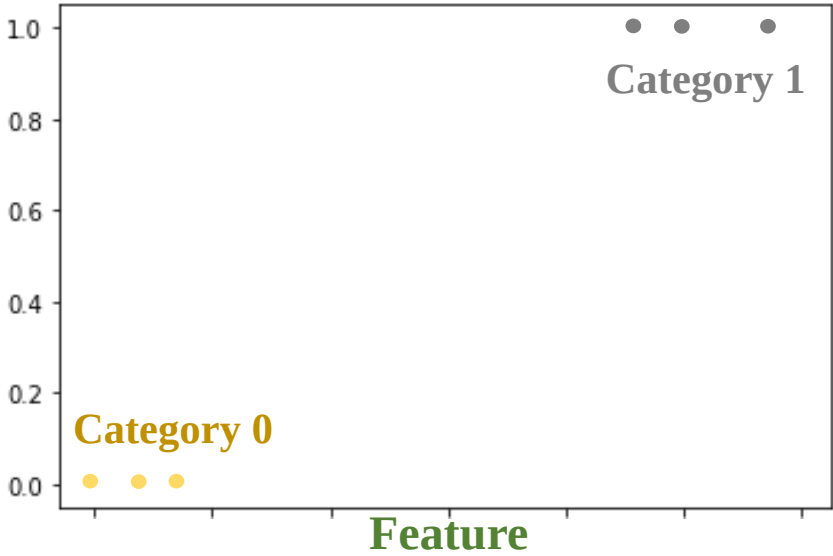
❖ Given a new kind of data

Feature	Label	
Petal_Length	Category	
1.4	Flower A	Category 0
1	Flower A	
1.5	Flower A	
3	Flower B	Category 1
3.8	Flower B	
4.1	Flower B	

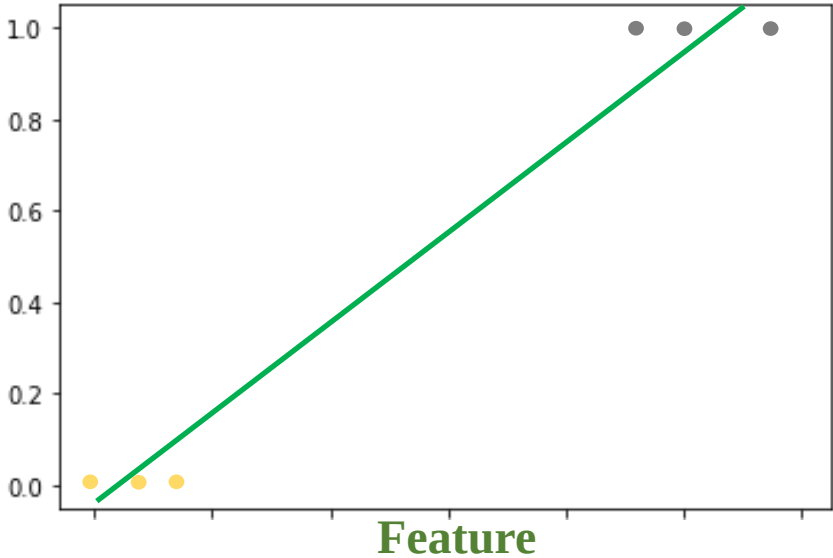
↓ Assign numbers
to categories

Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

Plot data

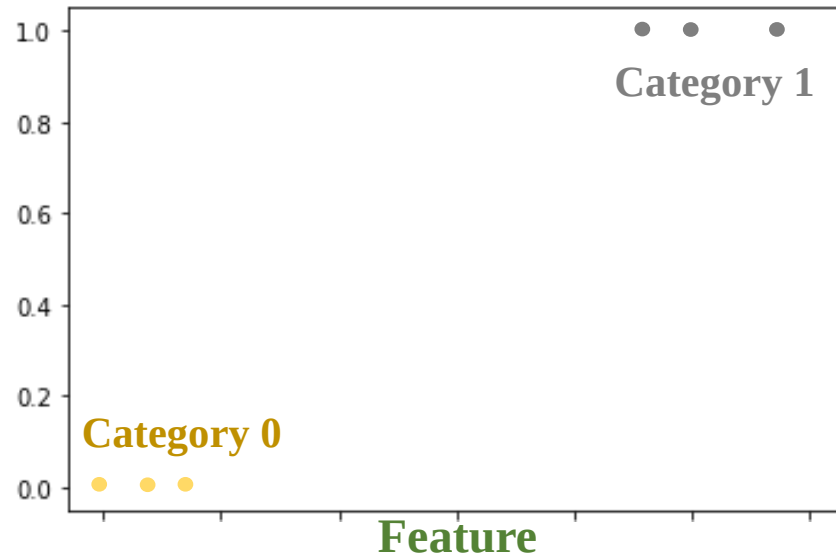


A line is not suitable for this data



What function?

Discussion



$$\hat{y} = \boldsymbol{\theta}^T \boldsymbol{x} = wx + b$$

$$\hat{y} \in (-\infty \quad +\infty)$$

❖ Look up what science has done!

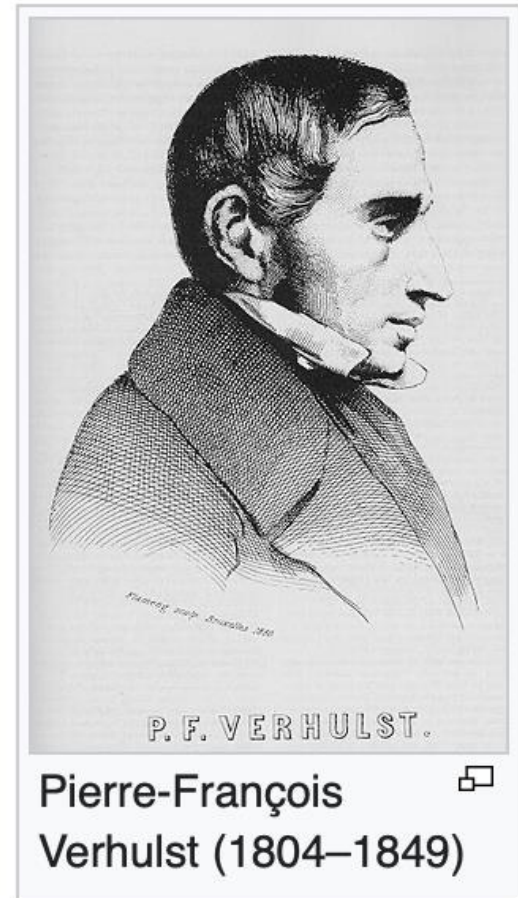
Phương trình vi phân của mô hình logistic growth có dạng:

$$\frac{dy}{dt} = ry \left(1 - \frac{y}{K} \right)$$

- ✓ $y^{(t)}$: Kích thước quần thể tại thời gian t
- ✓ r : Tốc độ tăng trưởng nội tại
- ✓ K : Sức chứa tối đa

Biểu diễn rằng sự thay đổi của quần thể theo thời gian phụ thuộc vào hai yếu tố:

- ✓ Tốc độ tăng trưởng tỉ lệ thuận với kích thước quần thể hiện tại.
- ✓ Hiệu ứng giảm tốc khi kích thước quần thể tiến gần tới giới hạn sức chứa K .



❖ Giải phương trình vi phân

Đầu tiên, chúng ta sẽ tách biến:

$$\frac{1}{y \left(1 - \frac{y}{K}\right)} dy = r dt$$

Để giải phương trình này, chúng ta sẽ phân tích mẫu số của tích phân:

$$\frac{1}{y \left(1 - \frac{y}{K}\right)} = \frac{1}{y} + \frac{1}{K - y}$$

Nên phương trình trở thành:

$$\left(\frac{1}{y} + \frac{1}{K - y} \right) dy = r dt$$

Giờ tích phân cả hai vế:

$$\int \left(\frac{1}{y} + \frac{1}{K - y} \right) dy = \int r dt$$

Tích phân của vế trái là:

$$\ln|y| - \ln|K - y| = rt + C$$

Ở đây, C là hằng số tích phân.

❖ Đưa về dạng hàm sigmoid

Bây giờ, chúng ta sẽ sắp xếp lại phương trình này để biểu diễn y :

$$\ln \left(\frac{y}{K - y} \right) = rt + C$$

Lấy số mũ của cả hai vế:

$$\frac{y}{K - y} = e^{rt+C}$$

Đặt $A = e^C$, ta có:

$$\frac{y}{K - y} = Ae^{rt}$$

Từ đây, chúng ta có thể giải y :

$$y = \frac{AKe^{rt}}{1 + Ae^{rt}}$$

Đặt $A = \frac{y_0}{K - y_0}$, trong đó y_0 là giá trị ban đầu của y khi $t = 0$.

Sau đó, ta có lời giải hoàn chỉnh của phương trình logistic:

$$y(t) = \frac{K}{1 + \left(\frac{K - y_0}{y_0} \right) e^{-rt}}$$

❖ Liên hệ với hàm sigmoid

Hàm sigmoid có dạng:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

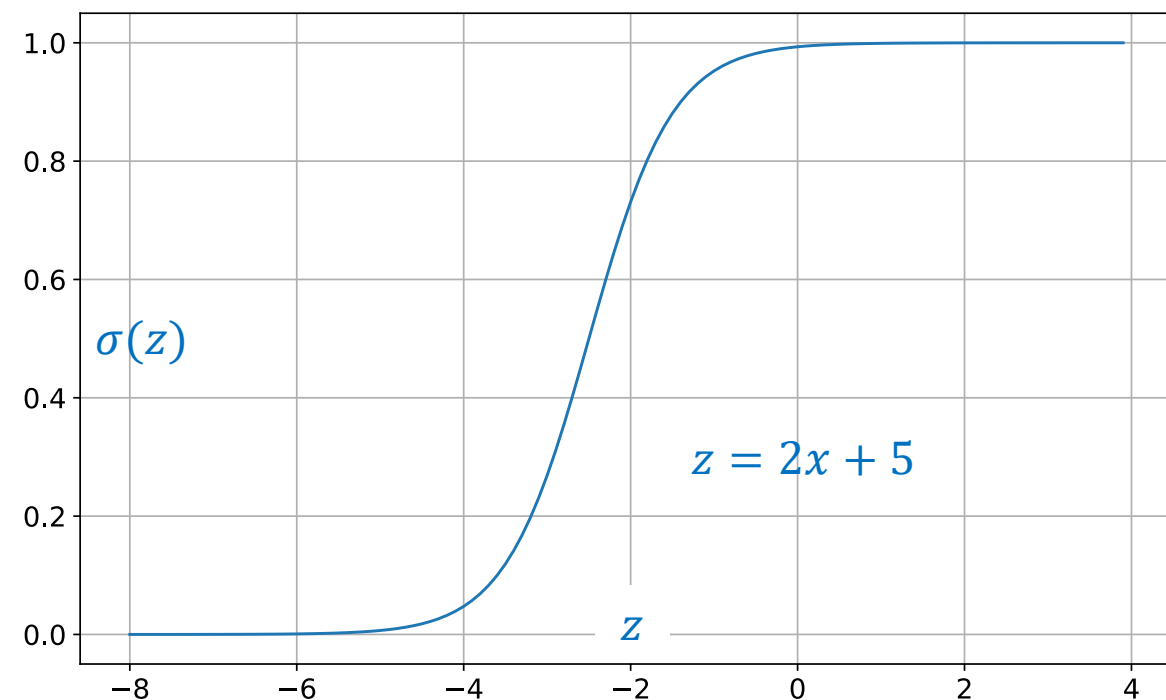
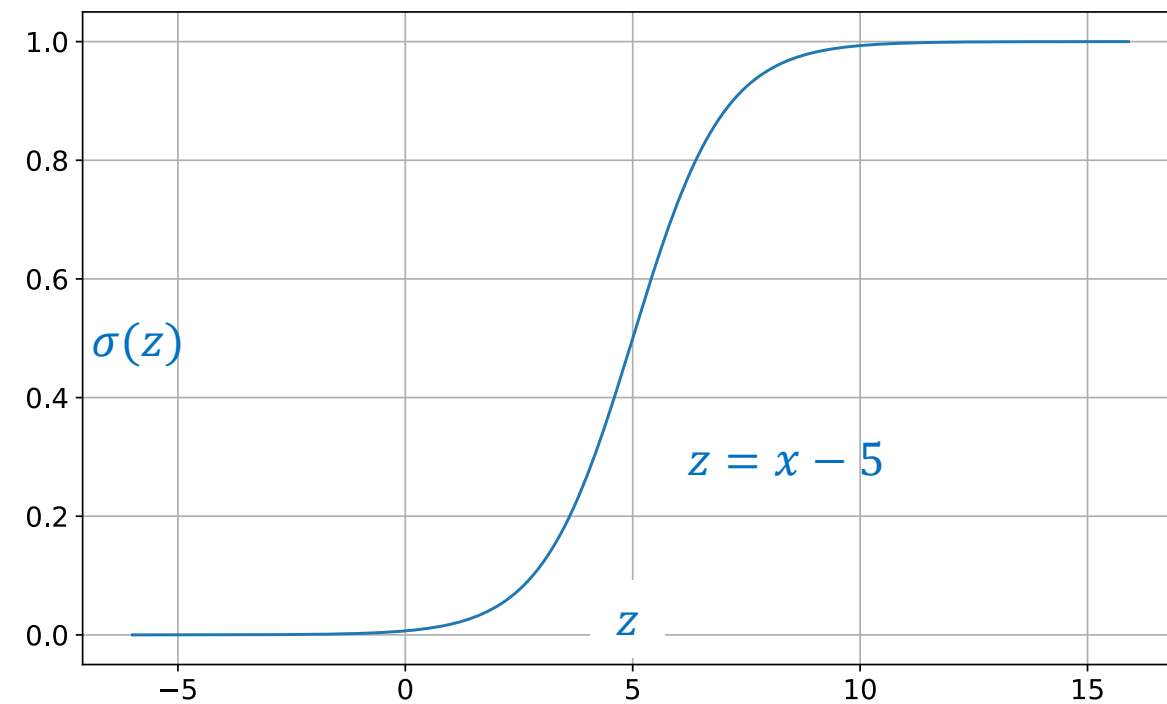
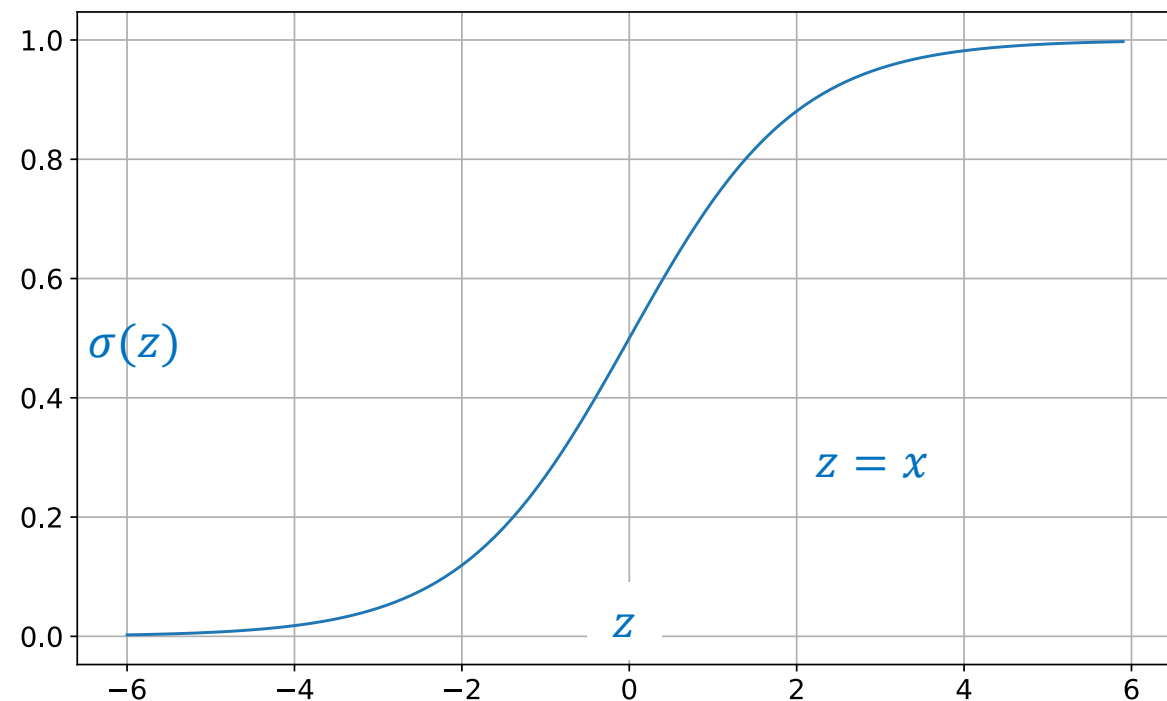
Để thấy sự tương đồng, chúng ta có thể viết lại lời giải của phương trình logistic như sau:

$$y(t) = \frac{K}{1 + e^{-rt} \left(\frac{K - y_0}{y_0} \right)}$$

Điều này cho thấy rằng hàm logistic thực chất là một dạng tổng quát của hàm sigmoid với giá trị K là giới hạn trên và rt điều khiển độ dốc.

Sigmoid function

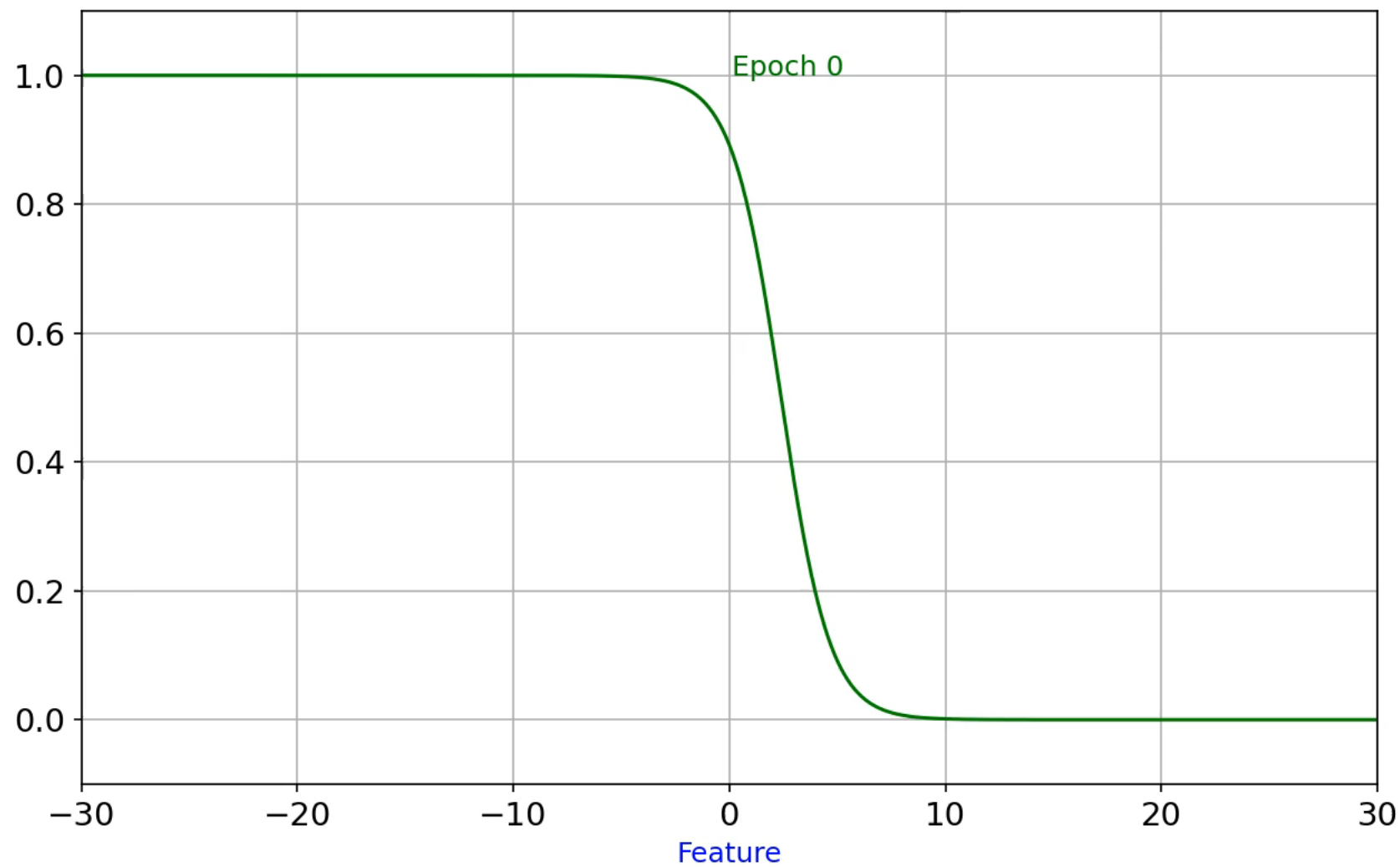
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

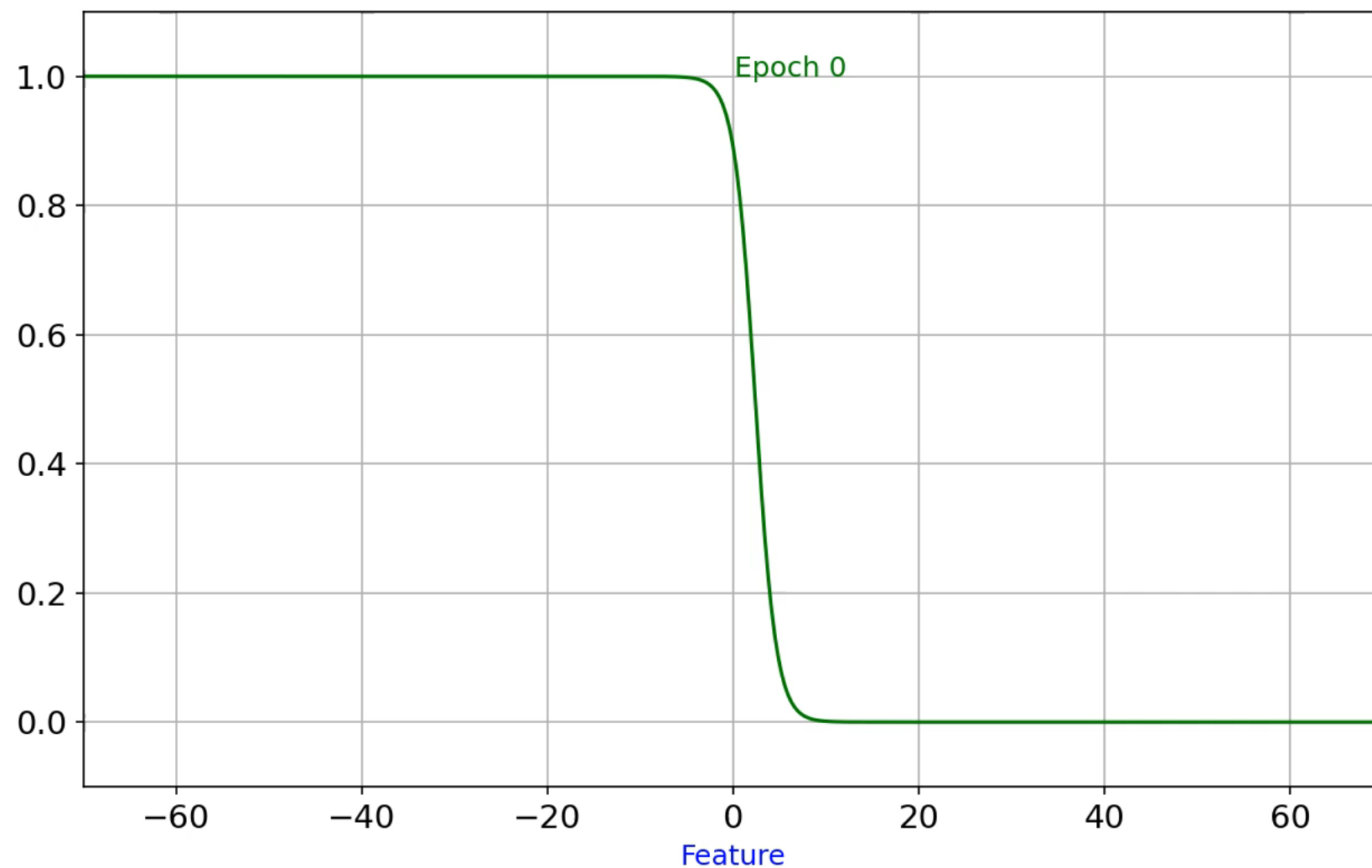
$$z = wx + b$$



Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

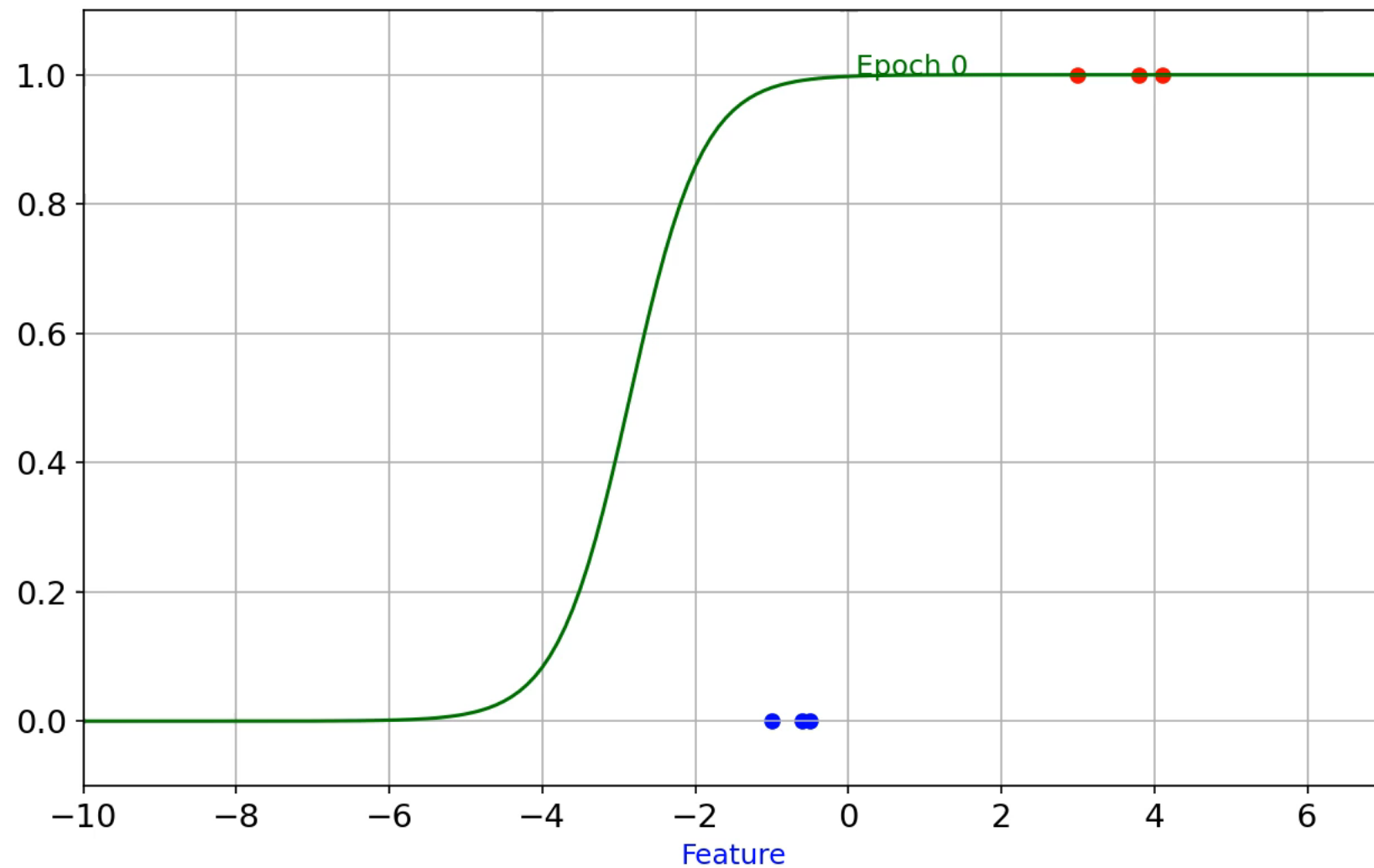
$$z = wx + b$$



Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = wx + b$$



❖ Using the Sigmoid function

Sigmoid function

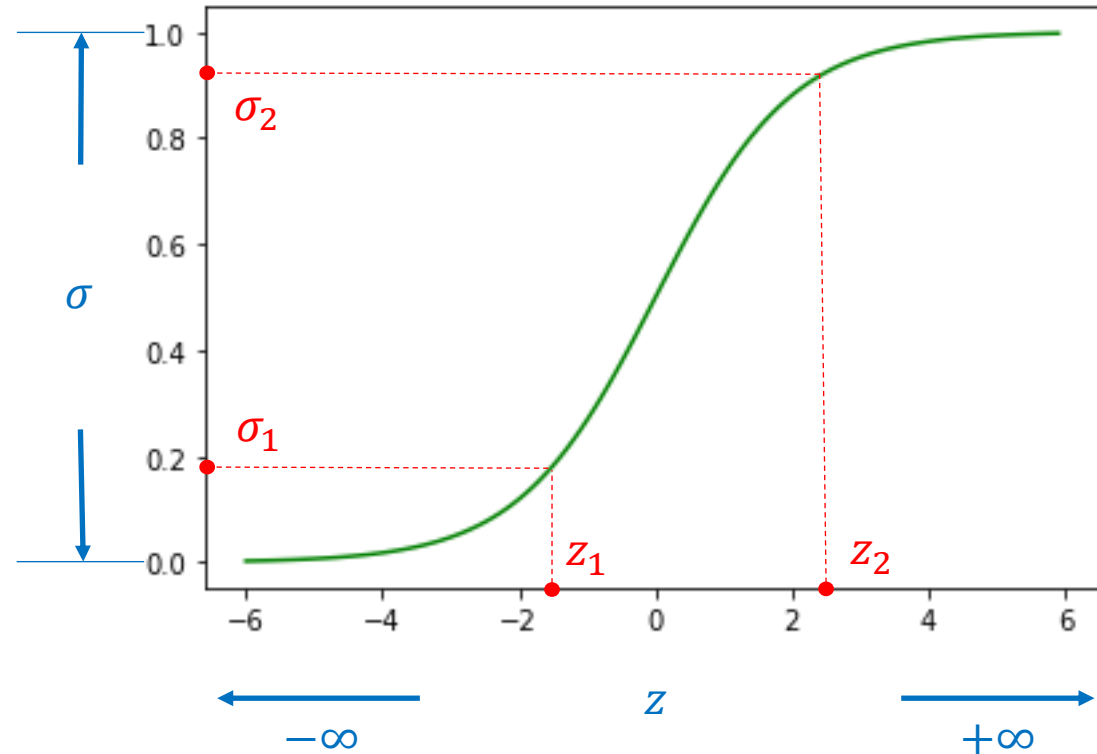
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z \in (-\infty + \infty)$$

$$\sigma(z) \in (0 \ 1)$$

Property

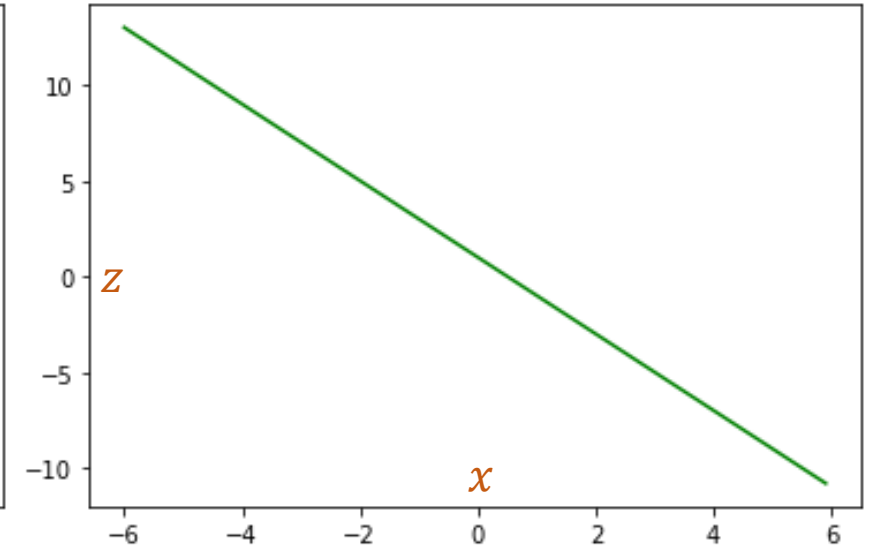
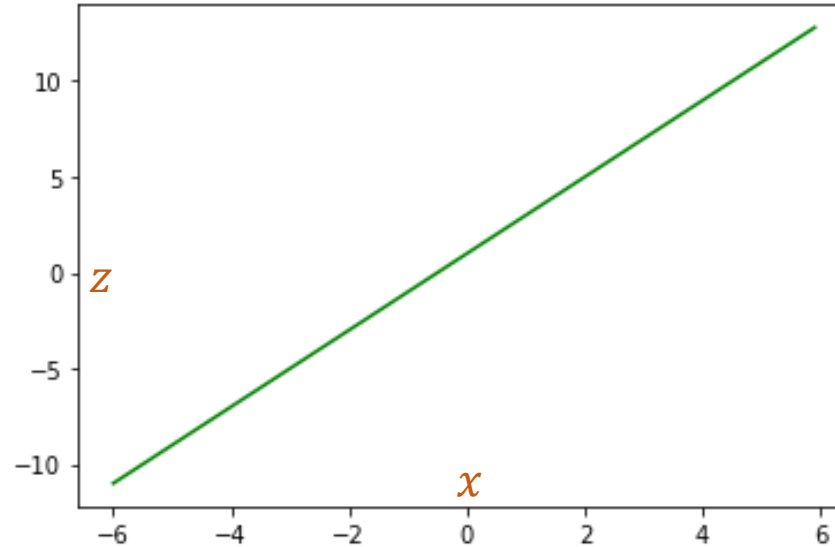
$$\forall z_1 z_2 \in [a \ b] \text{ and } z_1 \leq z_2 \\ \rightarrow \sigma(z_1) \leq \sigma(z_2)$$



❖ Using the Sigmoid function

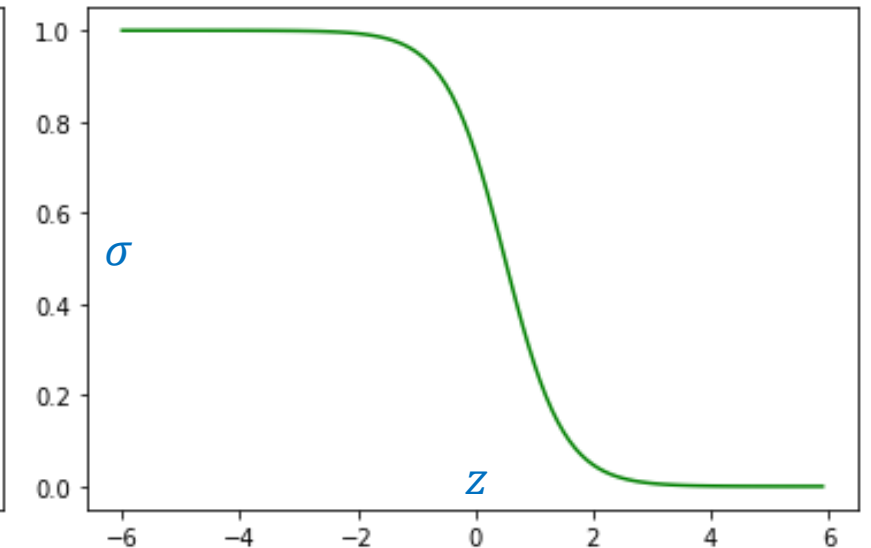
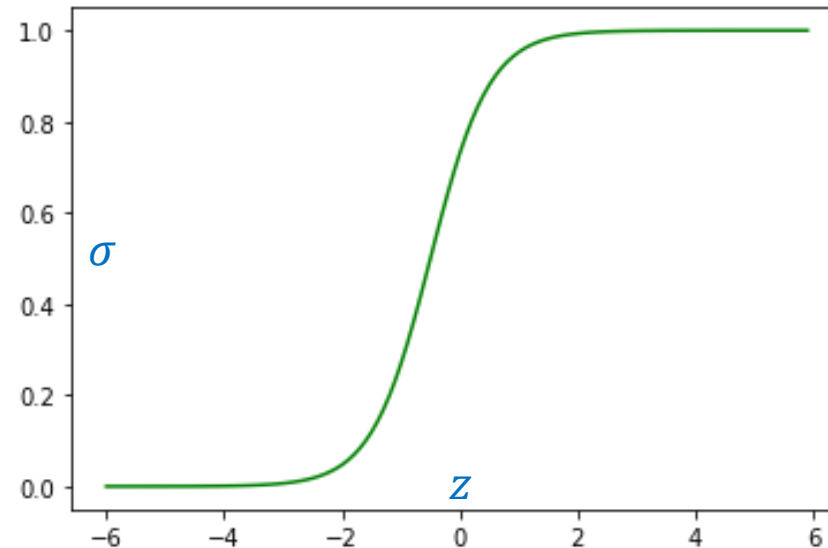
$$z = wx + b$$

$$z \in (-\infty + \infty)$$



$$z = wx + b$$
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \ 1)$$



❖ Using the Sigmoid function

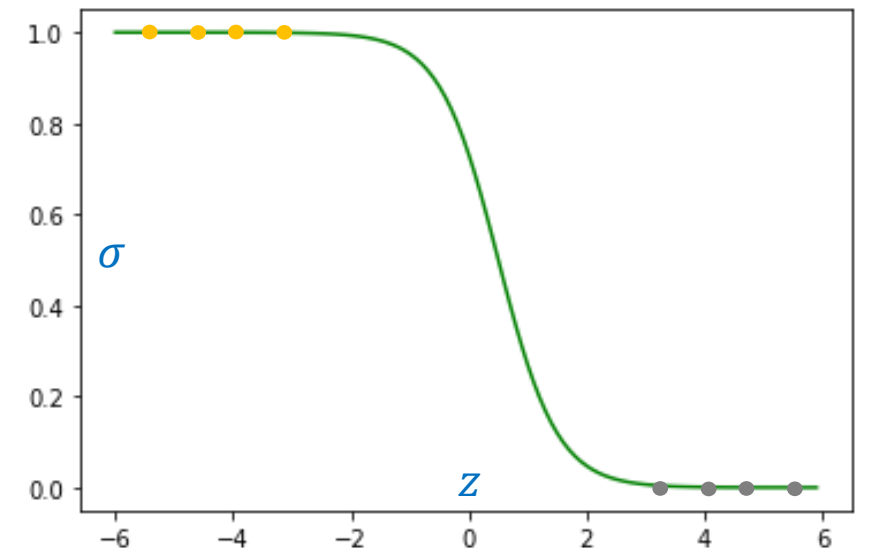
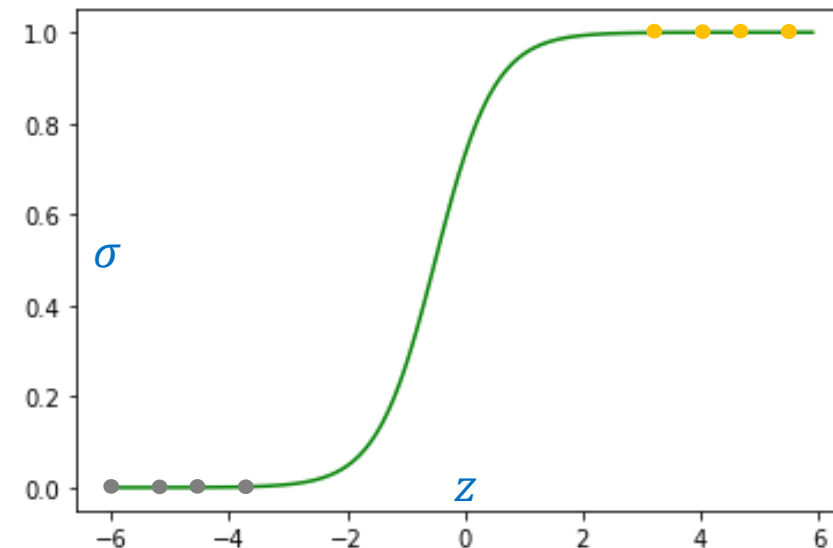
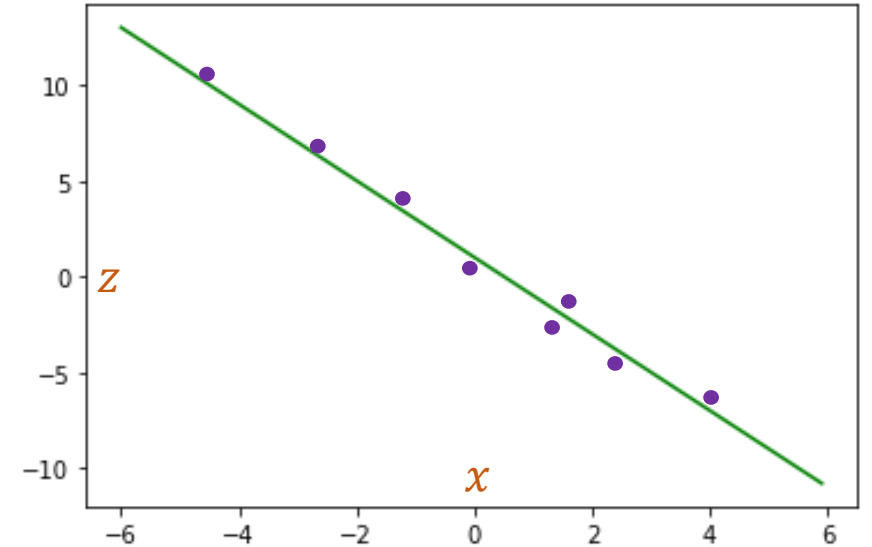
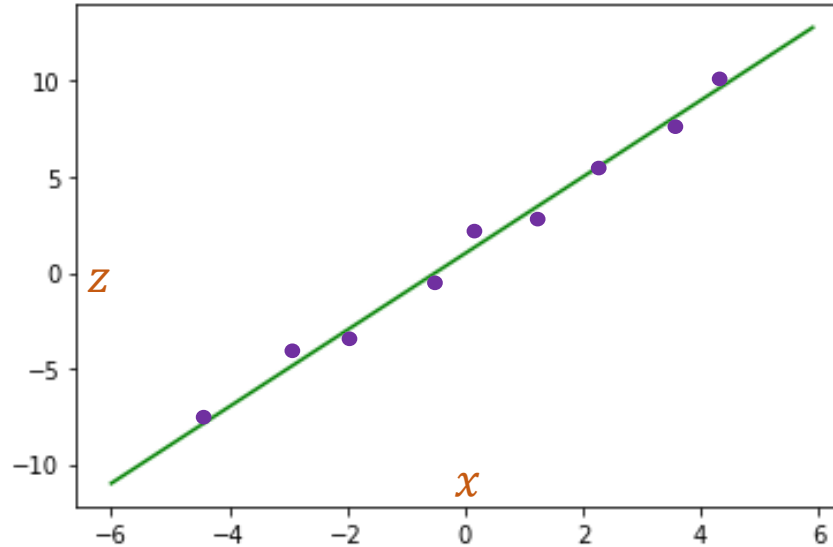
$$z = wx + b$$

$$z \in (-\infty + \infty)$$

$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \ 1)$$



❖ Using the Sigmoid function

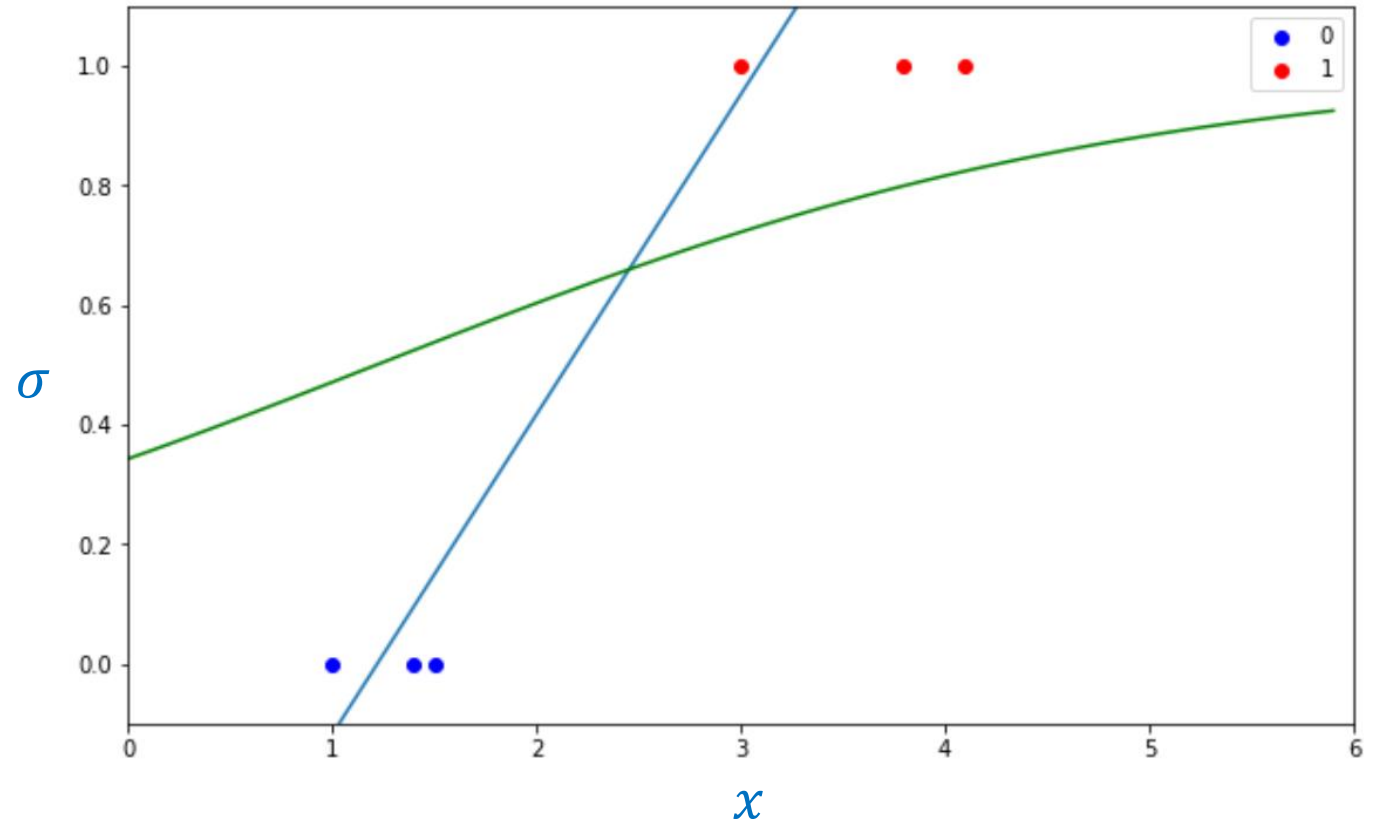
Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

z	$\sigma(z)$
0.095	0.52
-0.119	0.47
0.1485	0.53
0.951	0.72
1.379	0.79
1.5395	0.82

$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \ 1)$$



$$z = 0.535 * x - 0.654$$

❖ Using the Sigmoid function

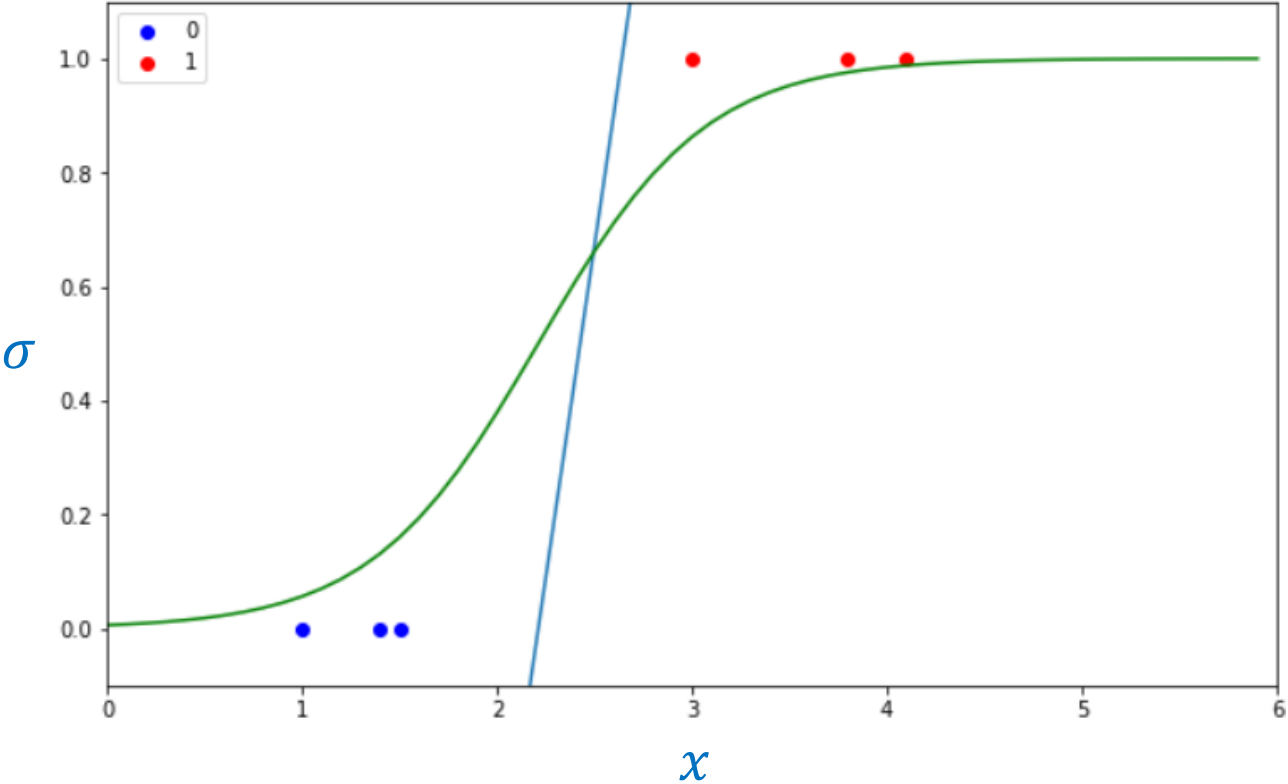
Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

z	$\sigma(z)$
-1.89	0.1309
-2.82	0.0559
-1.65	0.1598
1.837	0.8625
3.701	0.9759
4.401	0.9878

$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$



$$z = 2.331 * x - 5.156$$

❖ What about the loss function?

Feature	Label	
Petal_Length	Category	
1.4	Flower A	Category 0
1	Flower A	
1.5	Flower A	
3	Flower B	Category 1
3.8	Flower B	
4.1	Flower B	

Assign numbers to categories

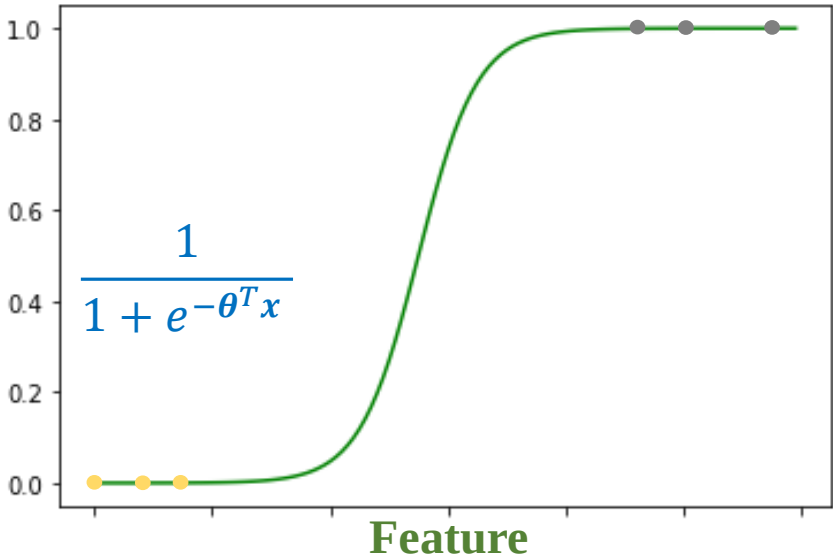
Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

Sigmoid function could fit the data

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\hat{y} \in (0 \quad 1)$$



Error

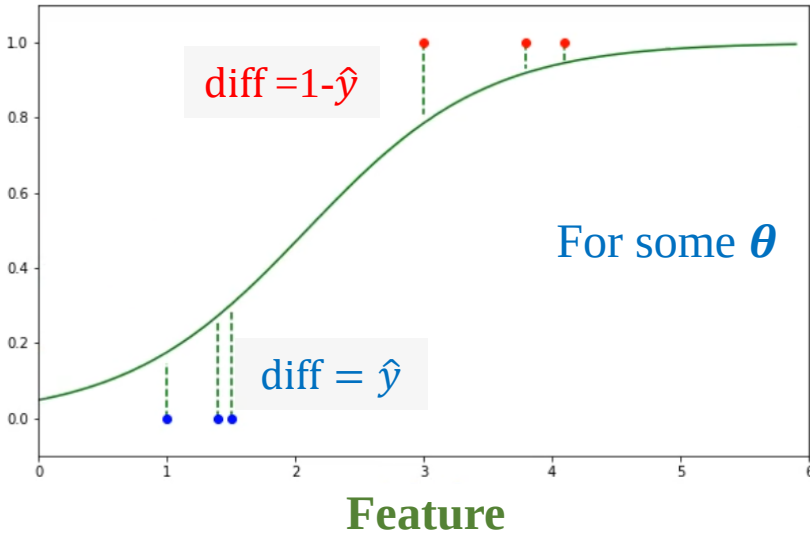
if $y = 1$

$\text{diff} = 1 - \hat{y}$

if $y = 0$

$\text{diff} = \hat{y}$

$$L(\hat{y}) = (\hat{y} - y)^2$$



❖ Construct loss

Model and Loss

$$z = \theta^T x = x^T \theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = (\hat{y} - y)^2$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i} \quad \frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) \quad \frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

Logistic Regression-MSE

❖ Implement

Model and Loss

$$z = \theta^T x = x^T \theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

```
def sigmoid_function(z):  
    return 1 / (1 + np.exp(-z))
```

```
def predict(x, theta):  
    y_hat = sigmoid_function(np.dot(x, theta))  
    return y_hat
```

```
def loss_function(y_hat, y):  
    return (y_hat - y)**2
```

```
def compute_gradient(x, y_hat, y):  
    return 2*x*(y_hat - y)*y_hat*(1 - y_hat)
```

```
# predict z  
y_hat = predict(xi, theta)
```

```
# compute loss  
loss = loss_function(y_hat, yi)
```

```
# compute gradient and update  
gradient = compute_gradient(xi, y_hat, yi)  
theta -= lr*gradient
```

Logistic Regression-MSE

❖ Result

Model and Loss

$$z = \theta^T x = x^T \theta$$

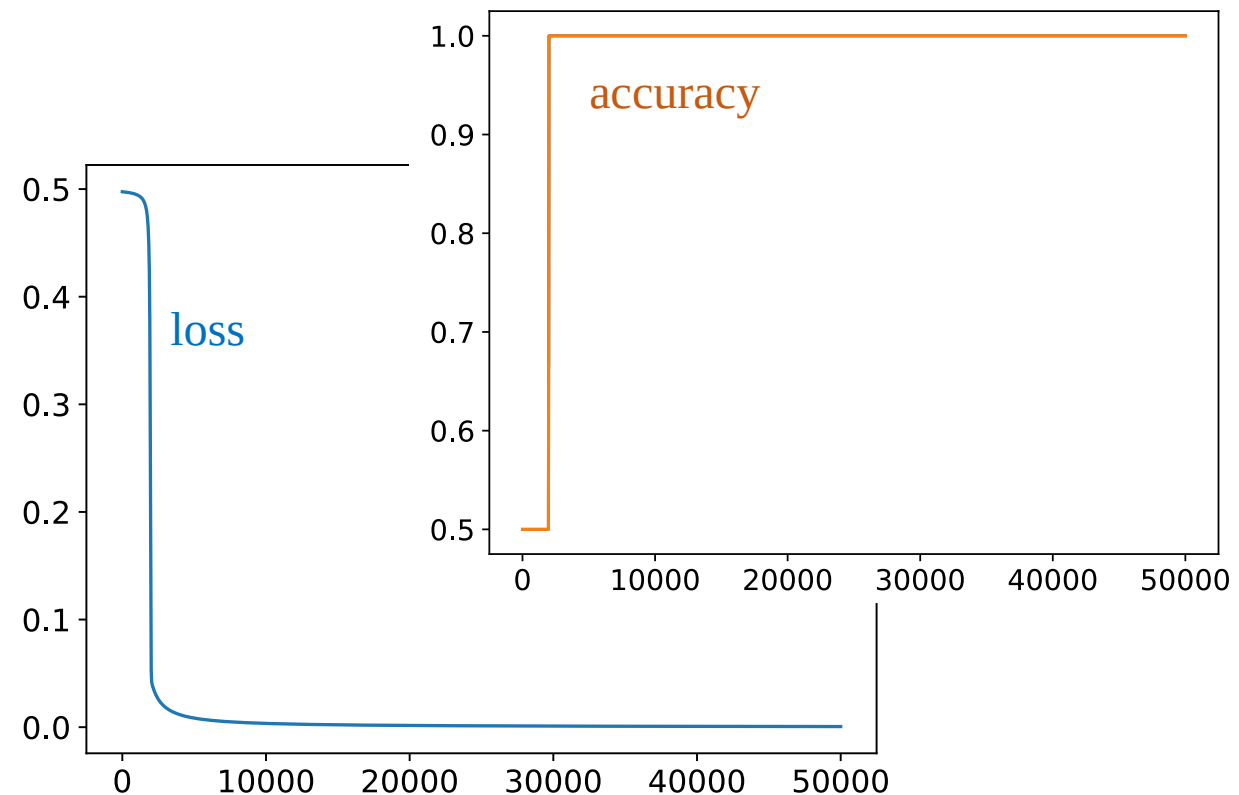
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

Done?

Feature	Label	
Petal Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	



Discussion

❖ Definition

The Hessian matrix or Hessian is a square matrix of second-order partial derivatives of a scalar-valued function

https://en.wikipedia.org/wiki/Hessian_matrix

Given $f(x, y)$

$f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

Given $f(x, y) = x^2 + 2x^2y + y^3$

$$\frac{\partial f}{\partial x} = 2x + 4xy$$

$$\frac{\partial f}{\partial y} = 2x^2 + 3y^2$$

$$H_f = \begin{bmatrix} 2 + 4y & 4x \\ 4x & 6y \end{bmatrix}$$

❖ Formulae

Model and Loss

$$z = \theta^T x = x^T \theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = (\hat{y} - y)^2$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i} \quad \frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) \quad \frac{\partial z}{\partial \theta_i} = x_i$$

First-order partial derivatives

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

Mean Squared Error

Model and Loss

$$z = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = (\hat{y} - y)^2$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i} \quad \frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) \quad \frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y}) = 2x_i(-\hat{y}^3 + \hat{y}^2 - y\hat{y} + y\hat{y}^2)$$

$$\frac{\partial^2 L}{\partial \theta_i^2} = \frac{\partial}{\partial \theta_i} [2x_i(-\hat{y}^3 + \hat{y}^2 - y\hat{y} + y\hat{y}^2)]$$

$$= 2x_i[-3\hat{y}^2 x_i \hat{y}(1 - \hat{y}) + 2x_i \hat{y} \hat{y}(1 - \hat{y}) - y x_i \hat{y}(1 - \hat{y}) + 2x_i y \hat{y} \hat{y}(1 - \hat{y})]$$

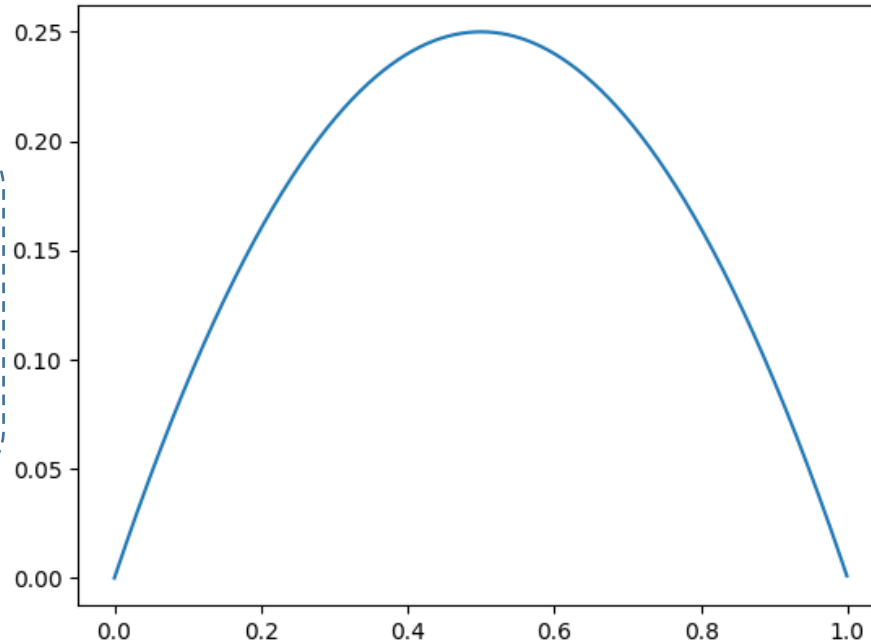
$$= 2x_i^2 \hat{y}(1 - \hat{y})[-3\hat{y}^2 + 2\hat{y} - y + 2y\hat{y}]$$

Mean Squared Error

$$\frac{\partial^2 L}{\partial \theta_i^2} = 2x_i^2 \hat{y}(1 - \hat{y})[-3\hat{y}^2 + 2\hat{y} - y + 2y\hat{y}]$$

$$x_i^2 \geq 0$$

$$\hat{y}(1 - \hat{y}) \in \left[0, \frac{1}{4}\right]$$

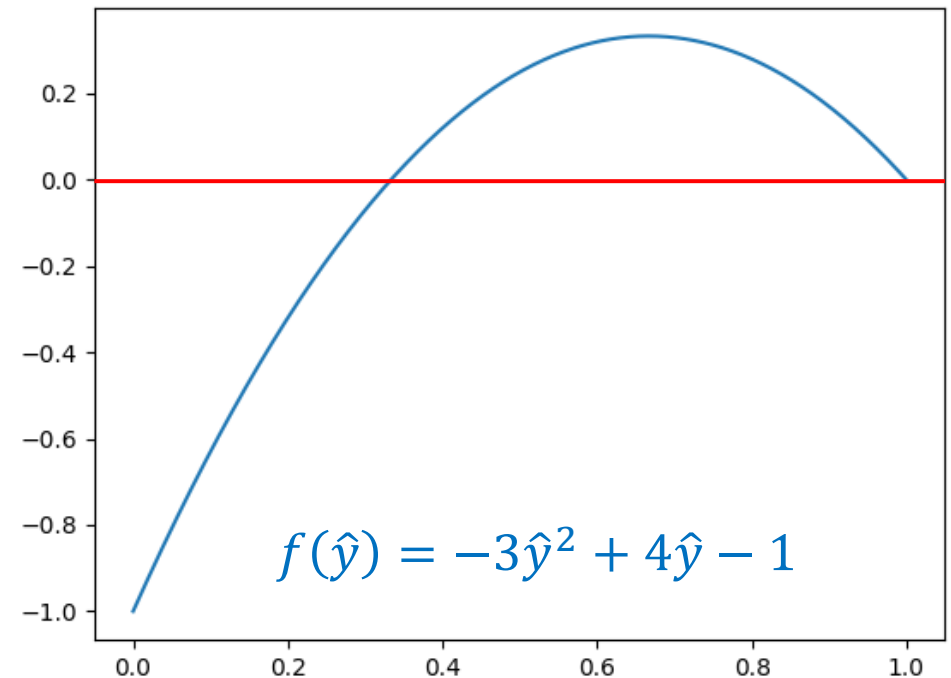
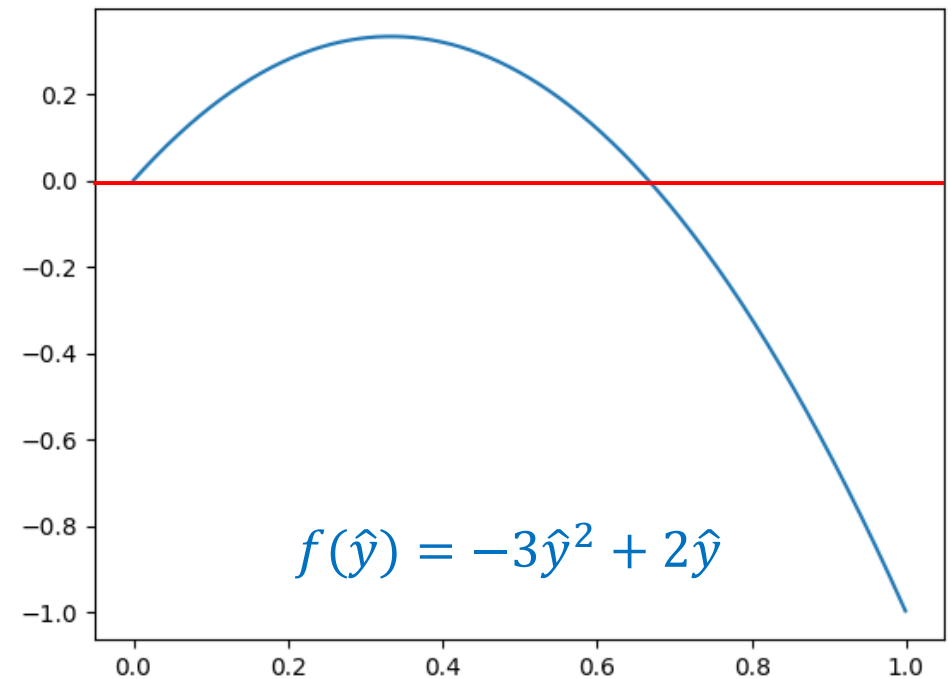


$$y = 0$$

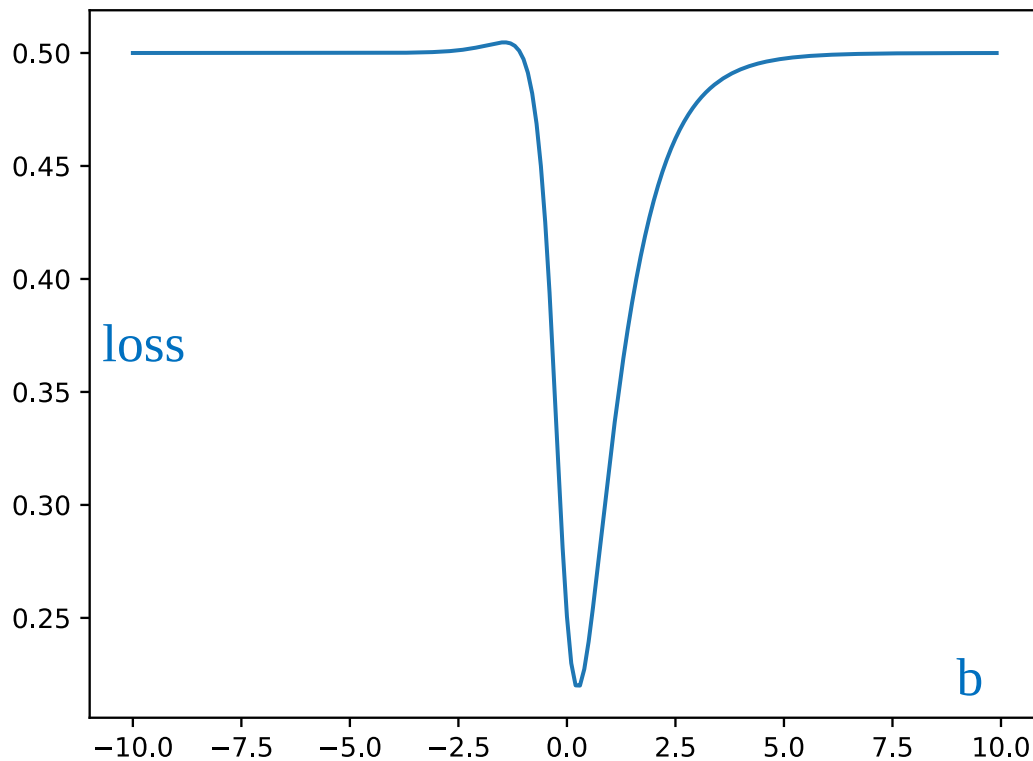
$$f(\hat{y}) = -3\hat{y}^2 + 2\hat{y}$$

$$y = 1$$

$$f(\hat{y}) = -3\hat{y}^2 + 4\hat{y} - 1$$



❖ Visualization



Mean Squared Error

Model and Loss

$$z = \theta^T x = x^T \theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

Outline

SECTION 1

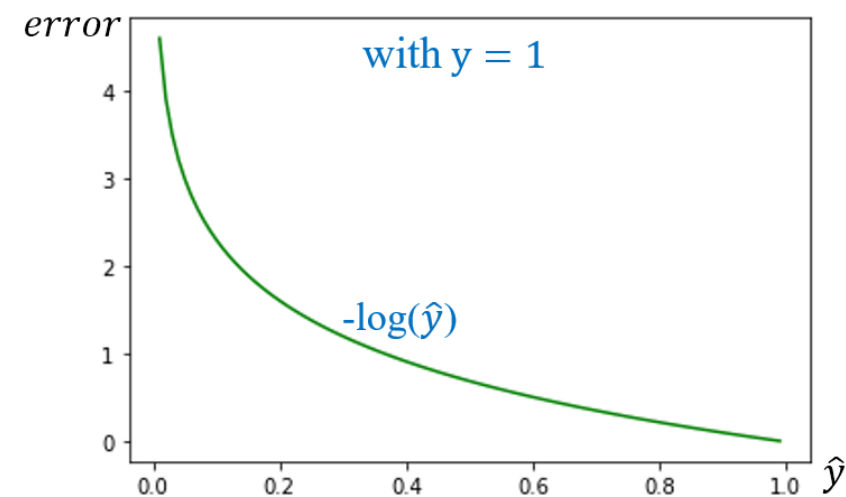
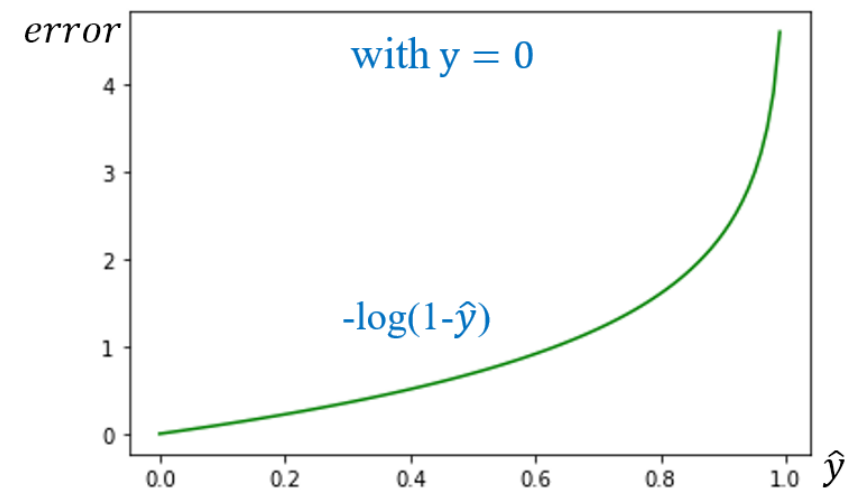
Review

SECTION 2

Lo.R. Using MSE

SECTION 3

Lo.R. Using BCE

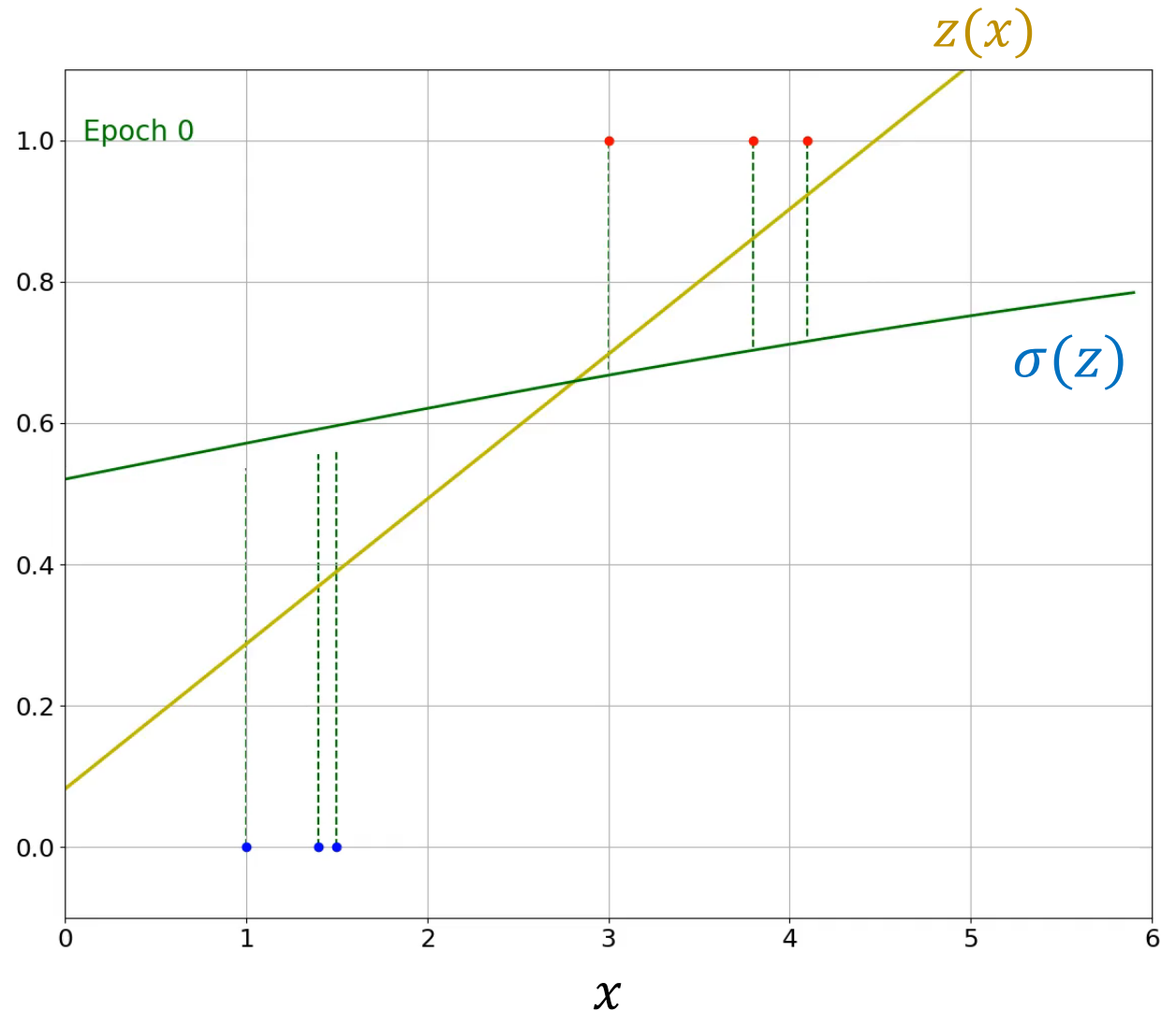


Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

$$z = wx + b$$

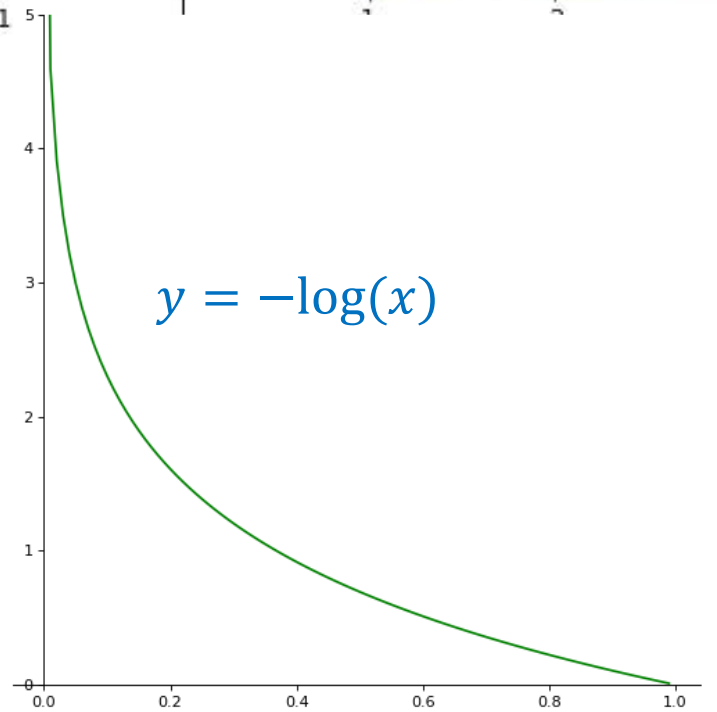
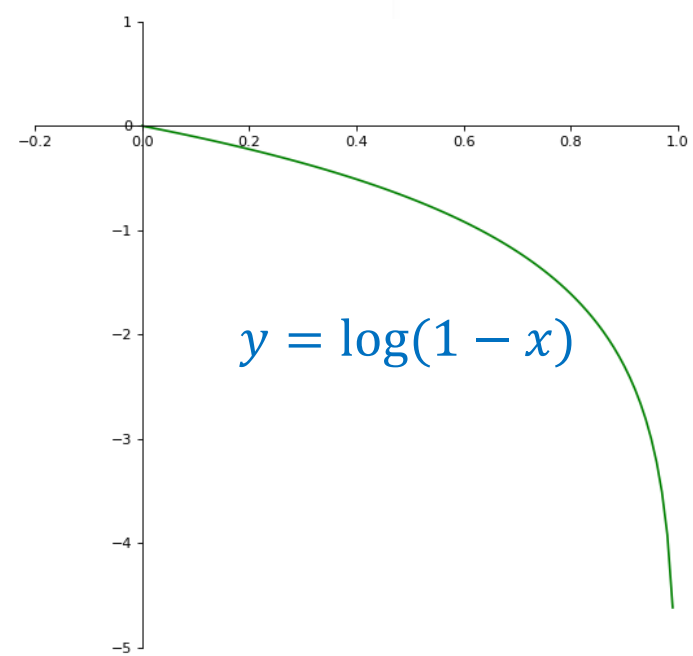
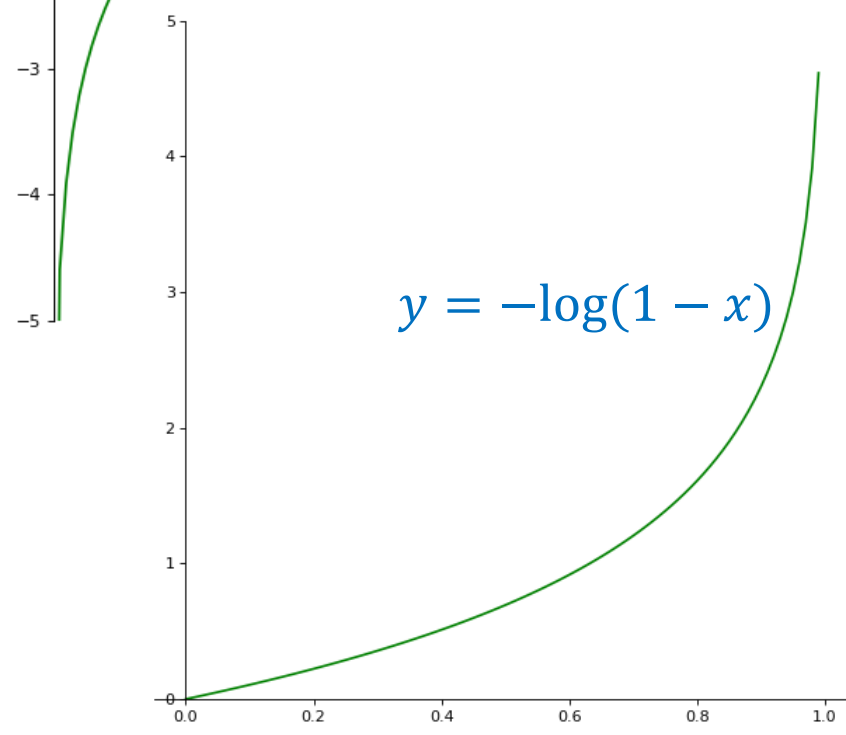
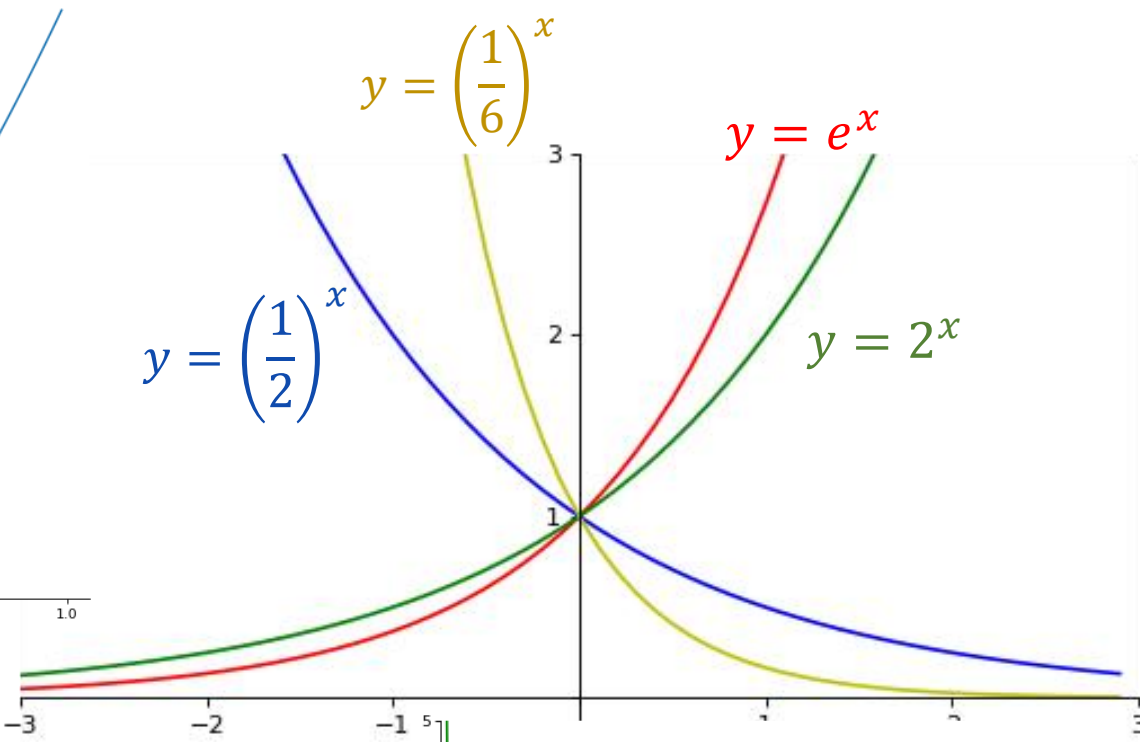
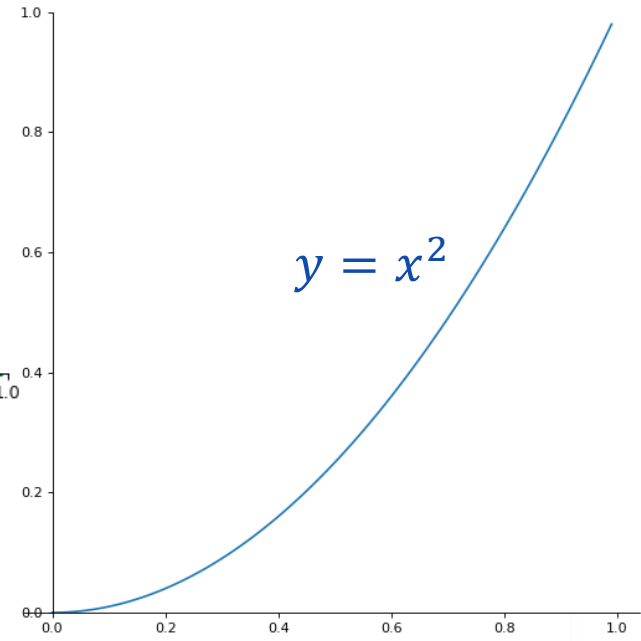
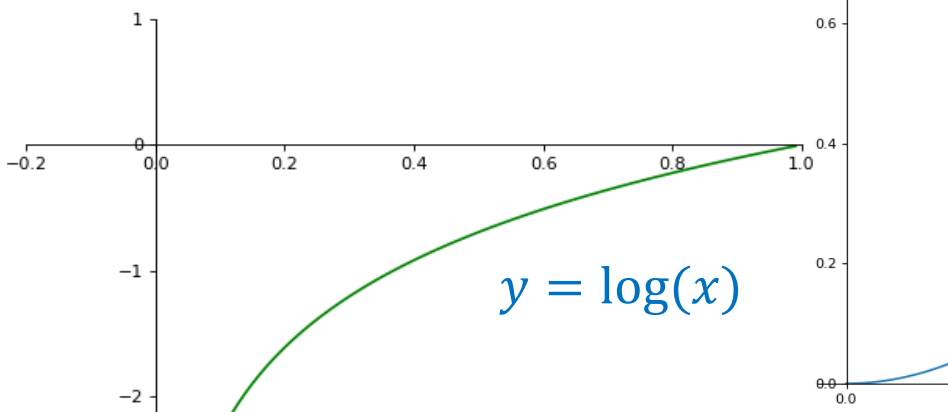
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$



How to evaluate the performance of a model?

❖ Suggested Functions



❖ Loss function

Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$

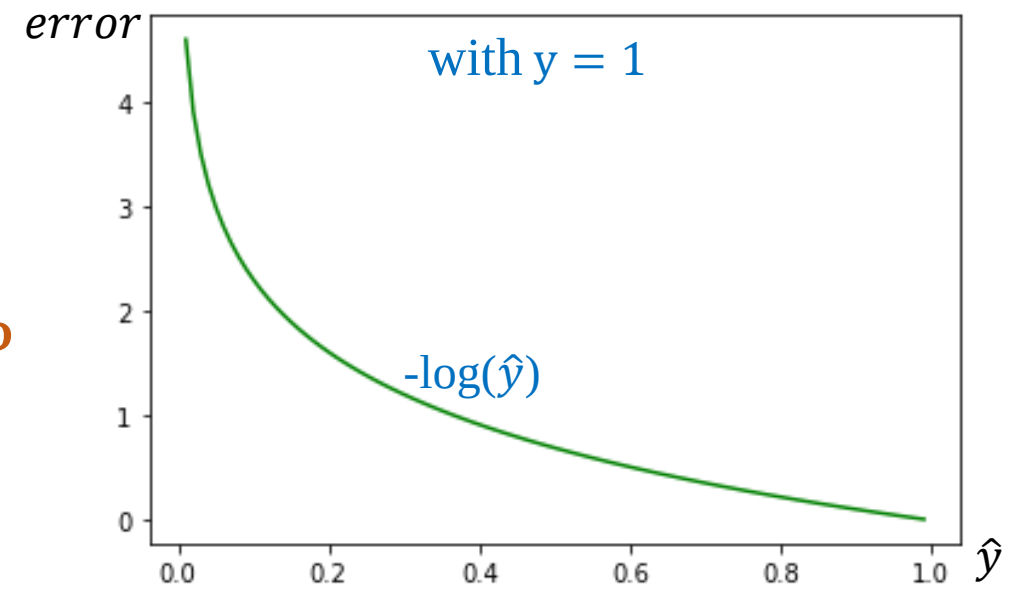
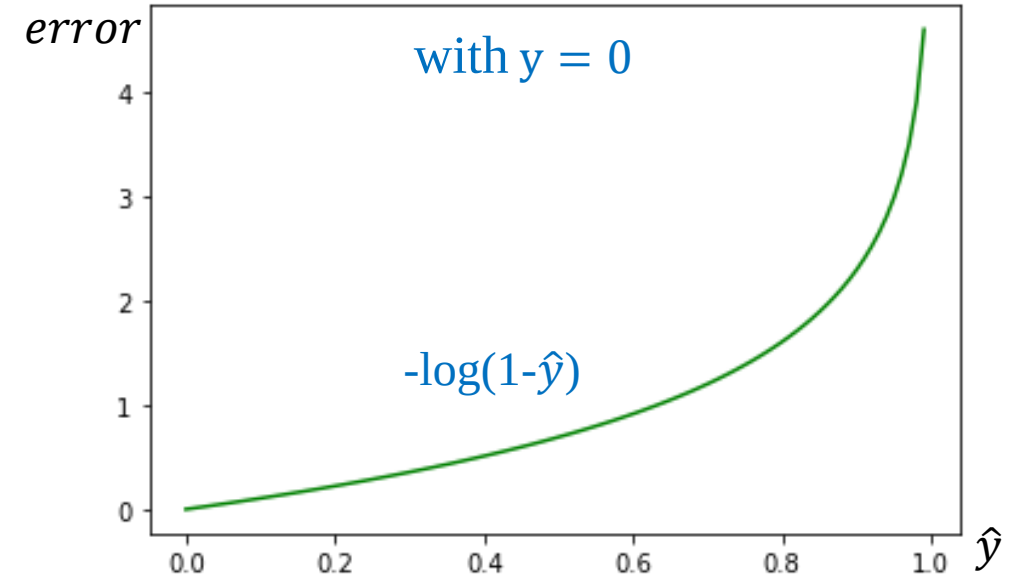
if $y = 1$

$$L(\hat{y}) = -\log(\hat{y})$$

if $y = 0$

$$L(\hat{y}) = -\log(1 - \hat{y})$$

How to
remove if?



Idea of Logistic Regression

40

❖ Loss function

Feature	Output	Label
Input	Output	Label
...	0.3	0
...	0.8	0
...	0.7	0
...	0.4	0
...	0.6	1
...	0.8	1
...	0.9	1
...	0.2	1

if $y = 0$

$$L(\hat{y}) = -\log(1 - \hat{y})$$

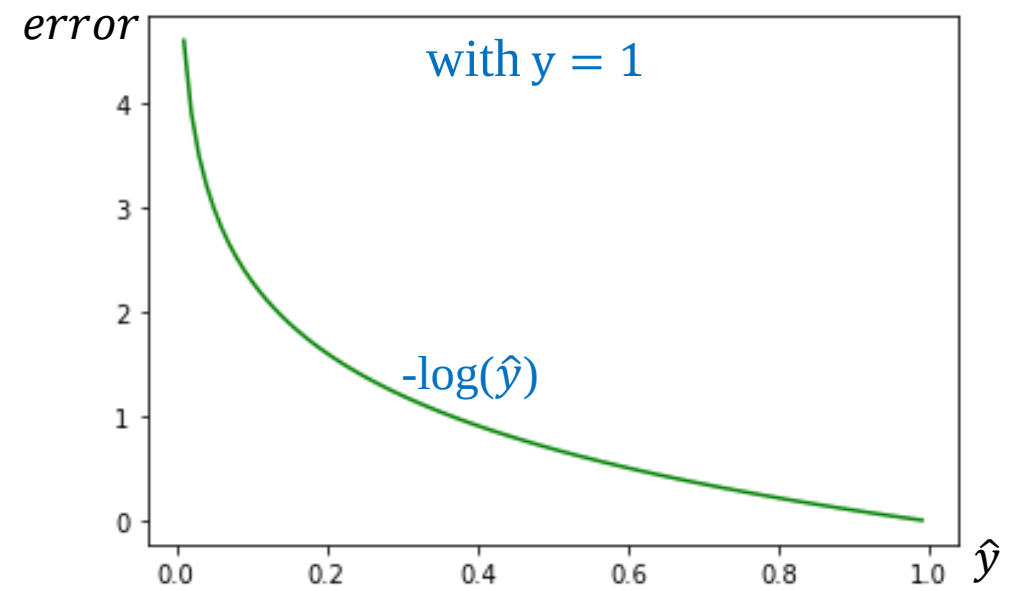
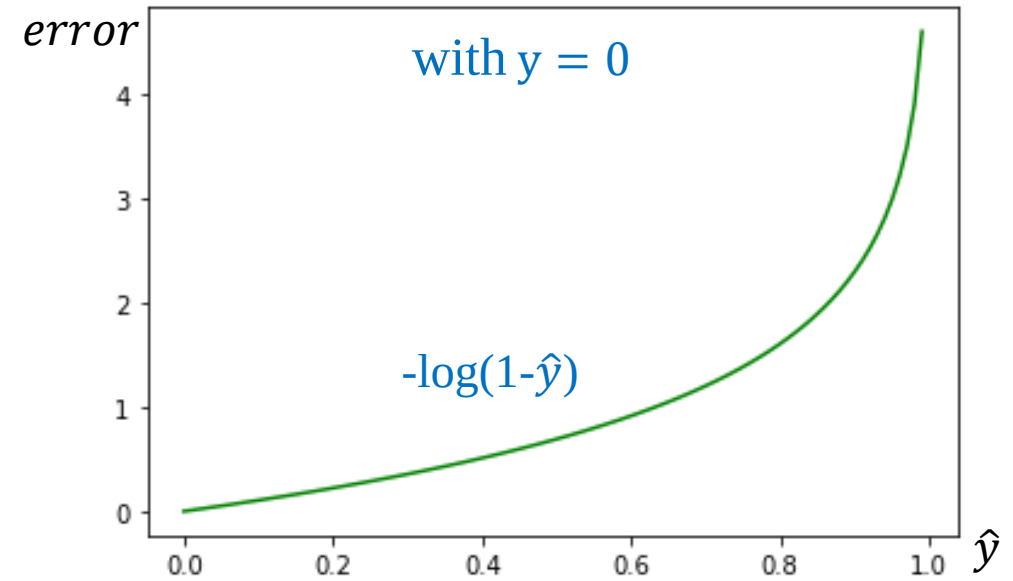
if $y = 1$

$$L(\hat{y}) = -\log(\hat{y})$$

one sample

Binary cross-entropy

$$L(y, \hat{y}) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$$



❖ Construct loss

$$z = wx + b$$

Model and Loss

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L(\hat{y}, y) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

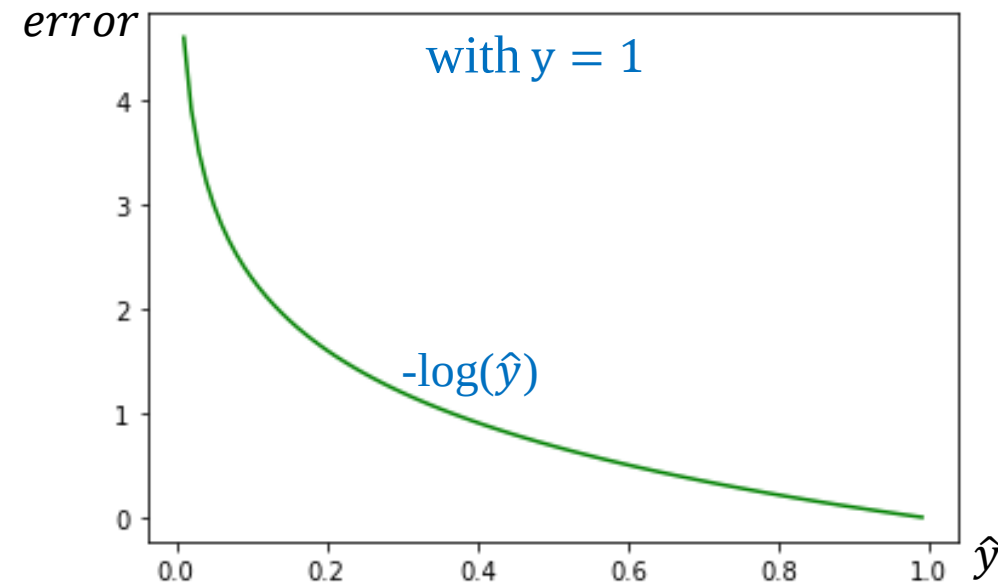
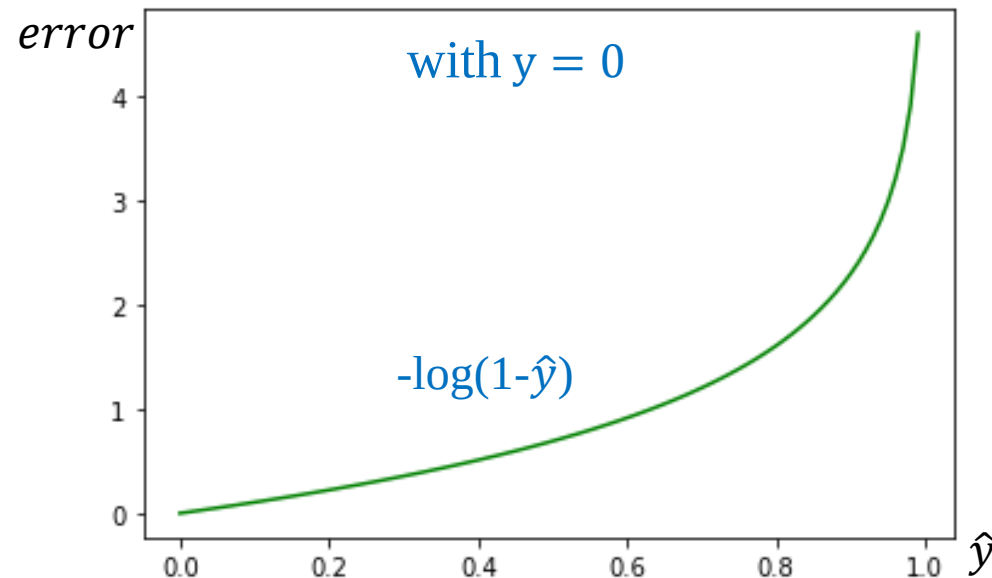
Derivative

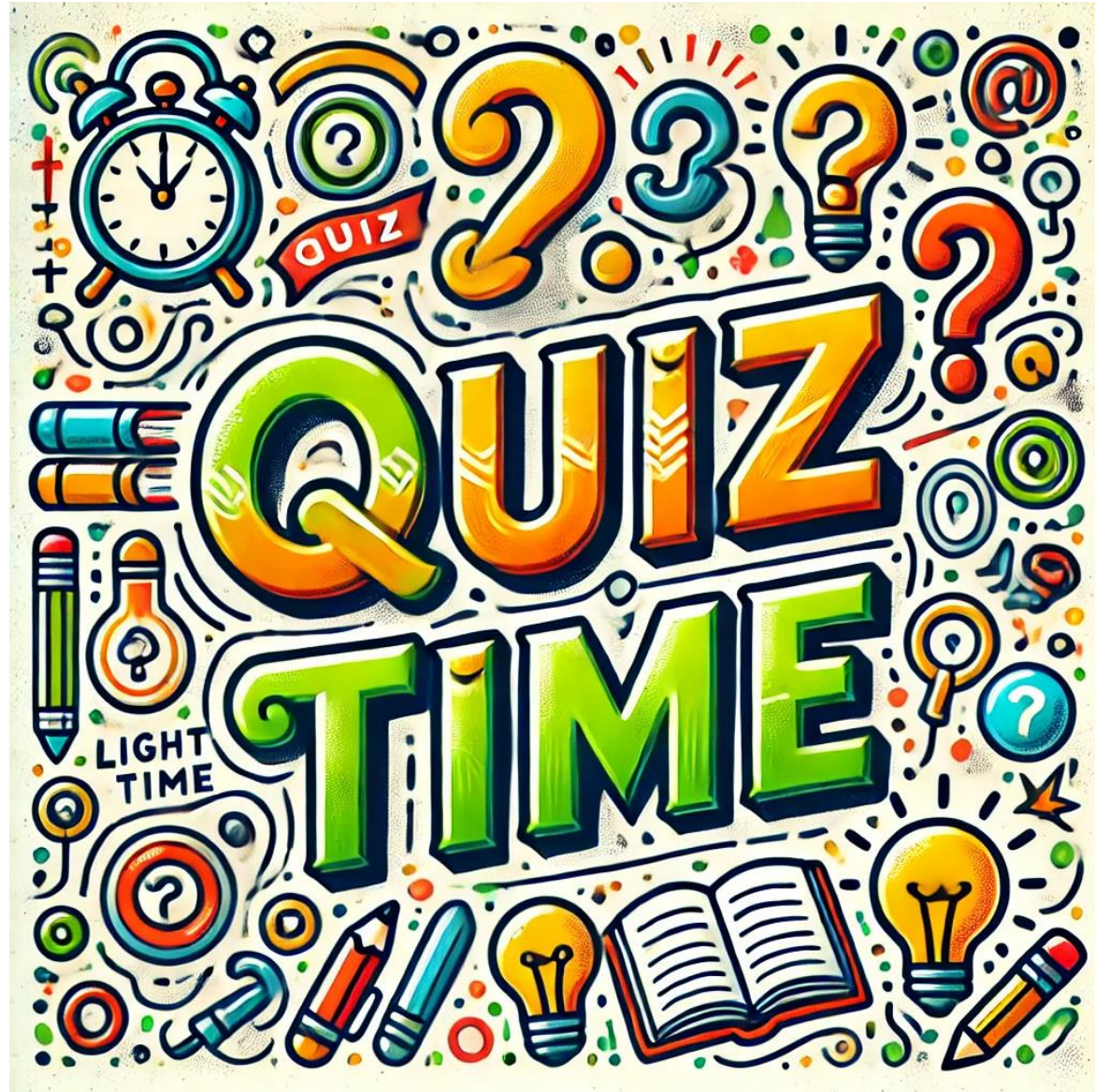
$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} - y)$$





❖ Hãy sắp xếp thứ tự đúng của các bước để thực hiện thuật toán Logistic Regression

1. Tính đạo hàm của hàm mất mát theo các tham số mô hình.
2. Khởi tạo các tham số ban đầu của mô hình (w và b).
3. Cập nhật các tham số mô hình sử dụng Gradient Descent.
4. Dự đoán đầu ra $\hat{y}$ bằng cách tính giá trị $z = \theta^T x$ và áp dụng hàm sigmoid.
5. Tính giá trị hàm mất mát dựa trên đầu ra dự đoán $\hat{y}$ và giá trị thực y .

a) 4, 5, 1, 2, 3

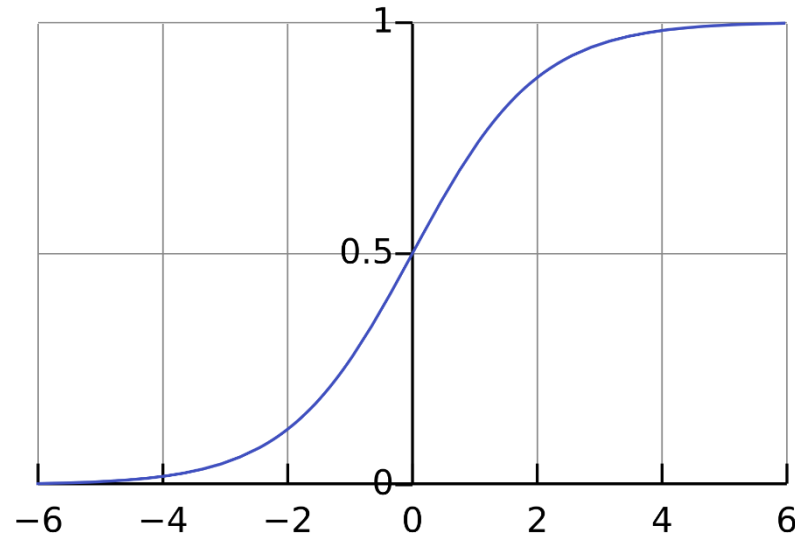
b) 2, 4, 5, 1, 3

c) 2, 5, 4, 1, 3

d) 2, 4, 3, 5, 1

Question 2

❖ Cho hàm sigmoid $\sigma(x) = \frac{1}{1+e^{-x}}$; Với $x_1 = 2$ và $x_2 = 10$ thì khẳng định nào sau đây đúng?



a) $\sigma(x_1) > \sigma(x_2)$

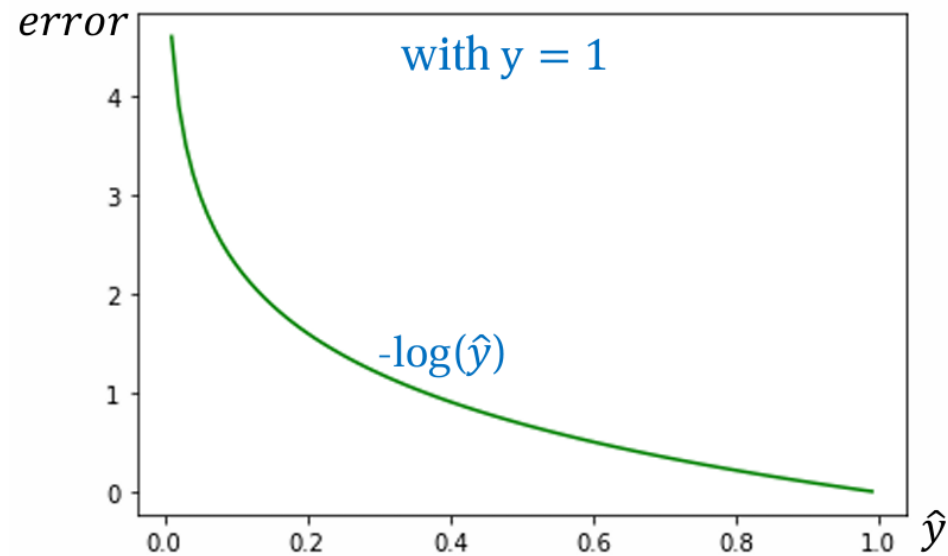
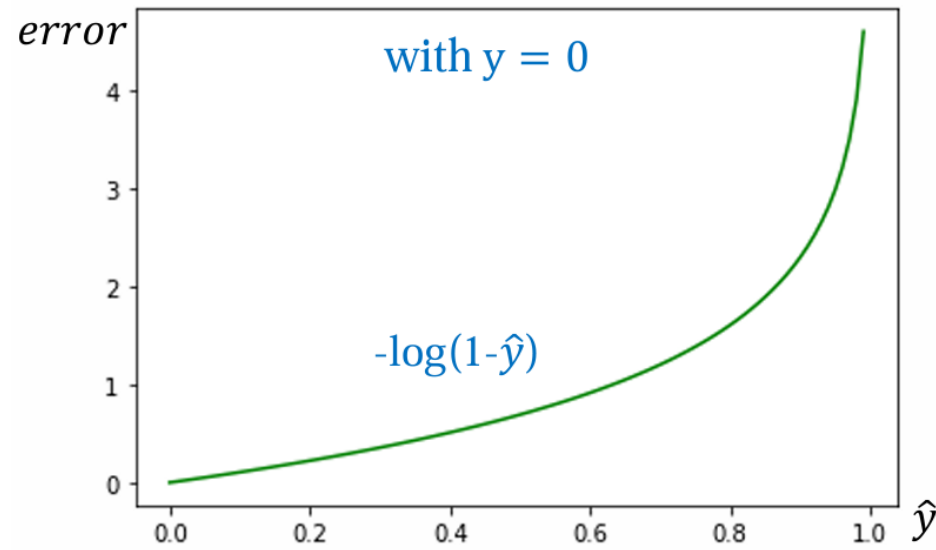
b) $\sigma(x_1) < \sigma(x_2)$

c) $\sigma(x_1) = \sigma(x_2)$

d) Chưa thể khẳng định được điều gì

Question 3

❖ Công thức của hàm mất mát Binary Cross-Entropy (BCE) trong Logistic Regression là gì?



a) $L(\hat{y}, y) = -y \log(1 - \hat{y}) - (1 - y) \log \hat{y}$

b) $L(\hat{y}, y) = -\hat{y} \log(1 - y) - (1 - \hat{y}) \log y$

c) $L(\hat{y}, y) = -\hat{y} \log(1 - \hat{y}) - (1 - y) \log y$

d) $L(\hat{y}, y) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$

Question 4

❖ Hãy tính đạo hàm bậc hai theo x của hàm $f(x, y) = x^2 + 3y^2 + 4xy$

a) $\frac{\partial^2 f}{\partial x^2} = 2x + 4y$

b) $\frac{\partial^2 f}{\partial x^2} = 2$

c) $\frac{\partial^2 f}{\partial x^2} = 6y + 4x$

d) $\frac{\partial^2 f}{\partial x^2} = 4$



Question 5

❖ Hãy tính đạo hàm bậc hai theo y của hàm $f(x, y) = x^2 + 3y^2 + 4xy$

a) $\frac{\partial^2 f}{\partial y^2} = 6y + 4x$

b) $\frac{\partial^2 f}{\partial y^2} = 2x + 4y$

c) $\frac{\partial^2 f}{\partial y^2} = 4$

d) $\frac{\partial^2 f}{\partial y^2} = 6$

Question 6

❖ Hãy tính đạo hàm hỗn hợp của hàm $f(x, y) = x^2 + 3y^2 + 4xy$

a) $\frac{\partial^2 f}{\partial x \partial y} = 6, \frac{\partial^2 f}{\partial y \partial x} = 2$

b) $\frac{\partial^2 f}{\partial x \partial y} = 2, \frac{\partial^2 f}{\partial y \partial x} = 6$

c) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 4$

d) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 2$

Feature	Label
Petal_Length	Category
1.4	0
1	0
1.5	0
3	1
3.8	1
4.1	1

Category 0

Category 1

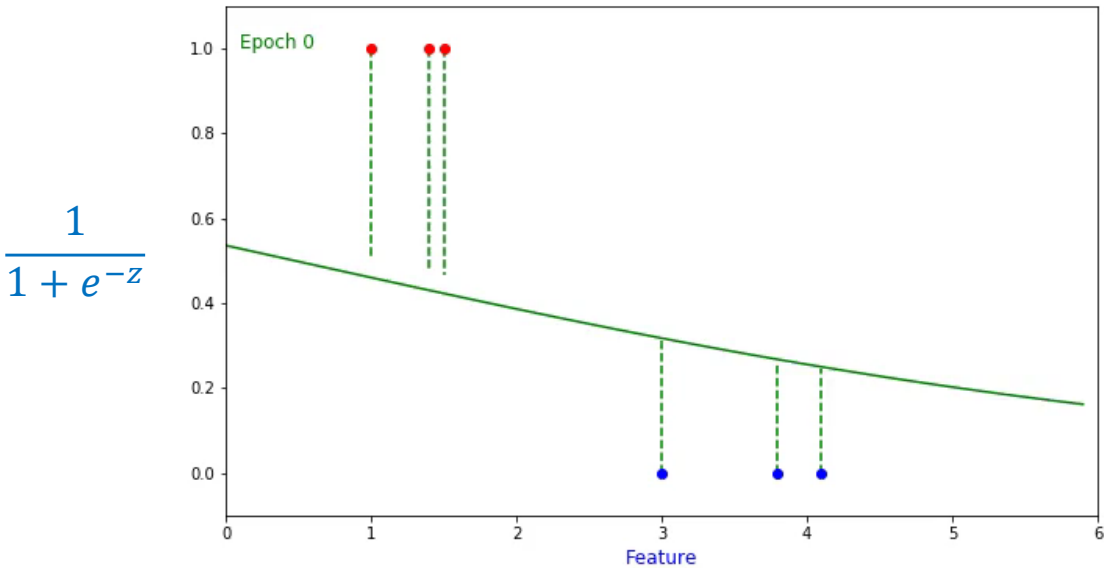
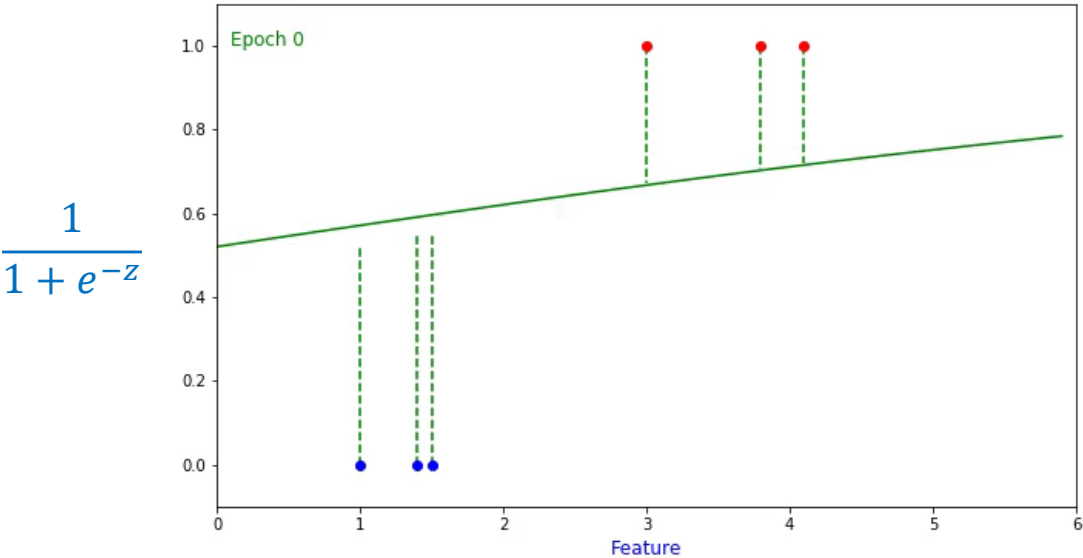
$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Feature	Label
Petal_Length	Category
1.4	1
1	1
1.5	1
3	0
3.8	0
4.1	0

Category 0

Category 1



Binary Cross-entropy

❖ Convex function

$$z = \theta^T x$$

Model and Loss

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

simplified version

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} - y)$$

$$\frac{\partial^2 L}{\partial \theta_i^2} = \frac{\partial}{\partial \theta_i} [x_i(\hat{y} - y)] = x_i^2(\hat{y} - \hat{y}^2) \geq 0$$

$$x_i^2 \geq 0 \quad \hat{y} - \hat{y}^2 \in \left[0, \frac{1}{4}\right]$$

Derivative

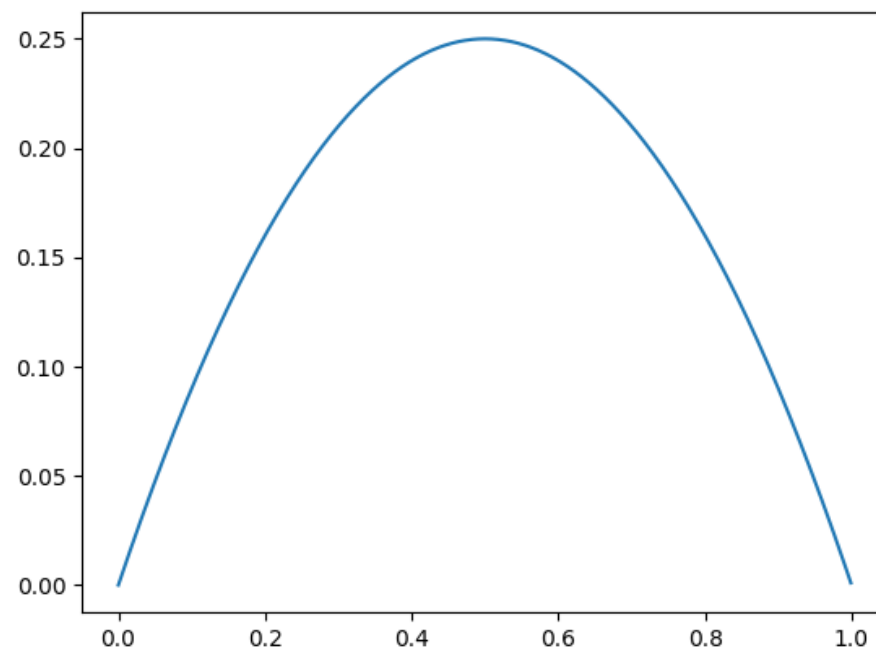
$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

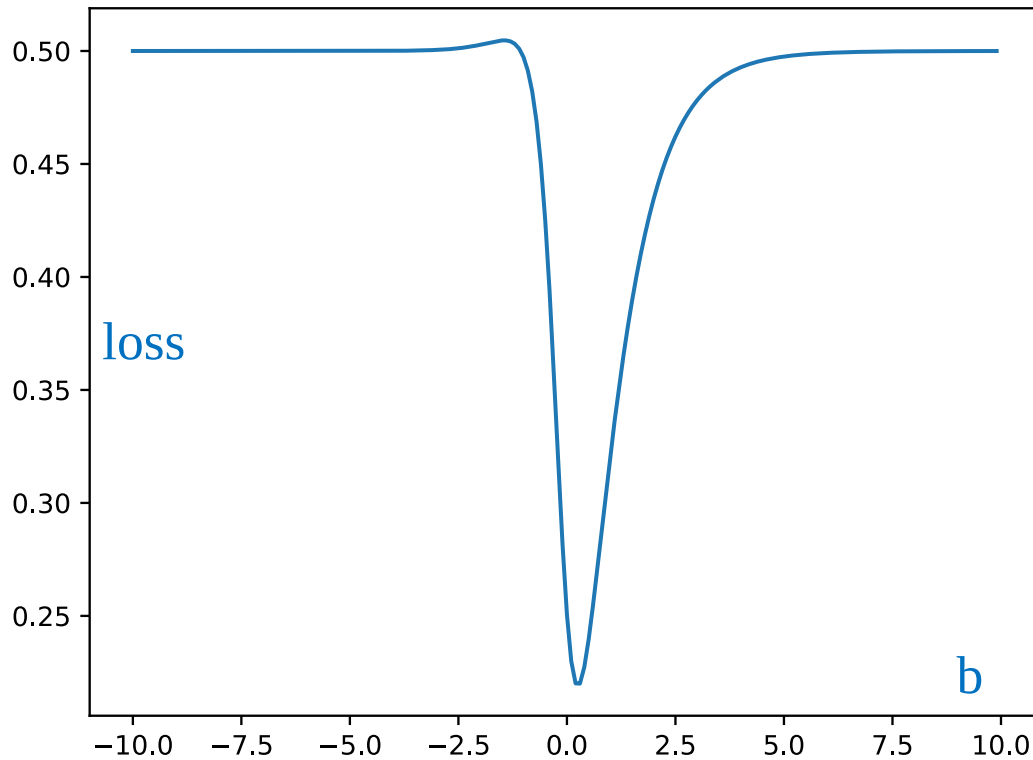
$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

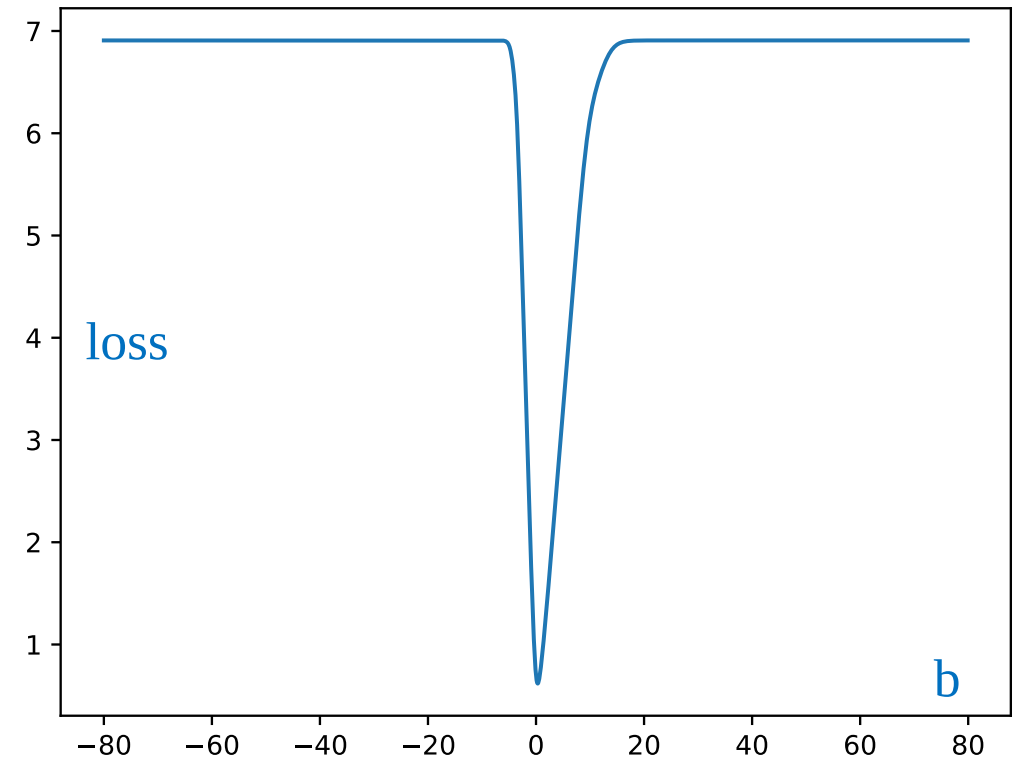
$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} - y)$$



❖ Visualization



Mean Squared Error



Binary Cross-Entropy

Example

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$z = wx + b$$
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

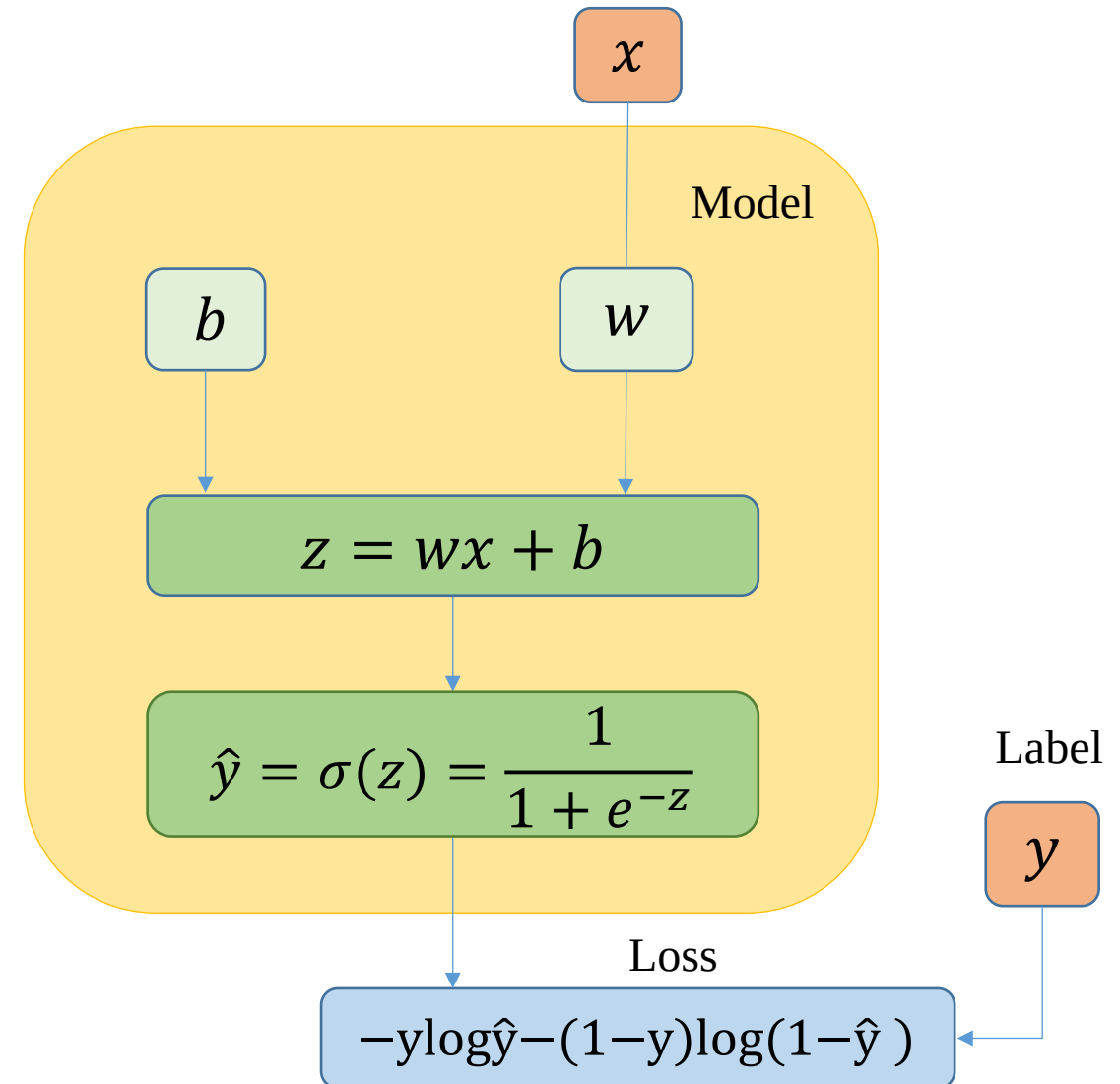
$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \quad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

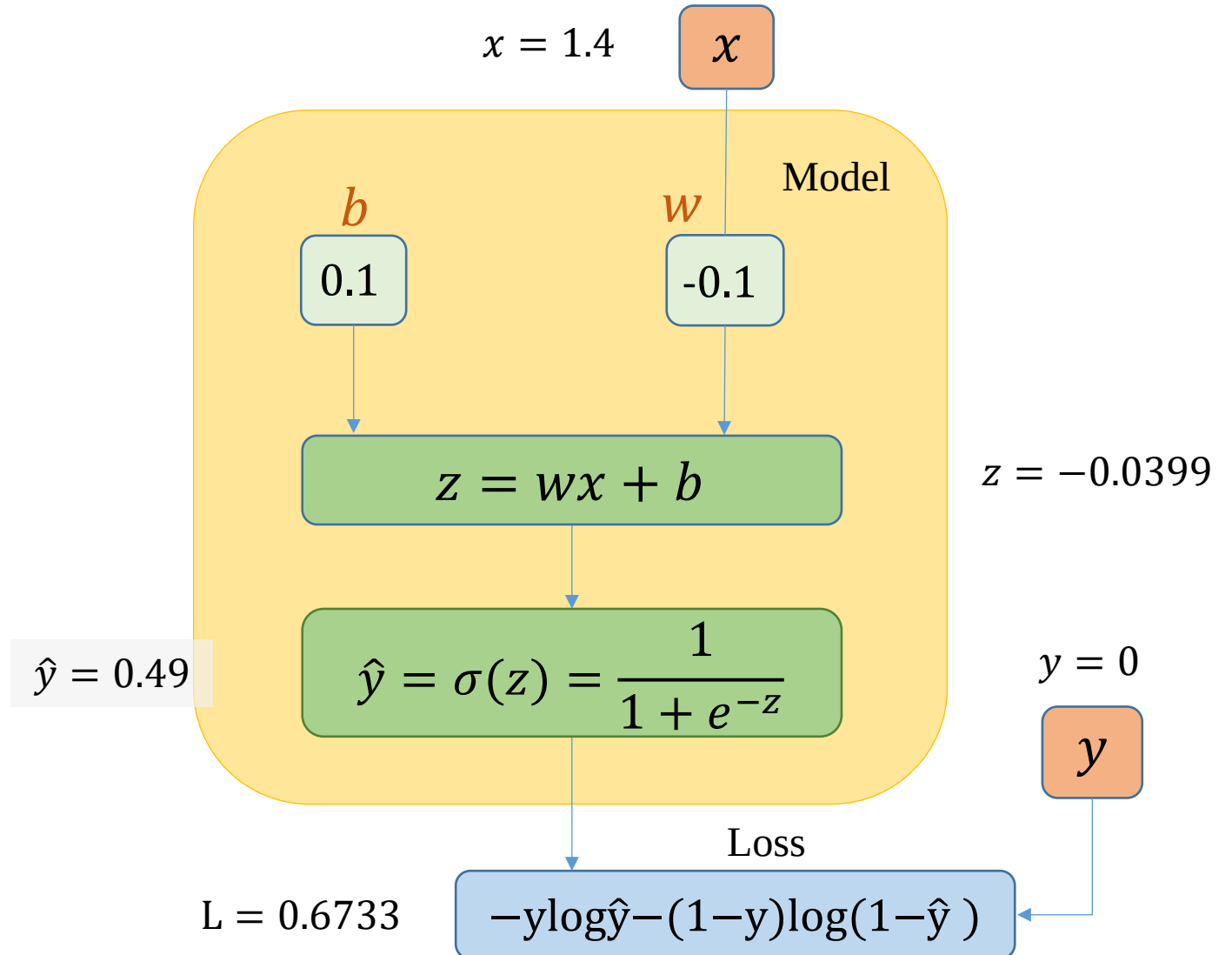
$$w = w - \eta \frac{\partial L}{\partial w} \quad b = b - \eta \frac{\partial L}{\partial b}$$



Dataset

Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

$x = 1.4$ $y = 0$



Dataset

Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

$x = 1.4$

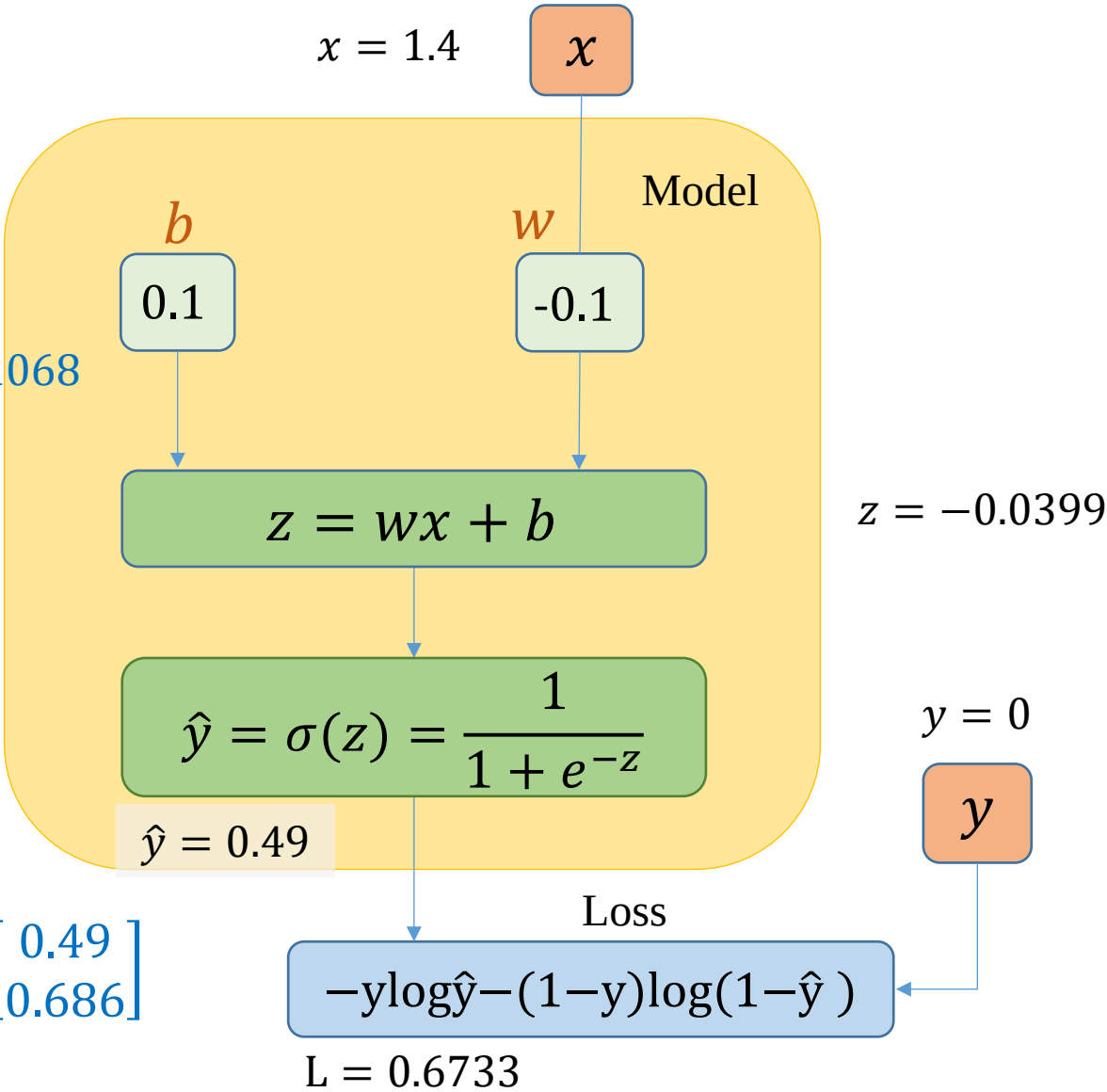
$y = 0$

$$\eta = 0.01$$

$$b = 0.1 - \eta 0.49 = 0.095$$

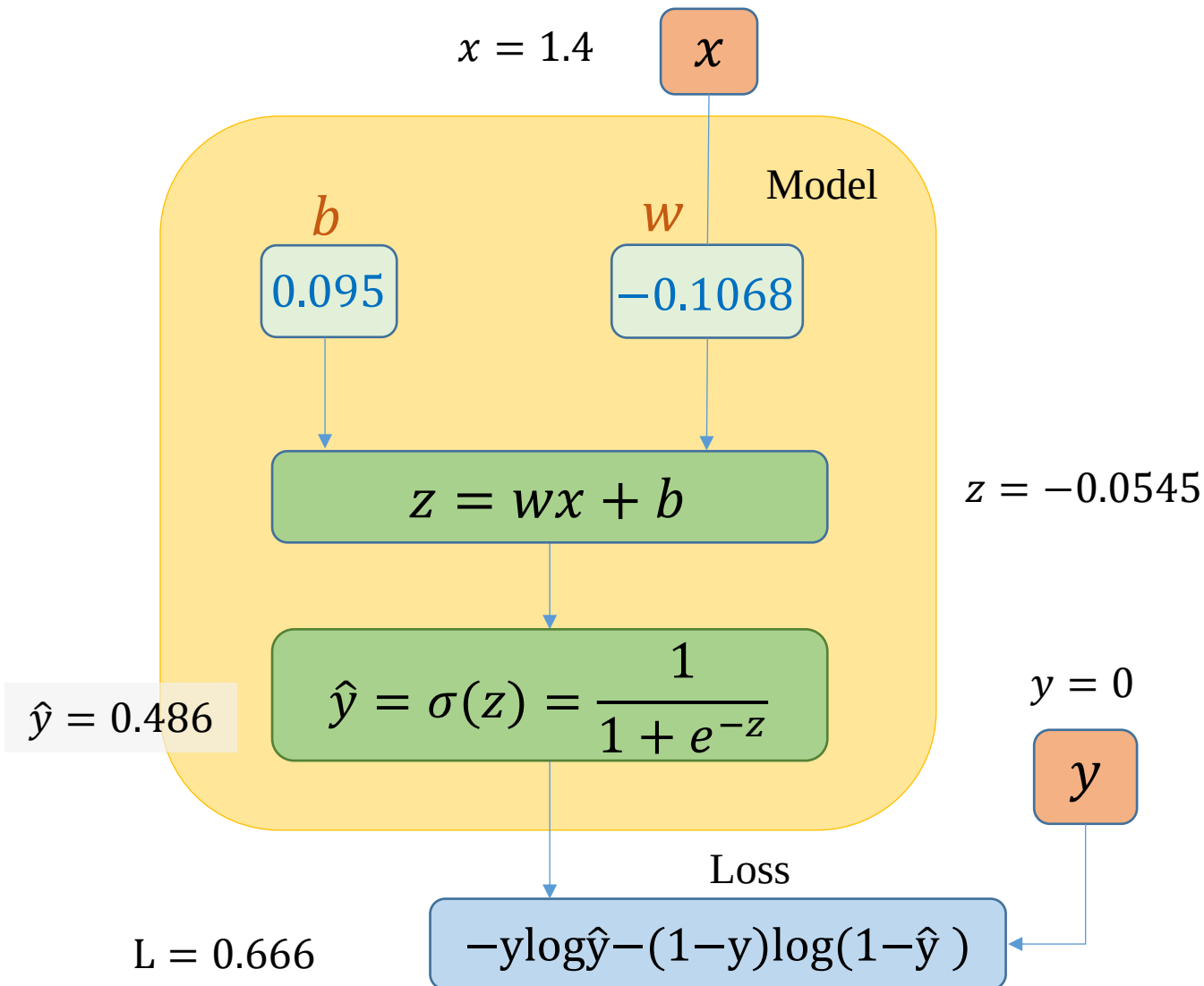
$$w = -0.1 - \eta 0.686 = -0.1068$$

$$\begin{bmatrix} L'_b \\ L'_w \end{bmatrix} = \begin{bmatrix} 1 * 0.49 \\ 1.4 * 0.49 \end{bmatrix} = \begin{bmatrix} 0.49 \\ 0.686 \end{bmatrix}$$



Dataset

Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

 $x = 1.4$ $y = 0$ previous $L = 0.6733$ 

Another Example

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$z = w_1x_1 + w_2x_2 + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

4) Compute derivative

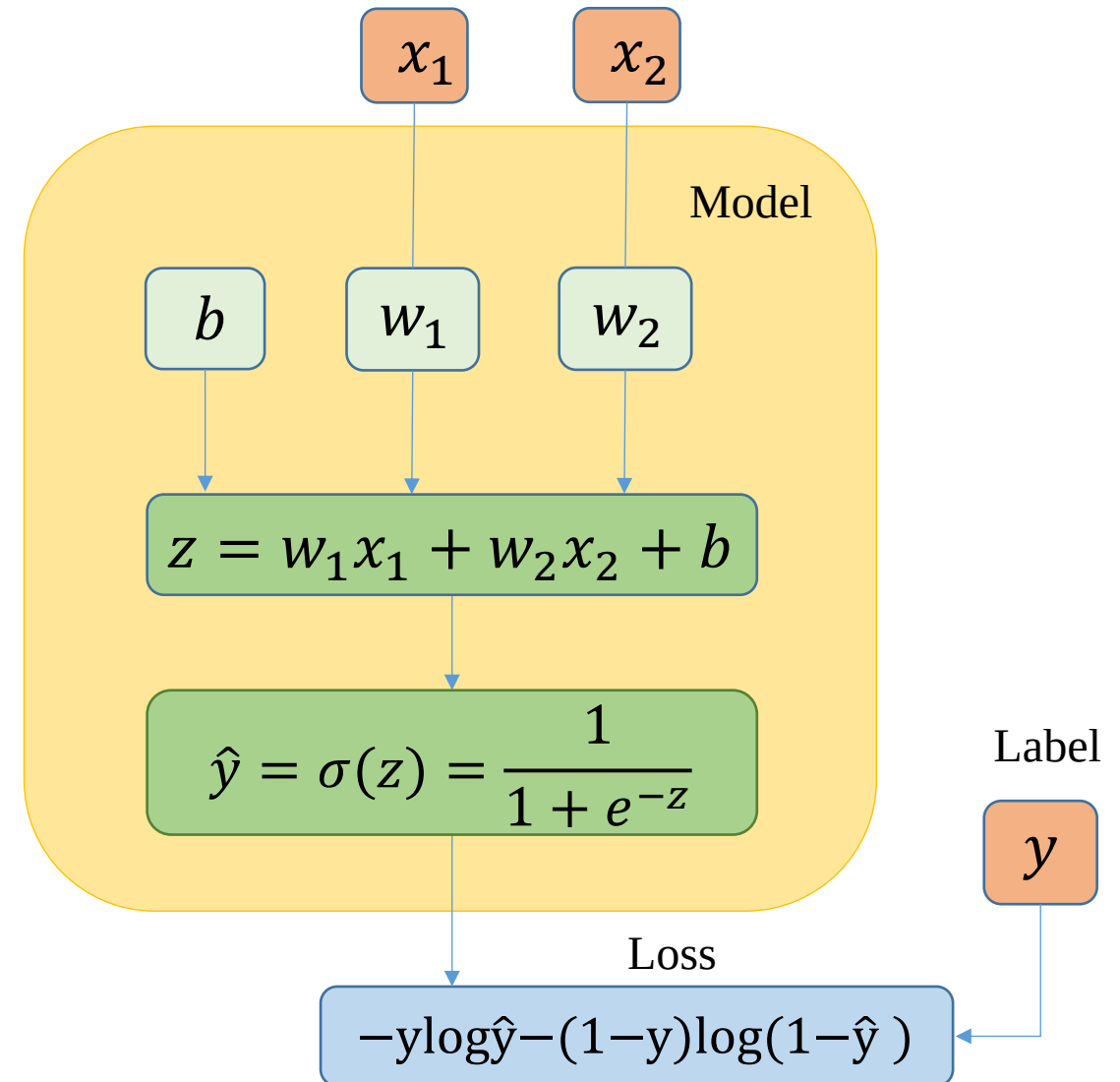
$$\frac{\partial L}{\partial w_i} = x_i(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w_i = w_i - \eta \frac{\partial L}{\partial w_i}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$



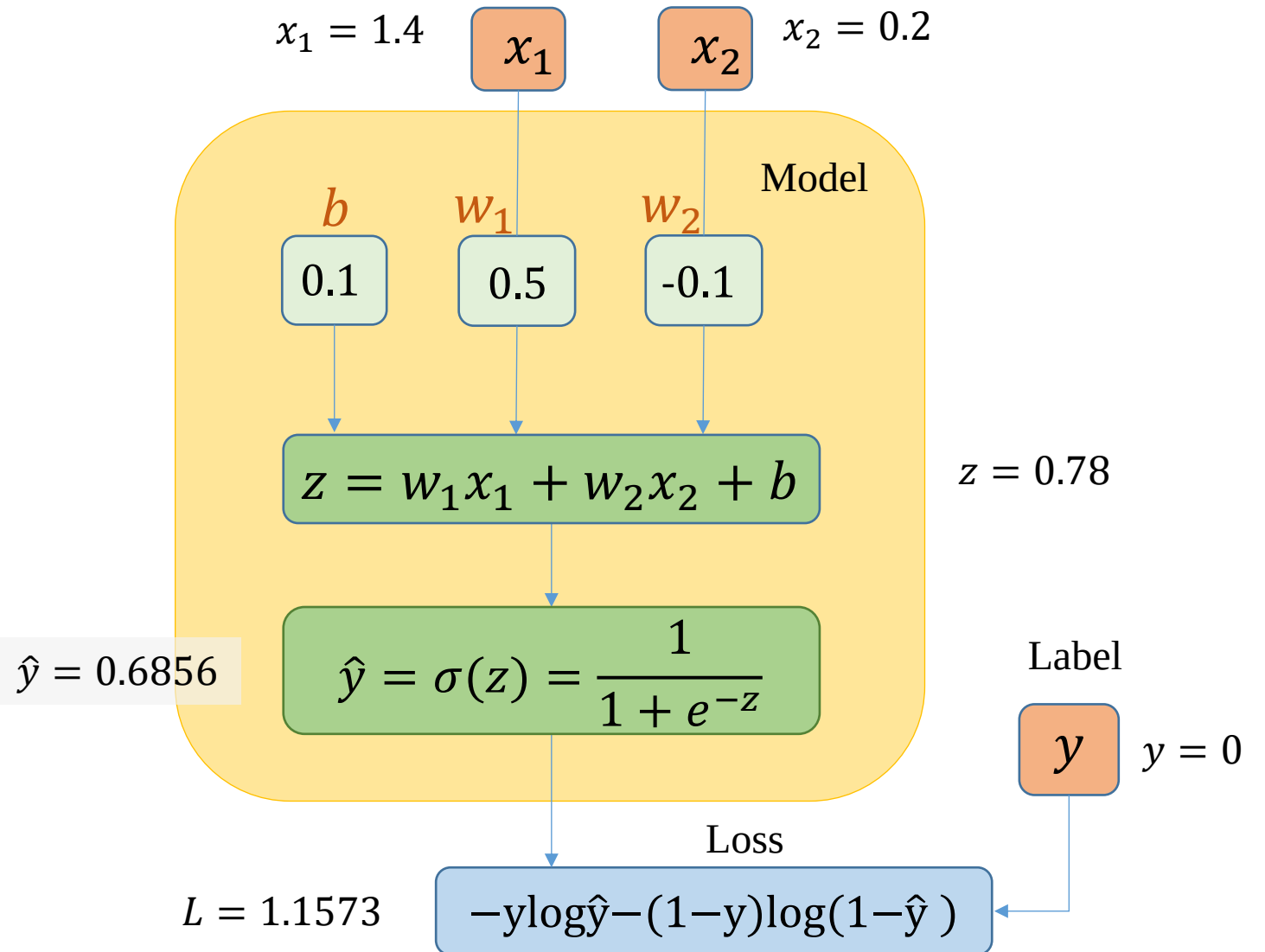
Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$x_1 = 1.4$$

$$x_2 = 0.2$$

$$y = 0$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$x_1 = 1.4$$

$$y = 0$$

$$x_2 = 0.2$$

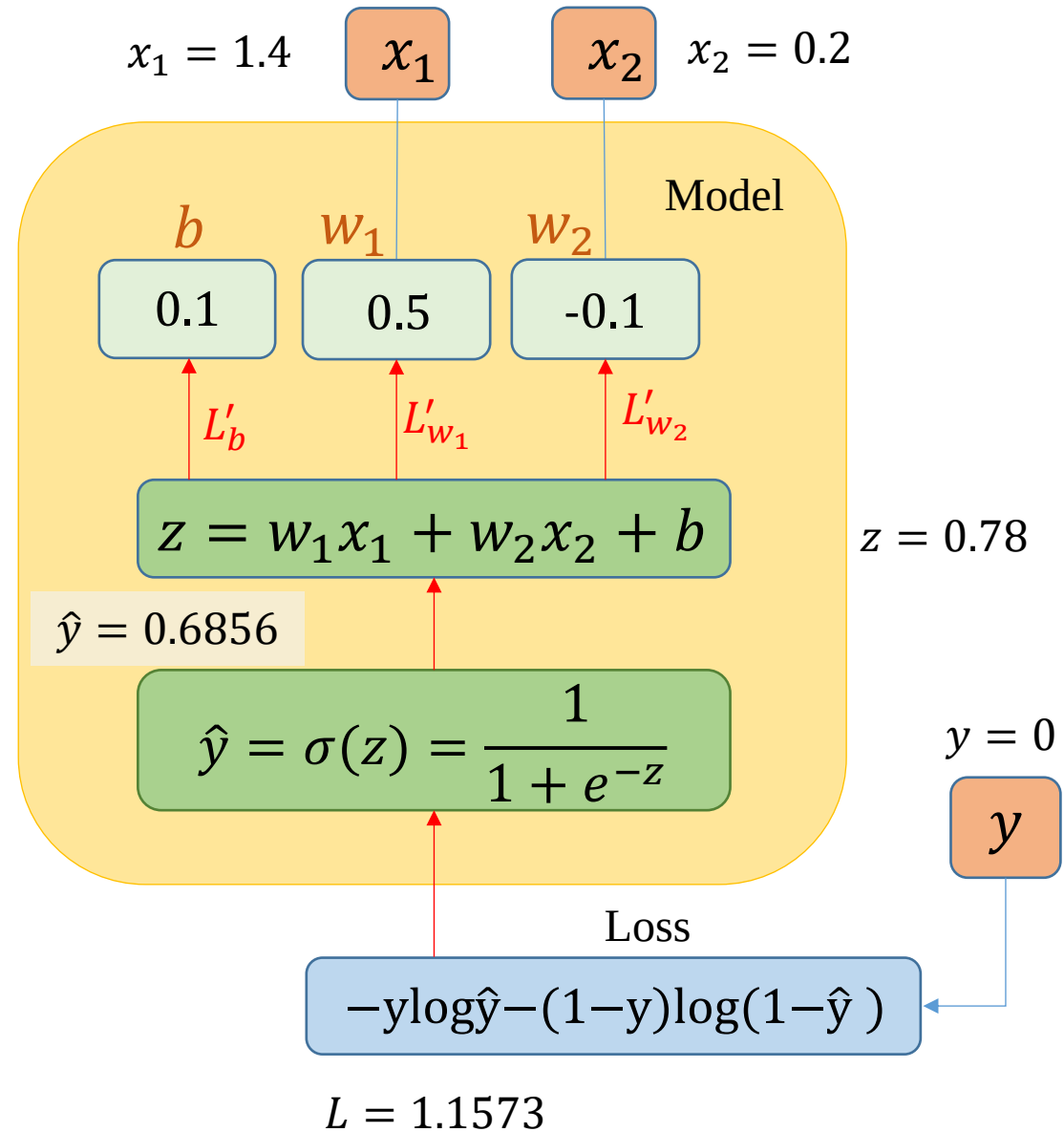
$$\begin{bmatrix} L'_b \\ L'_{w_1} \\ L'_{w_2} \end{bmatrix} = \begin{bmatrix} 1 * 0.6856 \\ 1.4 * 0.6856 \\ 0.2 * 0.6856 \end{bmatrix} = \begin{bmatrix} 0.6856 \\ 0.9599 \\ 0.1371 \end{bmatrix}$$

$$\eta = 0.01$$

$$\begin{aligned} b &= 0.1 - \eta 0.6856 \\ &= 0.0931 \end{aligned}$$

$$\begin{aligned} w_1 &= 0.5 - \eta 0.9598 \\ &= 0.4990 \end{aligned}$$

$$\begin{aligned} w_2 &= -0.1 + \eta 0.1371 \\ &= -0.1013 \end{aligned}$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$x_1 = 1.4$$

$$y = 0$$

$$x_2 = 0.2$$

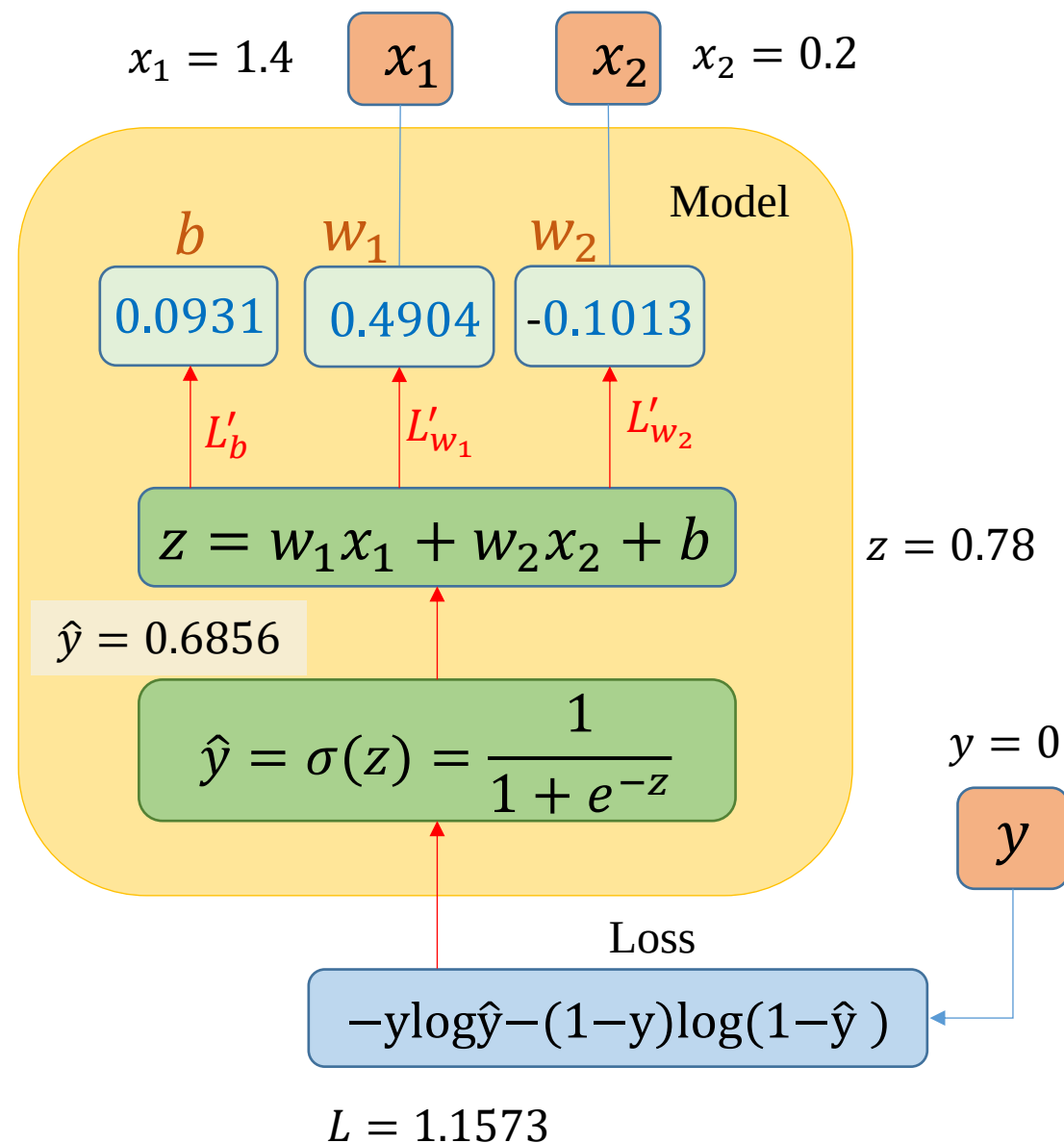
$$\begin{bmatrix} L'_b \\ L'_{w_1} \\ L'_{w_2} \end{bmatrix} = \begin{bmatrix} 1 * 0.6856 \\ 1.4 * 0.6856 \\ 0.2 * 0.6856 \end{bmatrix} = \begin{bmatrix} 0.6856 \\ 0.9599 \\ 0.1371 \end{bmatrix}$$

$$\eta = 0.01$$

$$\begin{aligned} b &= 0.1 - \eta 0.6856 \\ &= 0.0931 \end{aligned}$$

$$\begin{aligned} w_1 &= 0.5 - \eta 0.9598 \\ &= 0.4990 \end{aligned}$$

$$\begin{aligned} w_2 &= -0.1 + \eta 0.1371 \\ &= -0.1013 \end{aligned}$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

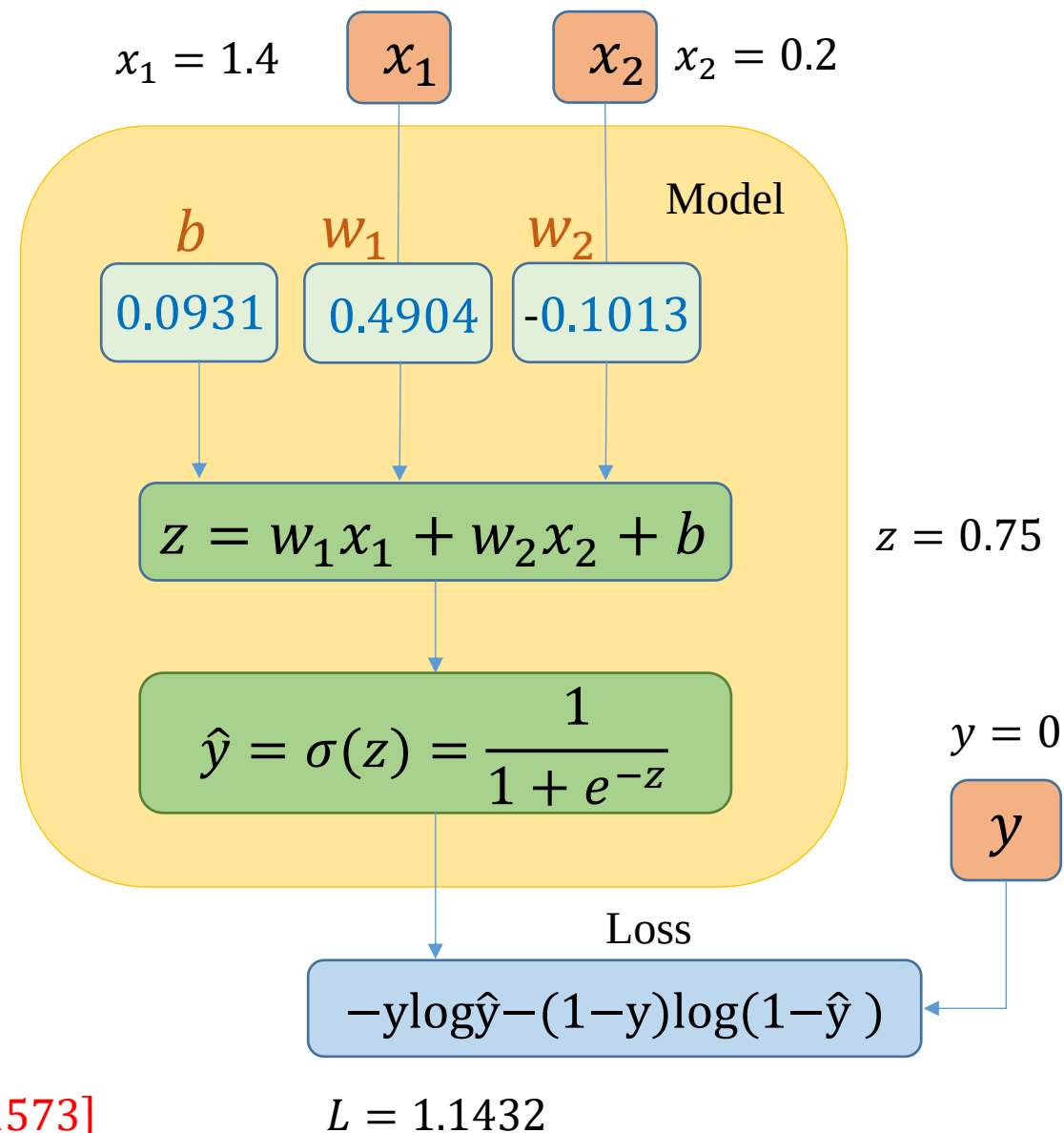
$x_1 = 1.4$

$x_2 = 0.2$

$y = 0$

$\hat{y} = 0.6812$

previous $L = [1.1573]$



Summary

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Traditional

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta} \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \mathbf{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate

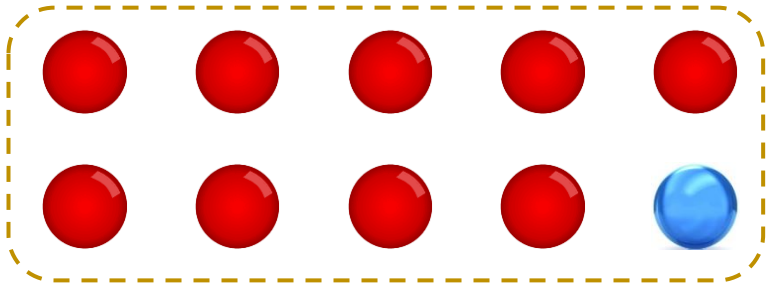
Vectorized



**Construct Logistic Regression
in a different way (optional)**

Entropy

❖ Motivation



A: Get a red ball

B: Get a blue ball

$$p(A) = \frac{9}{10} = 0.9$$

$$p(B) = \frac{1}{10} = 0.1$$

E: Pick a ball from the basket

Experiment 1

Got a red ball



Experiment 2

Got a blue ball



Which experiment makes you more surprised?

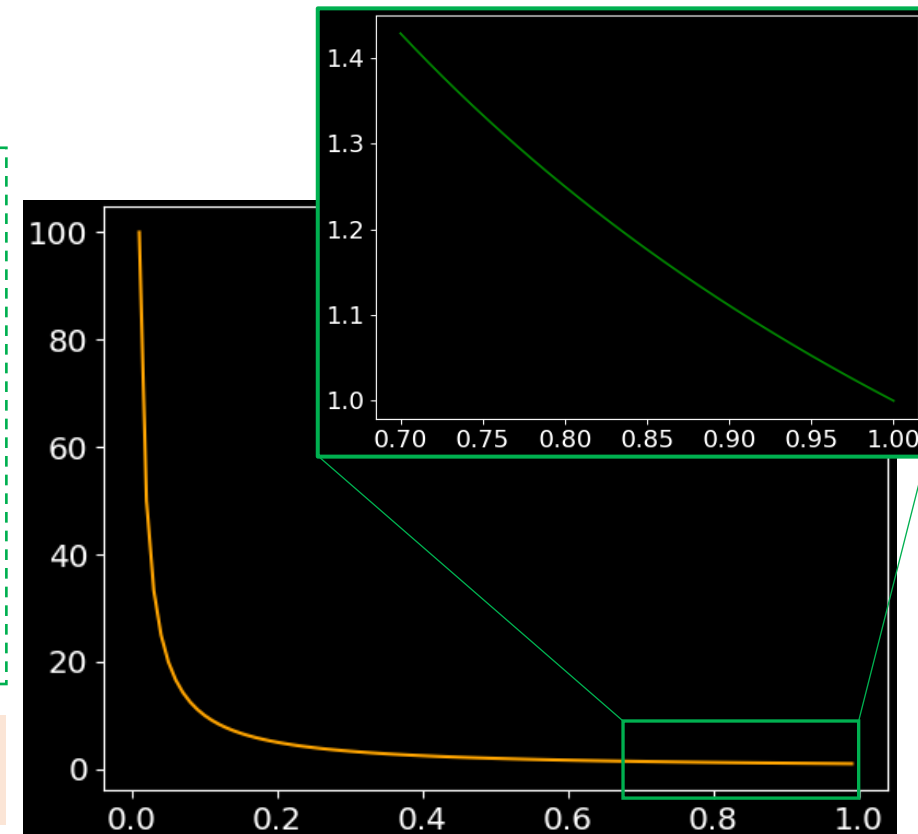
How to measure
the surprises?

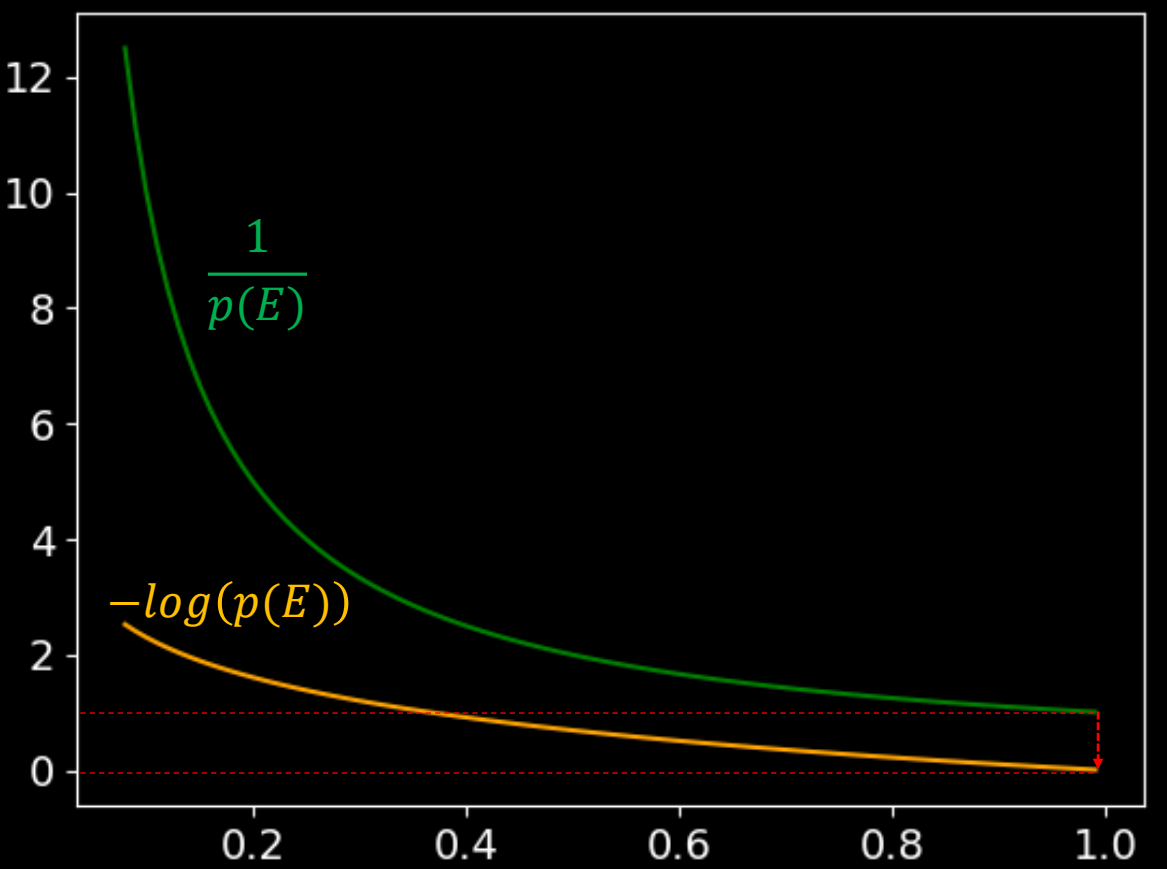
Observation

$Surprise(E) \updownarrow p(E)$

$$\rightarrow Surprise(E) = \frac{1}{p(E)}$$

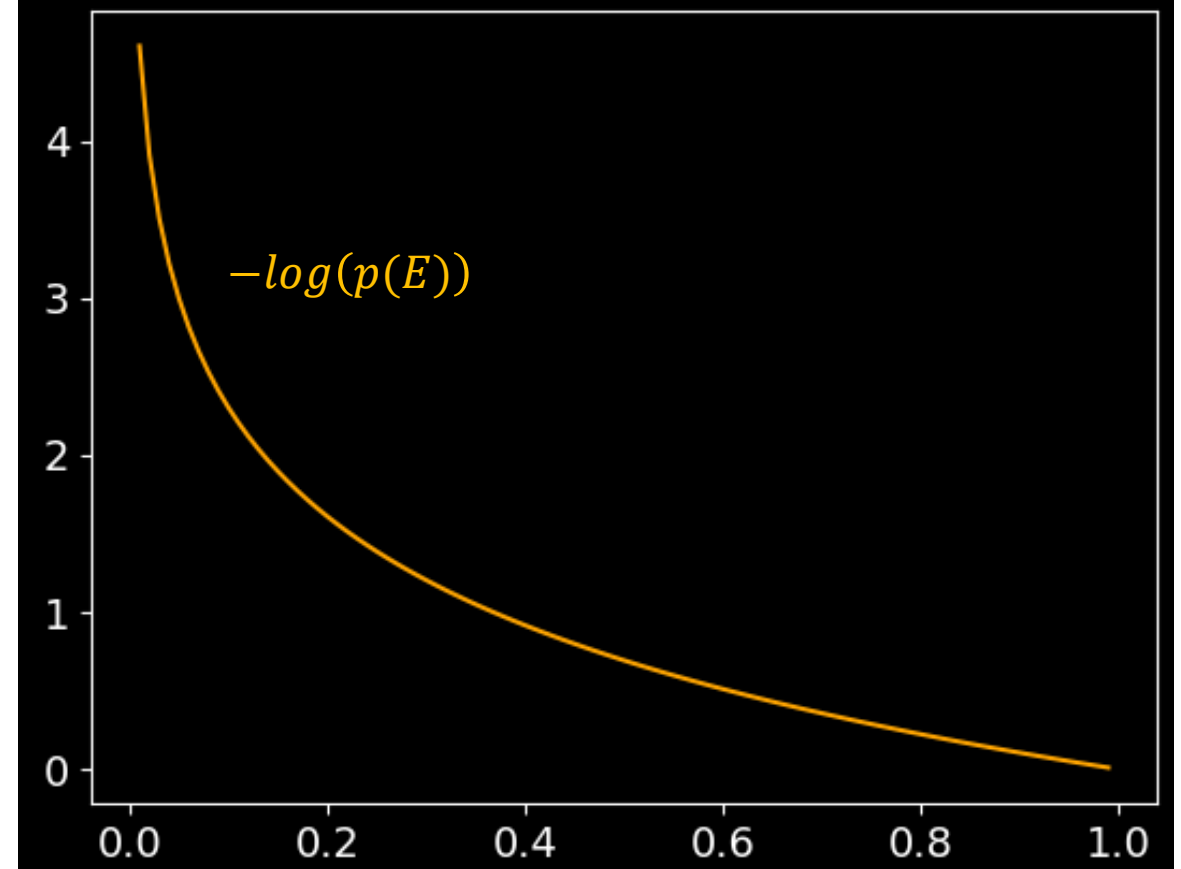
Problem?





Monotonic decrease of the function surprise(E)

$$\begin{aligned} \log(\text{Surprise}(E)) &= \log\left(\frac{1}{p(E)}\right) \\ &= -\log(p(E)) \end{aligned}$$

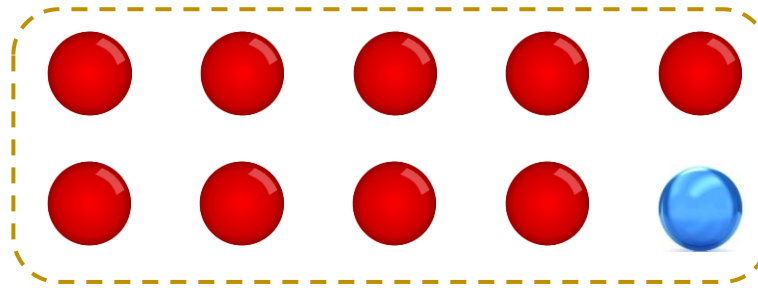
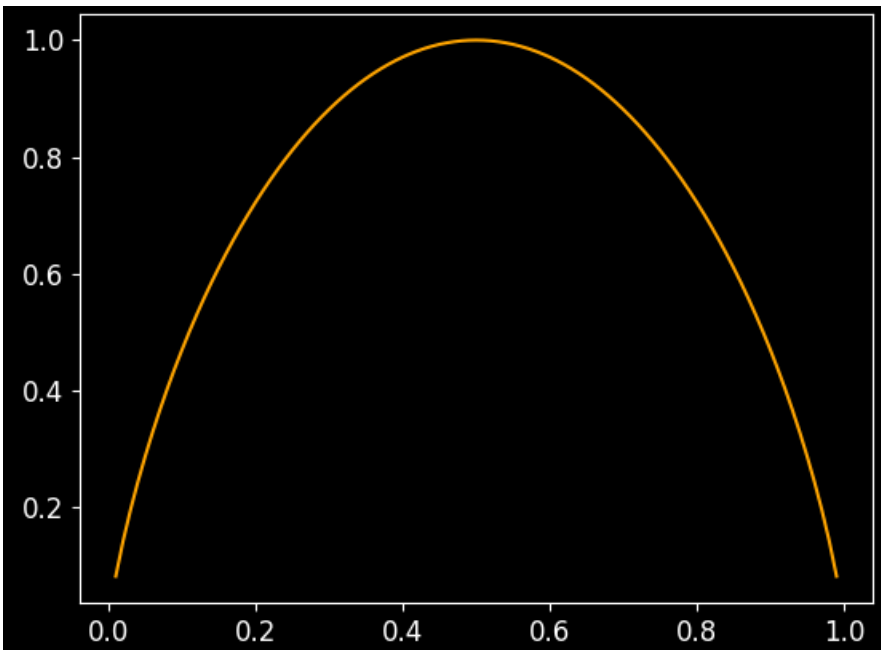


In information theory

$$\text{Information}(x) = -\log(p(x))$$

Entropy: Average of information

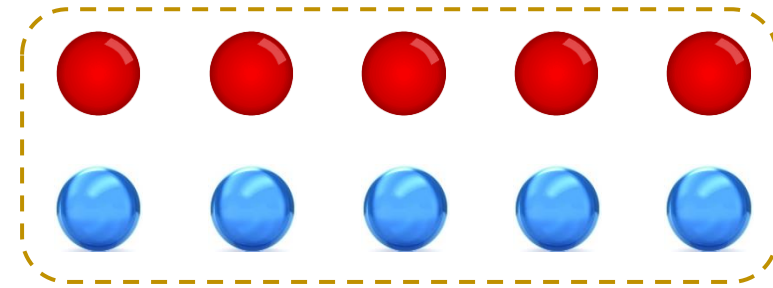
$$H(X) := - \sum_{x \in X} p(x) \log(p(x))$$



$$p(X = 0) = \frac{9}{10} = 0.9$$

$$p(X = 1) = \frac{1}{10} = 0.1$$

$$\begin{aligned} H(X) &= - \sum_{x \in X} p(x) \log(p(x)) \\ &= -0.9 \log(0.9) - 0.1 \log(0.1) \\ &= 0.468 \end{aligned}$$



$$p(X = 0) = \frac{5}{10} = 0.5$$

$$p(X = 1) = \frac{5}{10} = 0.5$$

$$\begin{aligned} H(X) &= - \sum_{x \in X} p(x) \log(p(x)) \\ &= -0.5 \log(0.5) - 0.5 \log(0.5) \\ &= 1.0 \end{aligned}$$

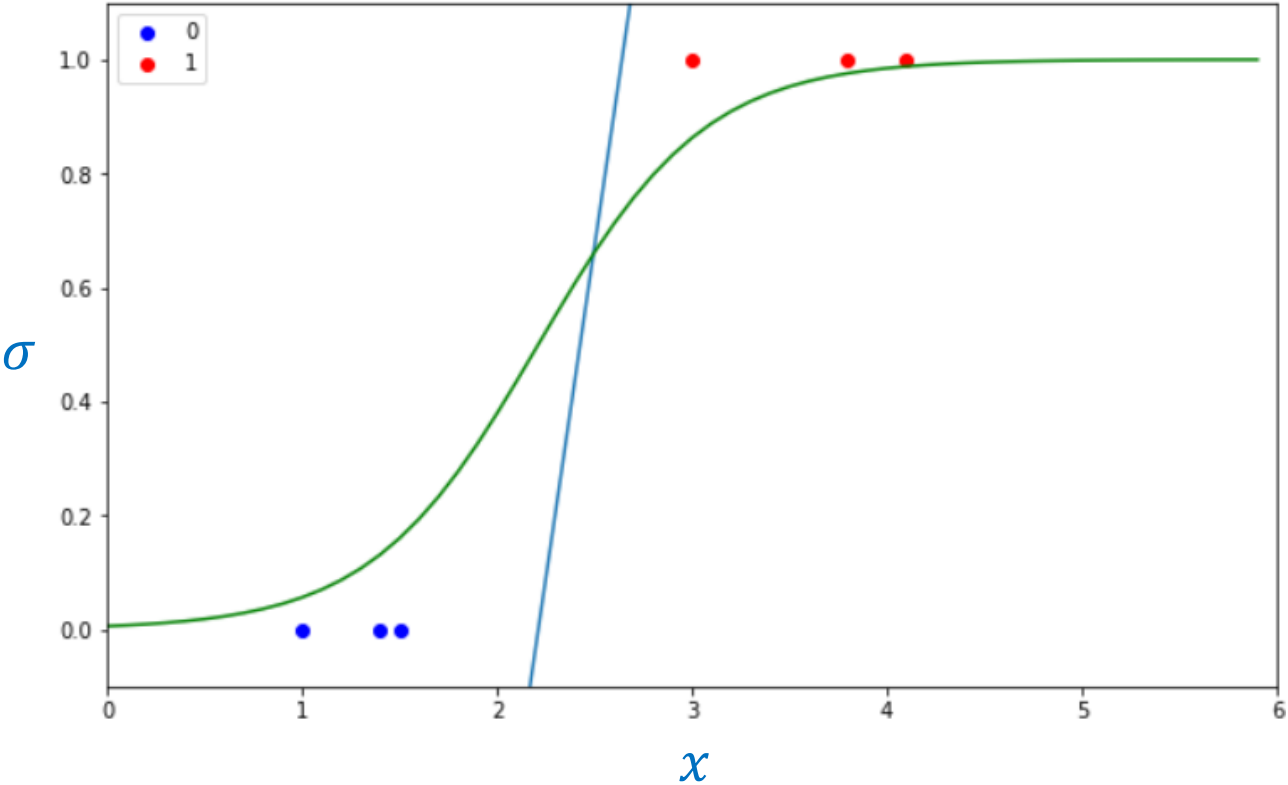
Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

z	$\sigma(z)$
-1.89	0.1309
-2.82	0.0559
-1.65	0.1598
1.837	0.8625
3.701	0.9759
4.401	0.9878

$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$

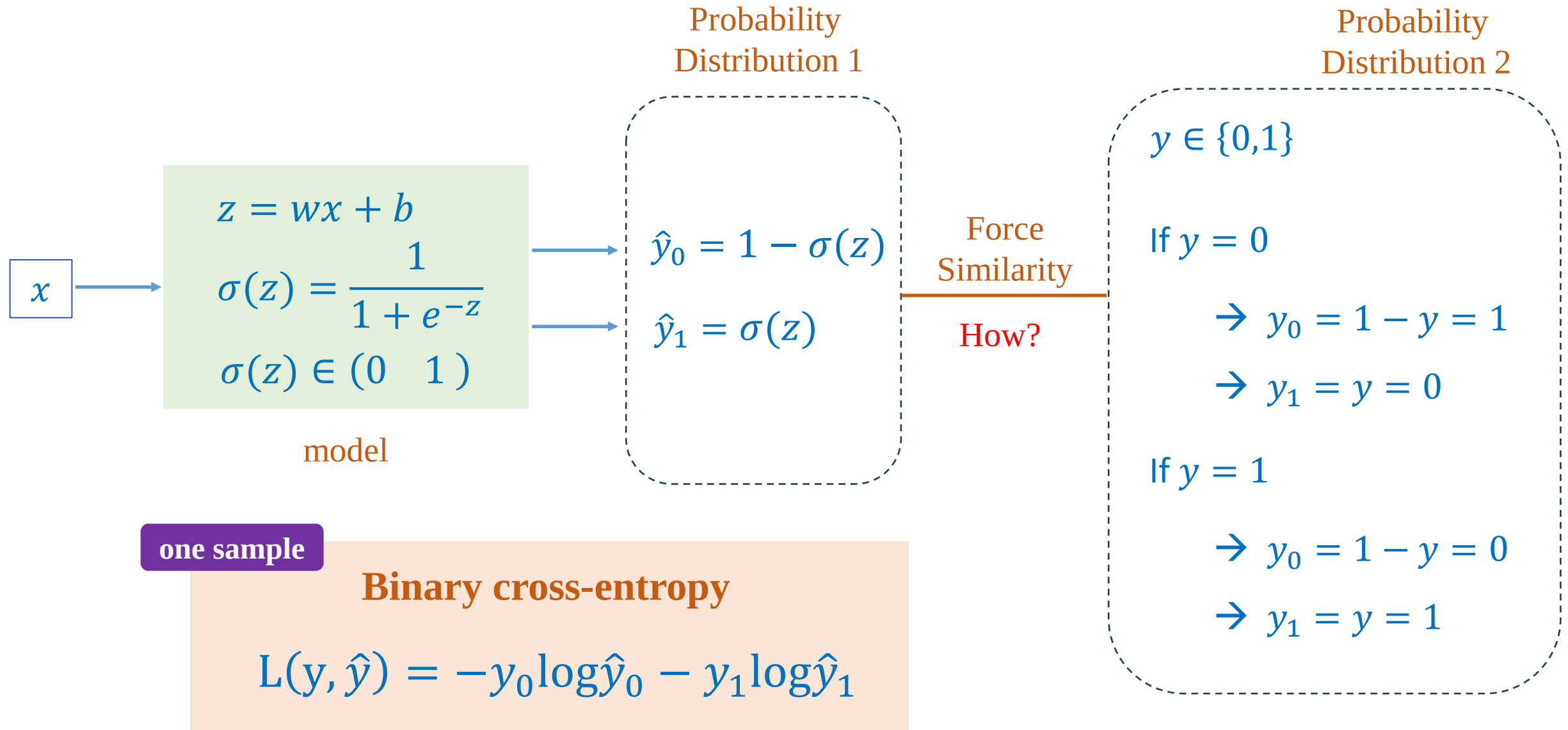


How to interpret $\sigma(z)$?

(Binary) Cross-Entropy

58

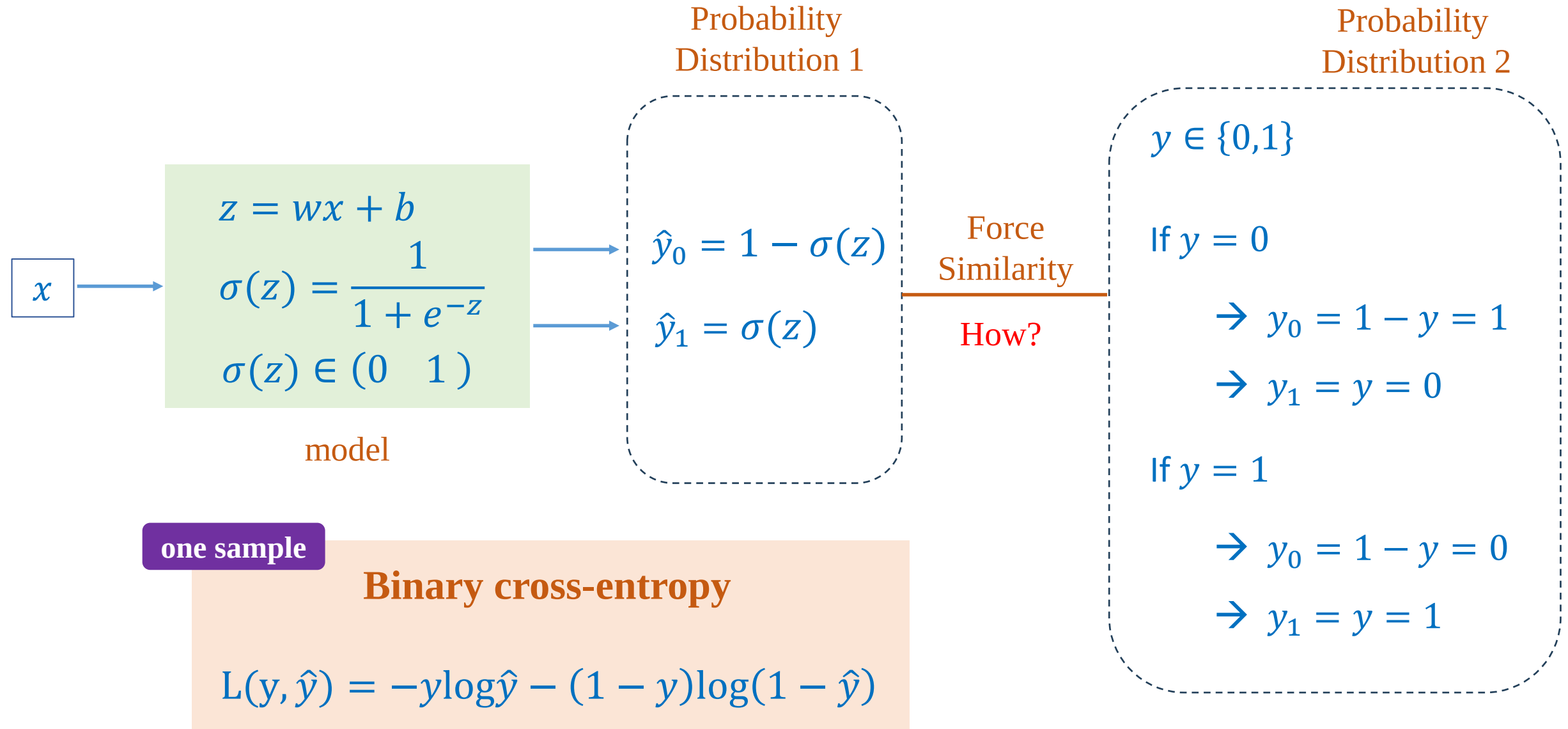
Given a sample (x, y)



(Binary) Cross-Entropy

58

Given a sample (x, y)



Prove convexity

Suppose we have:

- A single feature input x , and a target $y \in \{0, 1\}$.
- A logistic model with one feature, so $z = wx + b$, where w is the weight and b is the bias.

The model outputs the probability:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(wx+b)}}$$

The binary cross-entropy loss for this single point is:

$$L(w, b) = -(y \log \hat{y} + (1 - y) \log(1 - \hat{y}))$$

Step 1: Compute the Gradient of $L(w, b)$

1. Rewrite the loss in terms of $z = wx + b$:

$$L(z) = -(y \log(\sigma(z)) + (1 - y) \log(1 - \sigma(z)))$$

2. First Derivative with respect to z :

Since $\hat{y} = \sigma(z)$, the derivative of L with respect to z is:

$$\frac{dL}{dz} = \hat{y} - y$$

3. Gradient with respect to w and b :

Using the chain rule:

- $\frac{\partial L}{\partial w} = \frac{dL}{dz} \cdot \frac{dz}{dw} = (\hat{y} - y)x$
- $\frac{\partial L}{\partial b} = \frac{dL}{dz} \cdot \frac{dz}{db} = (\hat{y} - y)$

Step 2: Compute the Hessian Matrix of $L(w, b)$

Now, let's find the second derivatives to get the Hessian matrix of L with respect to w and b . The Hessian matrix H is:

$$H = \begin{bmatrix} \frac{\partial^2 L}{\partial w^2} & \frac{\partial^2 L}{\partial w \partial b} \\ \frac{\partial^2 L}{\partial b \partial w} & \frac{\partial^2 L}{\partial b^2} \end{bmatrix}$$

To compute each element:

1. Second derivative with respect to w :

$$\frac{\partial^2 L}{\partial w^2} = \frac{\partial}{\partial w} ((\hat{y} - y)x) = \hat{y}(1 - \hat{y})x^2$$

2. Second derivative with respect to b :

$$\frac{\partial^2 L}{\partial b^2} = \frac{\partial}{\partial b} (\hat{y} - y) = \hat{y}(1 - \hat{y})$$

3. Cross derivative with respect to w and b :

$$\frac{\partial^2 L}{\partial w \partial b} = \frac{\partial^2 L}{\partial b \partial w} = \frac{\partial}{\partial w} (\hat{y} - y) = \hat{y}(1 - \hat{y})x$$

So, the Hessian matrix H becomes:

$$H = \begin{bmatrix} \hat{y}(1 - \hat{y})x^2 & \hat{y}(1 - \hat{y})x \\ \hat{y}(1 - \hat{y})x & \hat{y}(1 - \hat{y}) \end{bmatrix}$$

Step 3: Check Positive Semi-Definiteness of H

To verify convexity, we need to confirm that H is positive semi-definite. For a 2×2 matrix, H is positive semi-definite if its determinant is non-negative and its diagonal entries are non-negative.

1. Determinant of H :

$$\begin{aligned} \det(H) &= (\hat{y}(1 - \hat{y})x^2) \cdot (\hat{y}(1 - \hat{y})) - (\hat{y}(1 - \hat{y})x)^2 \\ &= \hat{y}^2(1 - \hat{y})^2(x^2 - x^2) = 0 \end{aligned}$$

Since the determinant is zero, H is semi-definite.

2. Diagonal Entries of H :

Both diagonal entries, $\hat{y}(1 - \hat{y})x^2$ and $\hat{y}(1 - \hat{y})$, are non-negative (since $0 \leq p \leq 1$)

Prove using the eigenvalues of Hessian matrix

To prove that the Hessian matrix H has non-negative eigenvalues, let's explicitly analyze the eigenvalues of H for a 2×2 matrix.

The Hessian matrix we derived is:

$$H = \begin{bmatrix} \hat{y}(1 - \hat{y})x^2 & \hat{y}(1 - \hat{y})x \\ \hat{y}(1 - \hat{y})x & \hat{y}(1 - \hat{y}) \end{bmatrix}$$

Where:

- $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$,
- $0 \leq p \leq 1$, so $\hat{y}(1 - \hat{y}) \geq 0$

For a 2×2 matrix of the form:

$$H = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

The eigenvalues λ are given by the solutions to the characteristic polynomial:

$$\det(H - \lambda I) = 0$$

Where I is the identity matrix. For our Hessian matrix H , this becomes:

$$\det \begin{bmatrix} a - \lambda & b \\ b & d - \lambda \end{bmatrix} = 0$$

Expanding the determinant, we get:

$$(a - \lambda)(d - \lambda) - b^2 = 0$$

The eigenvalues λ are then:

$$\lambda = \frac{(a + d) \pm \sqrt{(a - d)^2 + 4b^2}}{2}$$

(Binary) Cross-Entropy

Applying to Our Hessian Matrix

For our Hessian matrix:

- $a = \hat{y}(1 - \hat{y})x^2$,
- $b = \hat{y}(1 - \hat{y})x$,
- $d = \hat{y}(1 - \hat{y})$.

Substitute these values into the formula for the eigenvalues:

1. Sum of the diagonal elements (trace of H):

$$a + d = \hat{y}(1 - \hat{y})x^2 + \hat{y}(1 - \hat{y}) = \hat{y}(1 - \hat{y})(x^2 + 1)$$

2. Difference of the diagonal elements:

$$a - d = \hat{y}(1 - \hat{y})(x^2 - 1)$$

3. Eigenvalues:

Using the eigenvalue formula:

$$\lambda = \frac{(a + d) \pm \sqrt{(a - d)^2 + 4b^2}}{2}$$

Substitute $a + d = \hat{y}(1 - \hat{y})(x^2 + 1)$, $a - d = \hat{y}(1 - \hat{y})(x^2 - 1)$, and $b = \hat{y}(1 - \hat{y})x$:

$$\lambda = \frac{\hat{y}(1 - \hat{y})(x^2 + 1) \pm \sqrt{(\hat{y}(1 - \hat{y})(x^2 - 1))^2 + 4(\hat{y}(1 - \hat{y})x)^2}}{2}$$

4. Simplifying the Square Root:

Notice that:

$$(\hat{y}(1 - \hat{y})(x^2 - 1))^2 + 4(\hat{y}(1 - \hat{y})x)^2 = \hat{y}^2(1 - \hat{y})^2((x^2 - 1)^2 + 4x^2)$$

Expanding $(x^2 - 1)^2 + 4x^2$, we get:

$$(x^4 - 2x^2 + 1) + 4x^2 = x^4 + 2x^2 + 1 = (x^2 + 1)^2$$

So:

$$\sqrt{(\hat{y}(1 - \hat{y})(x^2 - 1))^2 + 4(\hat{y}(1 - \hat{y})x)^2} = \hat{y}(1 - \hat{y})(x^2 + 1)$$

5. Eigenvalues:

Therefore:

$$\lambda = \frac{\hat{y}(1 - \hat{y})(x^2 + 1) \pm \hat{y}(1 - \hat{y})(x^2 + 1)}{2}$$

This yields two eigenvalues:

$$\lambda_1 = \hat{y}(1 - \hat{y})(x^2 + 1) \text{ and } \lambda_2 = 0$$



AI VIET NAM

@aivietnam.edu.vn

Insight into Logistic Regression

Quang-Vinh Dinh
Ph.D. in Computer Science

Objectives

One-Sample

$$\mathbf{x} = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\left\{ \begin{array}{l} b = b - \eta \frac{\partial L}{\partial b} \\ w = w - \eta \frac{\partial L}{\partial w} \\ \boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L \end{array} \right.$$

$$\rightarrow \boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

N-Samples

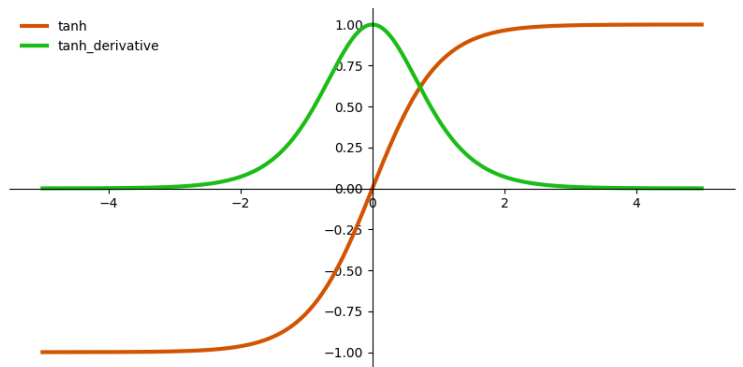
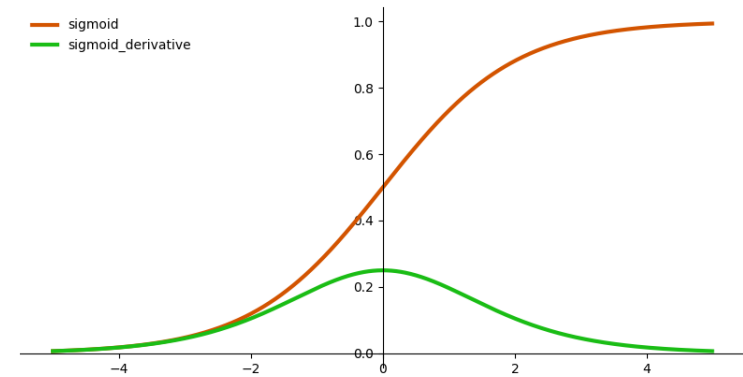
$$\mathbf{x} = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix} \quad \hat{\mathbf{y}} = \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix}$$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{y} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1 - \mathbf{y})^T \log(1 - \hat{\mathbf{y}}))$$

Sigmoid&Tanh



Outline

SECTION 1

Vectorization for 1-sample

SECTION 2

Vectorization for m-sample

SECTION 3

Vectorization for N-sample

SECTION 3

Sigmoid & Tanh Functions

$$z = \boldsymbol{\theta}^T \mathbf{x}$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\nabla_{\boldsymbol{\theta}} L = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w} \end{bmatrix}$$

$$\left\{ \begin{array}{l} b = b - \eta \frac{\partial L}{\partial b} \\ w = w - \eta \frac{\partial L}{\partial w} \end{array} \right. \quad \begin{array}{l} \boldsymbol{\theta} \\ \boldsymbol{\theta} \\ \nabla_{\boldsymbol{\theta}} L \end{array}$$

$$\rightarrow \boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

Implementation - One Sample

Feature	Label
Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

1) Pick a **sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

```
def sigmoid_function(z):  
    return 1 / (1 + math.exp(-z))  
  
def predict(x, w, b):  
    z = w*x + b  
    y_hat = sigmoid_function(z)  
  
    return y_hat  
  
def loss_function(y_hat, y):  
    return -y*math.log(y_hat) - (1 - y)*math.log(1 - y_hat)  
  
def compute_gradient(x, y_hat, y):  
    dw = x*(y_hat - y)  
    db = (y_hat - y)  
  
    return dw, db  
  
def update(w, b, dw, db, lr):  
    w = w - lr*dw  
    b = b - lr*db  
  
    return w, b
```

if #features changes, which functions are affected?

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w_i} = x_i(\hat{y} - y) \quad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w_i = w_i - \eta \frac{\partial L}{\partial w_i} \quad b = b - \eta \frac{\partial L}{\partial b}$$

```
def sigmoid_function(z):
    return 1 / (1 + np.exp(-z))
```

```
def predict(x1, x2, b, w1, w2):
    z = x1*w1 + x2*w2 + b
    y_hat = sigmoid_function(z)

    return y_hat
```

```
def loss_function(y_hat, y):
    return -y*np.log(y_hat) - (1 - y)*np.log(1 - y_hat)
```

```
def compute_gradient(x1, x2, y_hat, y):
    db = (y_hat - y)
    dw1 = x1*(y_hat - y)
    dw2 = x2*(y_hat - y)

    return (db, dw1, dw2)
```

How to solve the problem?

```
def update(b, w1, w2, lr, db, dw1, dw2):
    b = b - lr*db
    w1 = w1 - lr*dw1
    w2 = w2 - lr*dw2

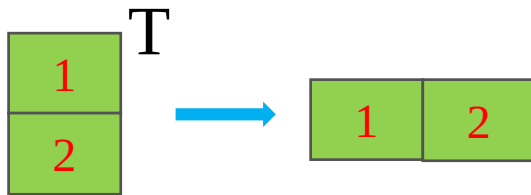
    return (b, w1, w2)
```


Vector/Matrix Operations

Transpose

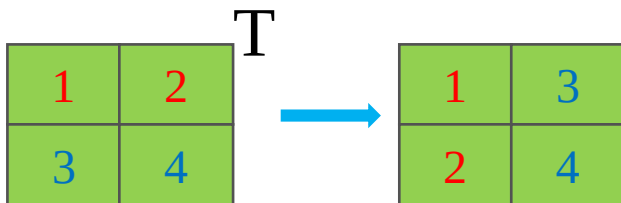
$$\vec{v} = \begin{bmatrix} v_1 \\ \dots \\ v_n \end{bmatrix}$$

$$\vec{v}^T = [v_1 \dots v_n]$$



$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

$$A^T = \begin{bmatrix} a_{11} & \dots & a_{m1} \\ \dots & \dots & \dots \\ a_{1n} & \dots & a_{mn} \end{bmatrix}$$

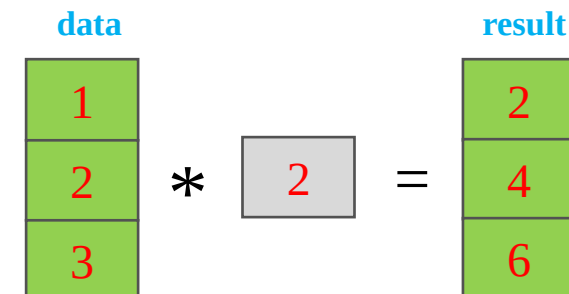


```
2 import numpy as np
3
4 # create data
5 data = np.array([1,2,3])
6 factor = 2
7
8 # broadcasting
9 result_multiplication = data*factor
```

```
[1 2 3]
[2 4 6]
```

Multiply with a number

$$\alpha \vec{u} = \alpha \begin{bmatrix} u_1 \\ \dots \\ u_n \end{bmatrix} = \begin{bmatrix} \alpha u_1 \\ \dots \\ \alpha u_n \end{bmatrix}$$

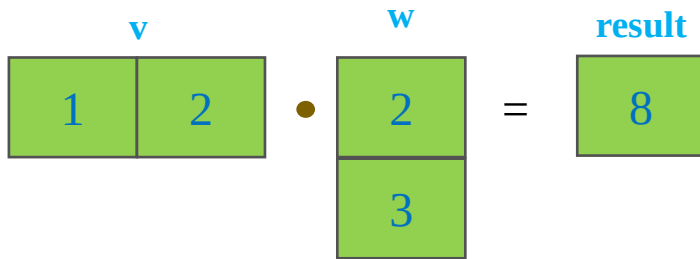


Vector/Matrix Operations

Dot product

$$\vec{v} = \begin{bmatrix} v_1 \\ \dots \\ v_n \end{bmatrix} \quad \vec{u} = \begin{bmatrix} u_1 \\ \dots \\ u_n \end{bmatrix}$$

$$\vec{v} \cdot \vec{u} = v_1 \times u_1 + \dots + v_n \times u_n$$



```
1 def dot_product(vector1, vector2):
2     '''
3     Compute dot product between two vectors
4     Output is a floating-point number
5     '''
6
7     return sum([v1*v2 for v1, v2 in zip(vector1, vector2)])
8
9 # test case
10 vector1 = [1, 2, 3]
11 vector2 = [2, 3, 4]
12
13 ouptut = dot_product(vector1, vector2)
14 print(ouptut)
15
16
17
18
19
20
```

```
2 import numpy as np
3
4 v = np.array([1, 2])
5 w = np.array([2, 3])
6
7 # Tính inner product giữa v và w
8 print('method 1 \n', v.dot(w))
9 print('method 2 \n', np.dot(v, w))
10
11
12
13
14
15
16
17
18
19
20
```

method 1
8

method 2
8

Vectorization

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7
x	y

1) Pick a **sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Traditional

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log (1 - \hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

$$z = wx + b \quad x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w \end{bmatrix} \rightarrow \theta^T = [b \quad w]$$

$$z = wx + b1 = [b \quad w] \begin{bmatrix} 1 \\ x \end{bmatrix} = \theta^T x$$

dot product

1) Pick a **sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

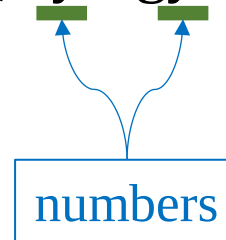
η is learning rate

Traditional

$$z = wx + b \quad x = \begin{bmatrix} 1 \\ x \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$z = \theta^T x \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$



What will we do?

1) Pick a **sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w} \qquad b = b - \eta \frac{\partial L}{\partial b}$$

$$\begin{bmatrix} (\hat{y} - y) \times 1 \\ (\hat{y} - y) \times x \end{bmatrix} = \underbrace{(\hat{y} - y)}_{\text{common factor}} \begin{bmatrix} 1 \\ x \end{bmatrix} = (\hat{y} - y) \mathbf{x} = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w} \end{bmatrix} = \nabla_{\theta} L \quad \rightarrow \quad \nabla_{\theta} L = \mathbf{x}(\hat{y} - y)$$

Traditional

Vectorization

$$z = \mathbf{w} \mathbf{x} + b \qquad \mathbf{x} = \begin{bmatrix} 1 \\ x \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\begin{cases} \frac{\partial L}{\partial b} = (\hat{y} - y) = (\hat{y} - y) \times 1 \\ \frac{\partial L}{\partial w} = x(\hat{y} - y) = (\hat{y} - y) \times x \end{cases}$$

Vectorization

1) Pick a **sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Traditional

$$z = \theta^T x$$

$$x = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\nabla_{\theta} L = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w} \end{bmatrix}$$

$$\left\{ \begin{array}{l} b = b - \eta \frac{\partial L}{\partial b} \\ w = w - \eta \frac{\partial L}{\partial w} \end{array} \right. \quad \nabla_{\theta} L$$

$$\rightarrow \theta = \theta - \eta \nabla_{\theta} L$$

Vectorization

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Traditional

1) Pick a sample $(\mathbf{x}, y)$ from training data

2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta} \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \mathbf{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate

Vectorized

❖ Implementation (using Numpy)

→ 1) Pick a sample $(\mathbf{x}, y)$ from training data



2) Compute output $\hat{y}$



$$z = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta} \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss



$$L(\hat{y}, y) = (-y \log \hat{y} - (1 - y) \log(1 - \hat{y}))$$

4) Compute derivative



$$\nabla_{\boldsymbol{\theta}} L = \mathbf{x}(\hat{y} - y)$$

→ 5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate

```
def sigmoid_function(z):  
    return 1 / (1 + np.exp(-z))
```

```
def predict(X, theta):  
    return sigmoid_function( np.dot(X.T, theta) )
```

```
def loss_function(y_hat, y):  
    return -y*np.log(y_hat) - (1 - y)*np.log(1 - y_hat)
```

```
def compute_gradient(X, y_hat, y):  
    return X*(y_hat - y)
```

```
def update(theta, lr, gradient):  
    return theta - lr*gradient
```

```
# compute output
```

```
y_hat = predict(X, theta)
```

```
# compute loss
```

```
loss = loss_function(y_hat, y)
```

```
# compute mean of gradient
```

```
gradient = compute_gradient(X, y_hat, y)
```

```
# update
```

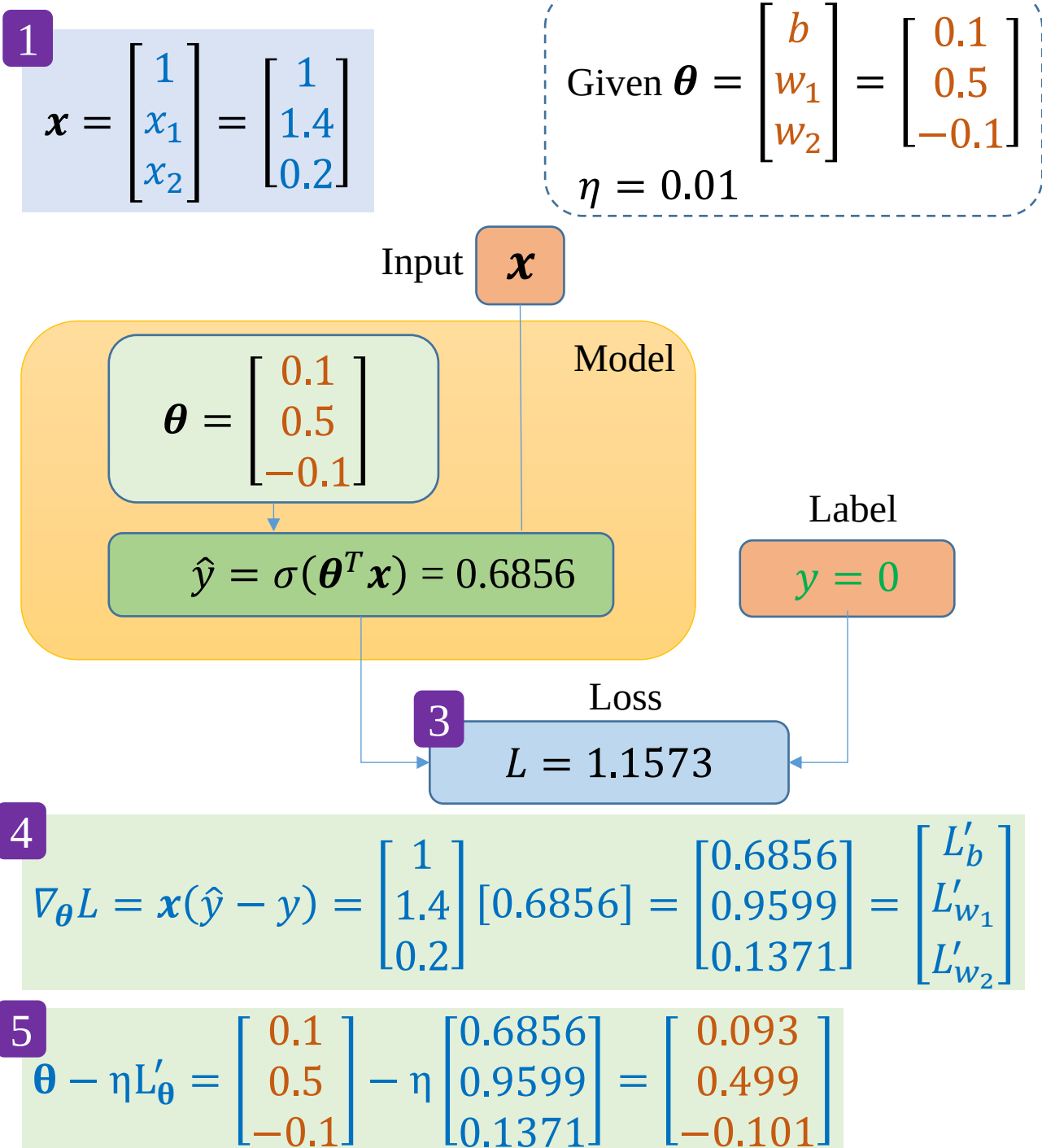
```
theta = update(theta, lr, gradient)
```

Given X and y

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

- 1) Pick a sample $(\mathbf{x}, y)$ from training data
- ↓
- 2) Compute output $\hat{y}$
- ↓ $z = \boldsymbol{\theta}^T \mathbf{x} = \mathbf{x}^T \boldsymbol{\theta} \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$
- 3) Compute loss
- ↓ $L(\hat{y}, y) = (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$
- 4) Compute derivative
- ↓ $\nabla_{\boldsymbol{\theta}} L = \mathbf{x}(\hat{y} - y)$
- 5) Update parameters
- $\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1.4 \\ 0.2 \end{bmatrix} \quad \mathbf{y} = [0]$$

Demo

- 1) Pick a sample $(\mathbf{x}, y)$ from training data
- 2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T \mathbf{x}$$
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

- 3) Compute loss

$$L(\boldsymbol{\theta}) = -y \log \hat{y} - (1-y) \log (1-\hat{y})$$

- 4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \mathbf{x}(\hat{y} - y)$$

- 5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate

Visualization – Decision Boundary

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T \mathbf{x}$$
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\boldsymbol{\theta}) = -y \log \hat{y} - (1-y) \log(1-\hat{y})$$

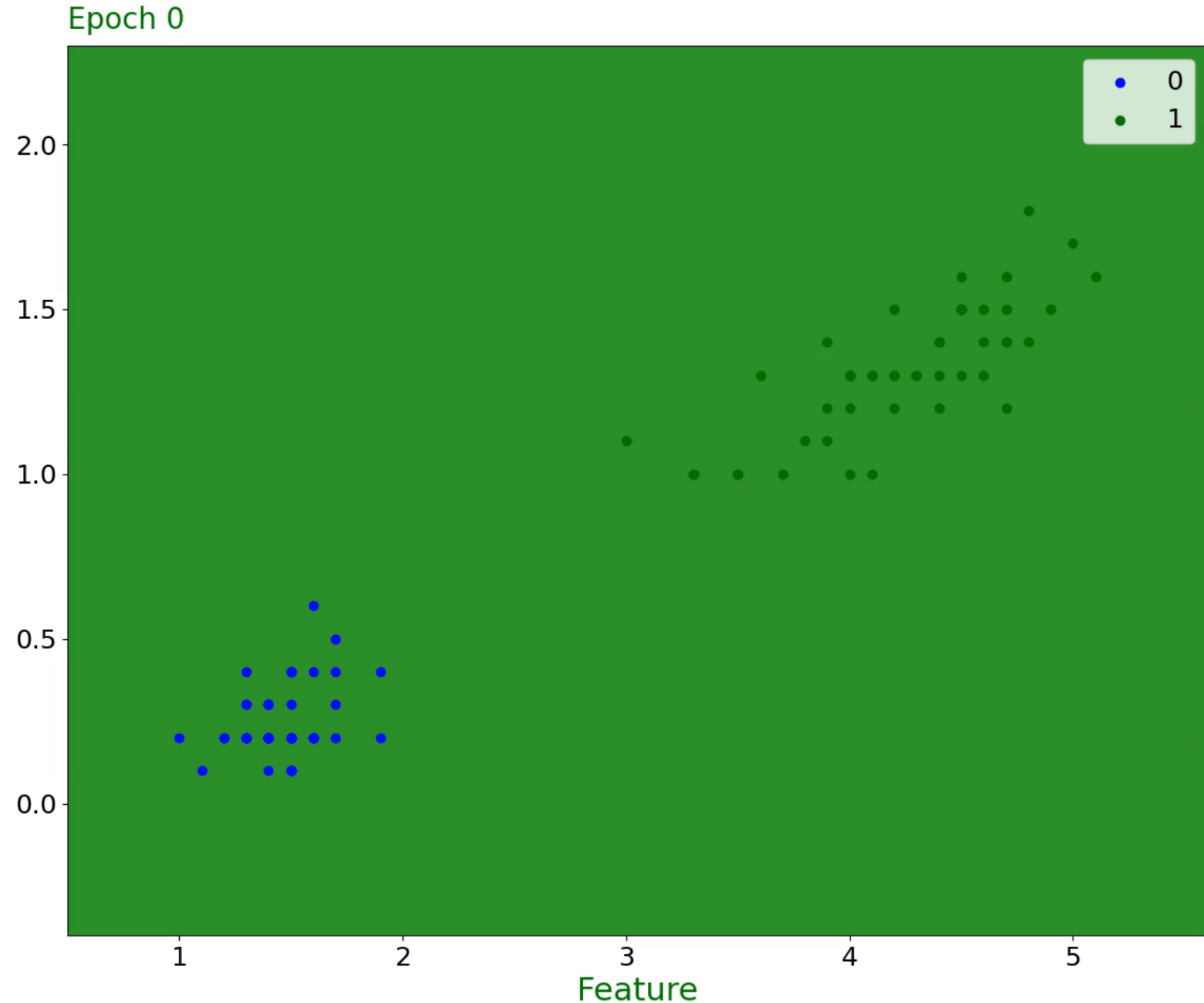
4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \mathbf{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate



Outline

SECTION 1

Vectorization for 1-sample

SECTION 2

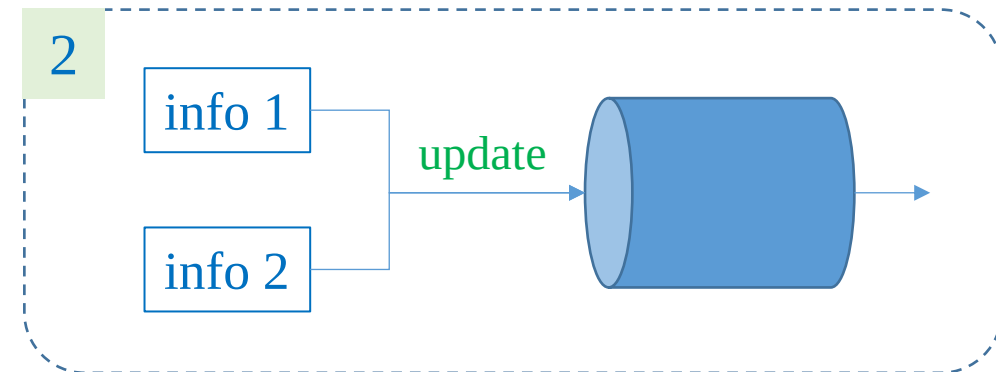
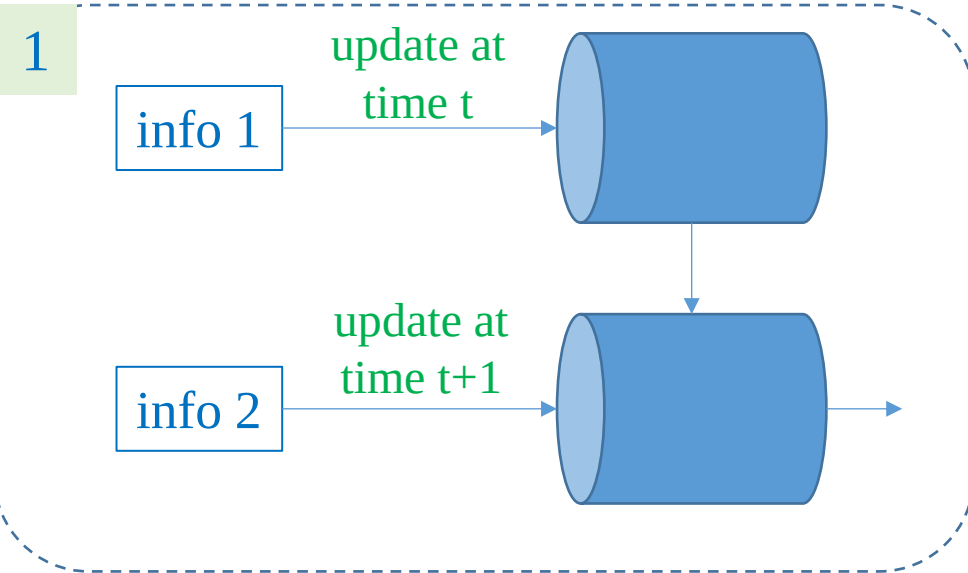
Vectorization for m-sample

SECTION 3

Vectorization for N-sample

SECTION 3

Sigmoid & Tanh Functions

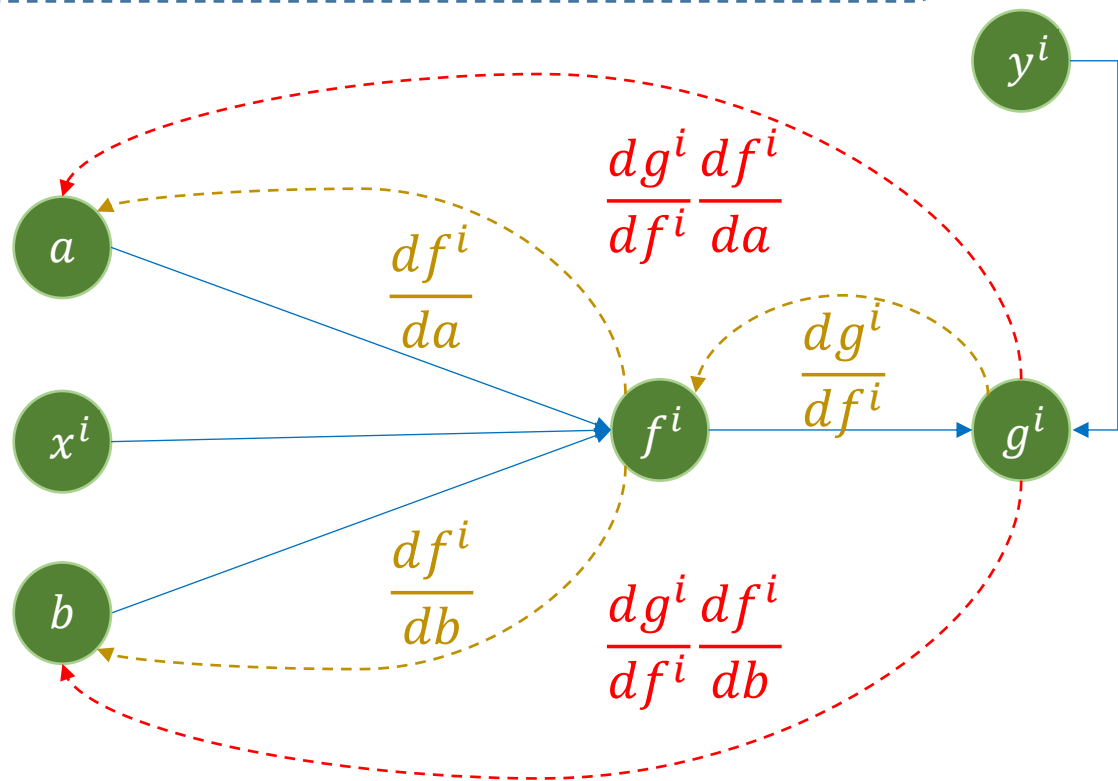


❖ Equations for partial gradients

$$f(x^i) = ax^i + b \qquad (x^1=1, y^1=5)$$

$$g(f^i) = (f^i - y^i)^2 \qquad (x^2=2, y^2=7)$$

illustration



$$\frac{df}{da} = x \qquad \frac{df}{db} = 1$$

$$\frac{dg}{df} = 2(f - y)$$

$$\frac{dg}{da} = \frac{dg}{df} \frac{df}{da} = 2x(f - y)$$

$$\frac{dg}{db} = \frac{dg}{df} \frac{df}{db} = 2(f - y)$$

During looking for optimal a and b , at a given time, a and b have concrete values

❖ Optimization for a composite function

Find a and b so that $g(f(x))$ is minimum

$$f(x^i) = ax^i + b \quad (x^1=1, y^1=5)$$

$$g(f^i) = (f^i - y^i)^2 \quad (x^2=2, y^2=7)$$

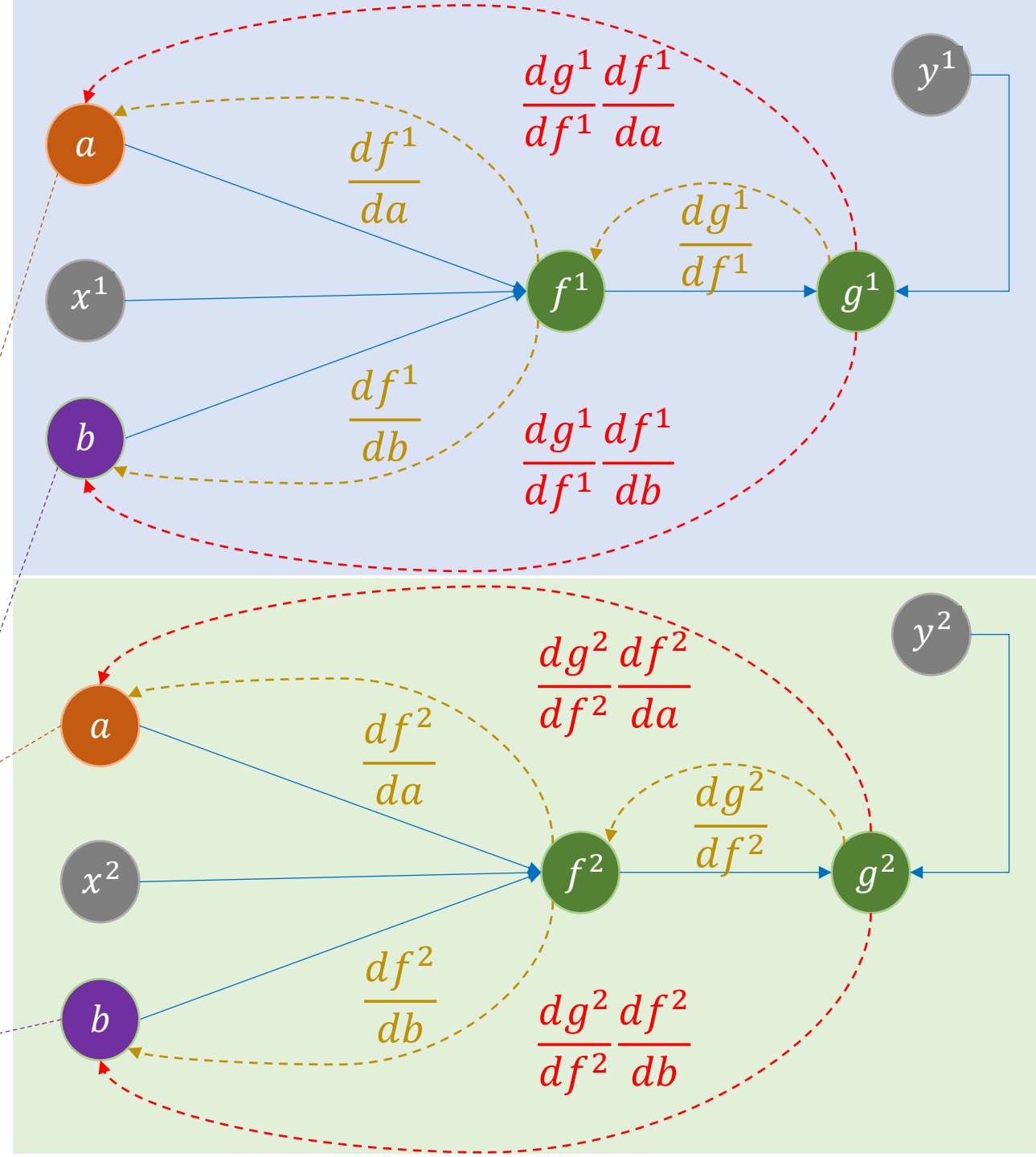
Partial derivative functions

$$\frac{dg}{da} = \frac{dg}{df} \frac{df}{da} = 2x(f - y)$$

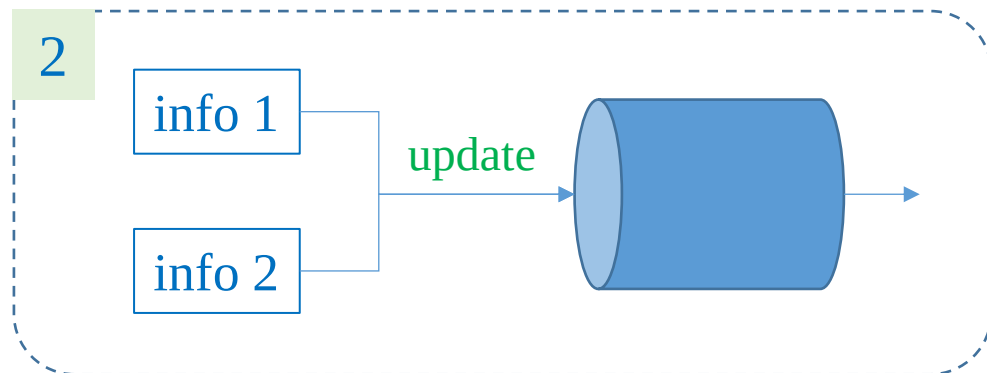
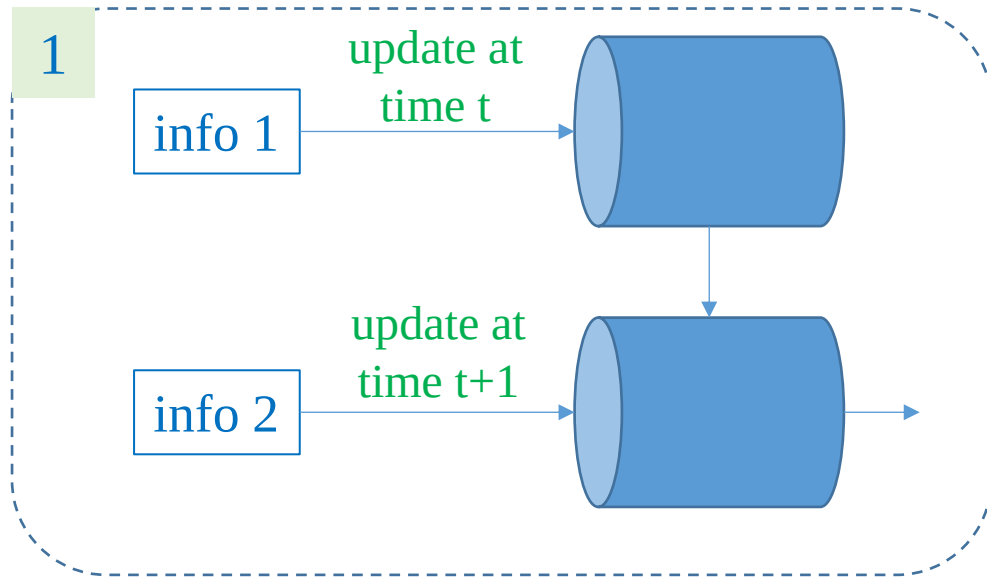
$$\frac{dg}{db} = \frac{dg}{df} \frac{df}{db} = 2(f - y)$$

$$\sum_i \frac{dg^i}{da} = \frac{dg^1}{df^1} \frac{df^1}{da} + \frac{dg^2}{df^2} \frac{df^2}{da}$$

$$\sum_i \frac{dg_i}{db} = \frac{dg^1}{df^1} \frac{df^1}{db} + \frac{dg^2}{df^2} \frac{df^2}{db}$$



❖ How to use gradient information



❖ Construct formulas

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

2) Compute output $\hat{y}$

$$\mathbf{x} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$\mathbf{z} = \begin{bmatrix} z^{(1)} \\ z^{(2)} \end{bmatrix} = \begin{bmatrix} w_1 x_1^{(1)} + w_2 x_2^{(1)} + b \\ w_1 x_1^{(2)} + w_2 x_2^{(2)} + b \end{bmatrix}$$

$$= \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix} \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \mathbf{x}\boldsymbol{\theta} = \begin{bmatrix} 0.83 \\ 2.02 \end{bmatrix}$$

❖ Construct formulas

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

2) Compute output $\hat{y}$

$$\mathbf{x} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta} = \begin{bmatrix} 0.83 \\ 2.02 \end{bmatrix}$$

$$\hat{\mathbf{y}} = \sigma(\mathbf{z}) = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{1}{1 + e^{-z^{(1)}}} \\ \frac{1}{1 + e^{-z^{(2)}}} \end{bmatrix} = \frac{1}{1 + e^{-z}} = \begin{bmatrix} 0.69 \\ 0.88 \end{bmatrix}$$

❖ Construct formulas

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1-\mathbf{y})^T \log(1-\hat{\mathbf{y}}))$$

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{L^{(1)}(\hat{y}^{(1)}, y^{(1)}) + L^{(2)}(\hat{y}^{(2)}, y^{(2)})}{m}$$

$$L^{(1)}(\hat{y}^{(1)}, y^{(1)}) = -y^{(1)} \log \hat{y}^{(1)} - (1-y^{(1)}) \log(1-\hat{y}^{(1)})$$

+

$$L^{(2)}(\hat{y}^{(2)}, y^{(2)}) = -y^{(2)} \log \hat{y}^{(2)} - (1-y^{(2)}) \log(1-\hat{y}^{(2)})$$

$$\mathbf{y}^T \log \hat{\mathbf{y}}$$

$$(1-\mathbf{y})^T \log(1-\hat{\mathbf{y}})$$

4) Compute derivative

sample 1

$$\frac{\partial L^{(1)}}{\partial b} = (\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_1} = x_1^{(1)} (\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_2} = x_2^{(1)} (\hat{y}^{(1)} - y^{(1)})$$

sample 2

$$\frac{\partial L^{(2)}}{\partial b} = (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_1} = x_1^{(2)} (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_2} = x_2^{(2)} (\hat{y}^{(2)} - y^{(2)})$$

$$\begin{aligned} \frac{\partial L}{\partial b} &= \frac{\partial L^{(1)}}{\partial b} + \frac{\partial L^{(2)}}{\partial b} = \frac{(\hat{y}^{(1)} - y^{(1)}) + (\hat{y}^{(2)} - y^{(2)})}{m} \\ &= \frac{1}{m} [1 * (\hat{y}^{(1)} - y^{(1)}) + 1 * (\hat{y}^{(2)} - y^{(2)})] \\ &= \frac{1}{m} \begin{bmatrix} x_0^{(1)} & x_0^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix} \end{aligned}$$

$$x_0^{(1)} = 1$$

$$x_0^{(2)} = 1$$

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \frac{\partial L^{(1)}}{\partial w_1} + \frac{\partial L^{(2)}}{\partial w_1} = \frac{x_1^{(1)} (\hat{y}^{(1)} - y^{(1)}) + x_1^{(2)} (\hat{y}^{(2)} - y^{(2)})}{m} \\ &= \frac{1}{m} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w_2} &= \frac{\partial L^{(1)}}{\partial w_2} + \frac{\partial L^{(2)}}{\partial w_2} = \frac{x_2^{(1)} (\hat{y}^{(1)} - y^{(1)}) + x_2^{(2)} (\hat{y}^{(2)} - y^{(2)})}{m} \\ &= \frac{1}{m} \begin{bmatrix} x_2^{(1)} & x_2^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix} \end{aligned}$$

4) Compute derivative

sample 1

$$\frac{\partial L^{(1)}}{\partial b} = (\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_1} = x_1^{(1)} (\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_2} = x_2^{(1)} (\hat{y}^{(1)} - y^{(1)})$$

sample 2

$$\frac{\partial L^{(2)}}{\partial b} = (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_1} = x_1^{(2)} (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_2} = x_2^{(2)} (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L}{\partial b} = \frac{1}{m} \begin{bmatrix} x_0^{(1)} & x_0^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{m} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_2} = \frac{1}{m} \begin{bmatrix} x_2^{(1)} & x_2^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\nabla_{\theta} L = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \frac{1}{m} \begin{bmatrix} x_0^{(1)} & x_0^{(2)} \\ x_1^{(1)} & x_1^{(2)} \\ x_2^{(1)} & x_2^{(2)} \end{bmatrix} \begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$(\mathbf{x}^{(1)})^T$ $(\mathbf{x}^{(2)})^T$ $\hat{\mathbf{y}}$ $\mathbf{y}$

$$\nabla_{\theta} L = \frac{1}{m} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

5) Update parameters

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$x = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$\nabla_{\theta} L = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \frac{1}{m} x^T (\hat{y} - y)$$

$$\begin{array}{l} b \\ w_1 \\ w_2 \\ \theta \end{array} = \begin{array}{l} b \\ w_1 \\ w_2 \\ \theta \end{array} - \eta \begin{array}{l} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \nabla_{\theta} L \end{array}$$



$$\theta = \theta - \eta \nabla_{\theta} L$$

Logistic Regression - Minibatch

1) Pick m samples from training data

2) Compute output $\hat{y}$

$$z = x\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = \frac{1}{m} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{m} x^T (\hat{y} - y)$$

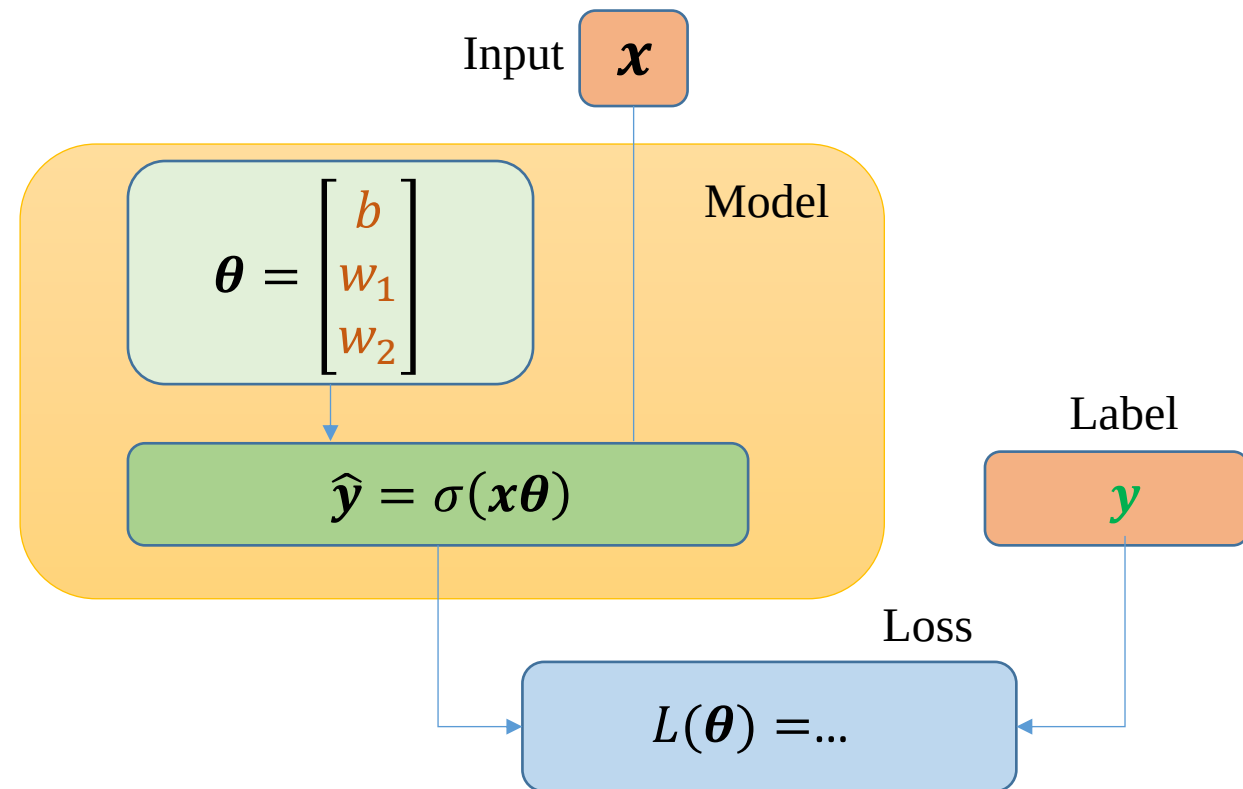
5) Update parameters

$$\theta = \theta - \eta \nabla_{\theta} L$$

η is learning rate

Mini-batch $m=2$

$$x = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix} \quad \theta = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick m samples from training data

2) Compute output $\hat{y}$

Mini-batch m=2

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{y} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1-\mathbf{y})^T \log(1-\hat{\mathbf{y}}))$$

4) Compute derivative

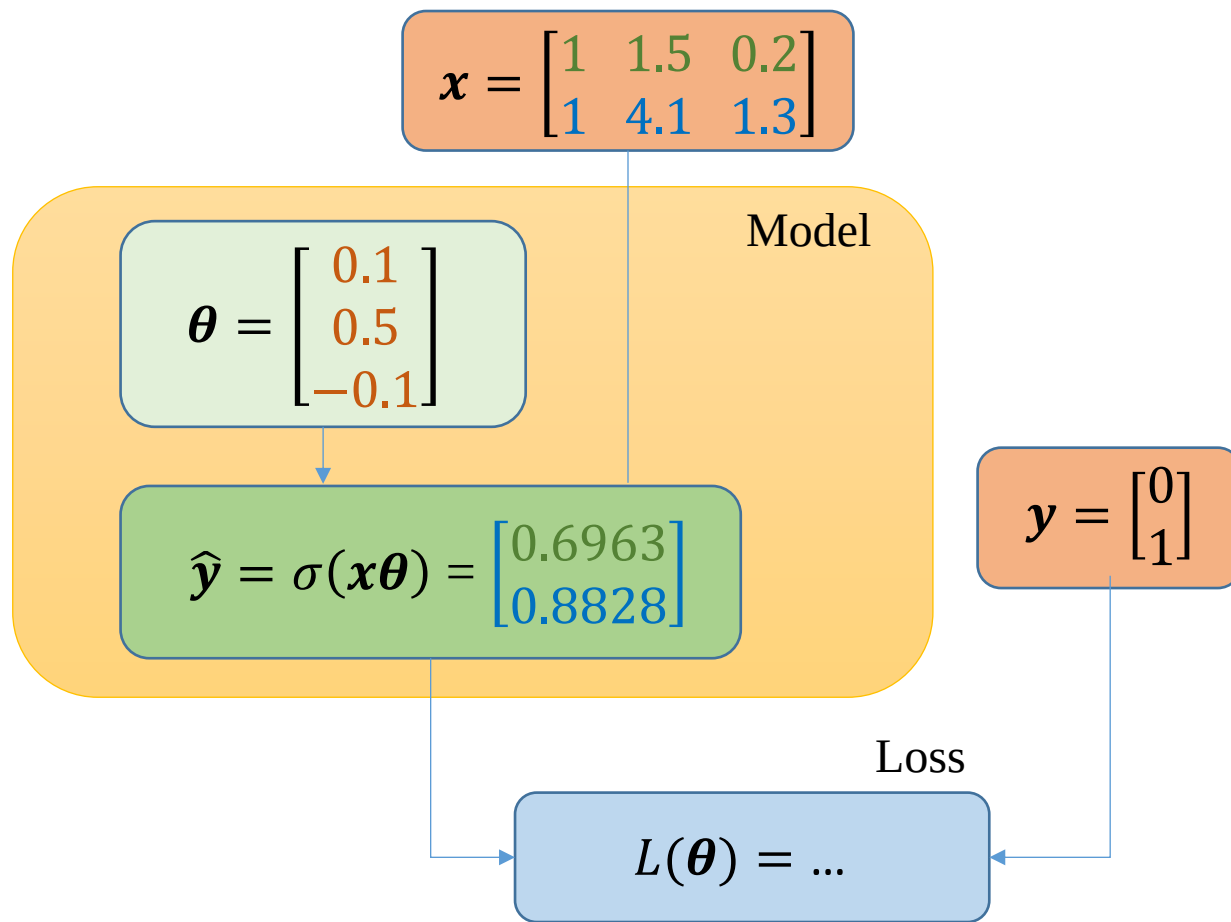
$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{m} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

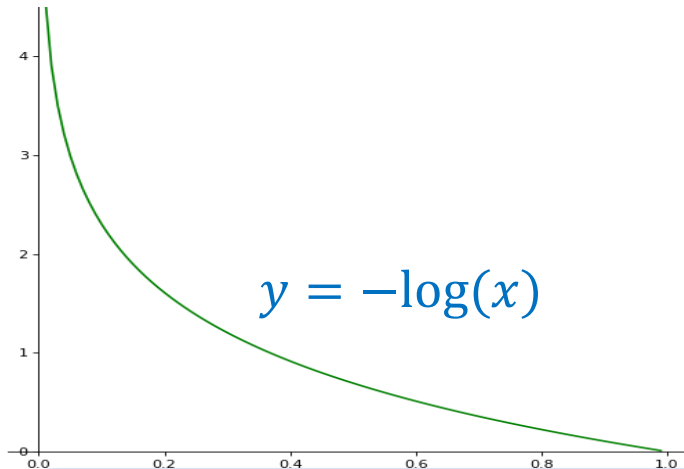
5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

2

$$\begin{aligned} \hat{y} &= \sigma(\mathbf{x}\boldsymbol{\theta}) = \sigma \left(\begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix} \right) \\ &= \sigma \left(\begin{bmatrix} 0.83 \\ 2.02 \end{bmatrix} \right) = \begin{bmatrix} 0.6963 \\ 0.8828 \end{bmatrix} \end{aligned}$$





1) Pick m samples from training data

2) Compute output $\hat{y}$

Mini-batch $m=2$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{y} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1 - \mathbf{y})^T \log(1 - \hat{\mathbf{y}}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{m} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

3

$$\begin{aligned} L(\boldsymbol{\theta}) &= \frac{1}{m} \left(-[0 \quad 1] \begin{bmatrix} \log 0.6963 \\ \log 0.8828 \end{bmatrix} - [1 \quad 0] \begin{bmatrix} \log(1 - 0.6963) \\ \log(1 - 0.8828) \end{bmatrix} \right) \\ &= \frac{1}{m} (-\log 0.8828 - \log(1 - 0.6963)) \\ &= \frac{0.1246 + 1.1917}{m} = 0.65815 \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

Model

$$\hat{\mathbf{y}} = \sigma(\mathbf{x}\boldsymbol{\theta}) = \begin{bmatrix} 0.6963 \\ 0.8828 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Loss

$$L(\boldsymbol{\theta}) = 0.65815$$

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick m samples from training data

2) Compute output $\hat{y}$

Mini-batch m=2

$$z = x\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

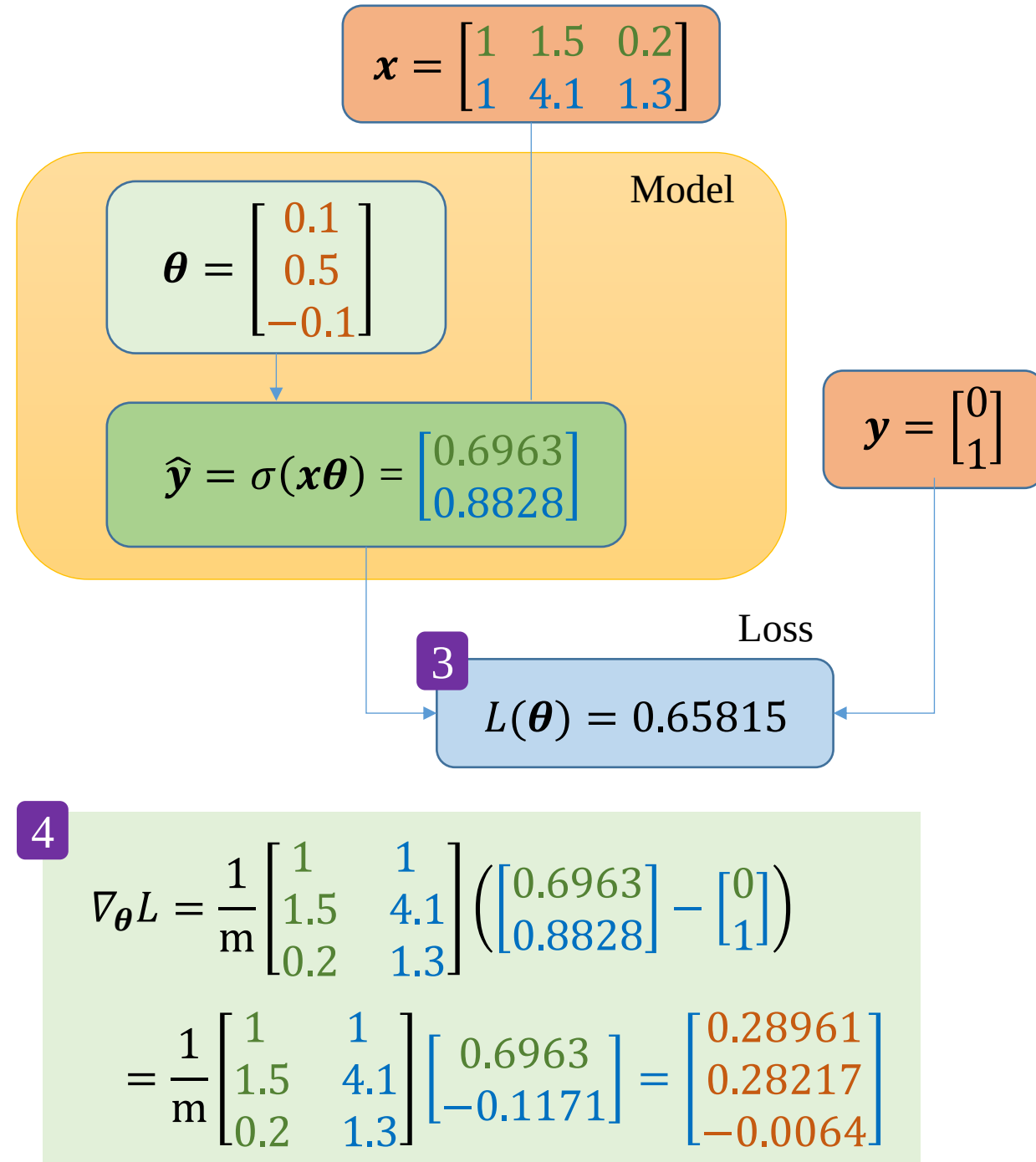
$$L(\hat{y}, y) = \frac{1}{m} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{m} x^T (\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta \nabla_{\theta} L$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick m samples from training data

2) Compute output $\hat{y}$

Mini-batch m=2

$$z = x\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

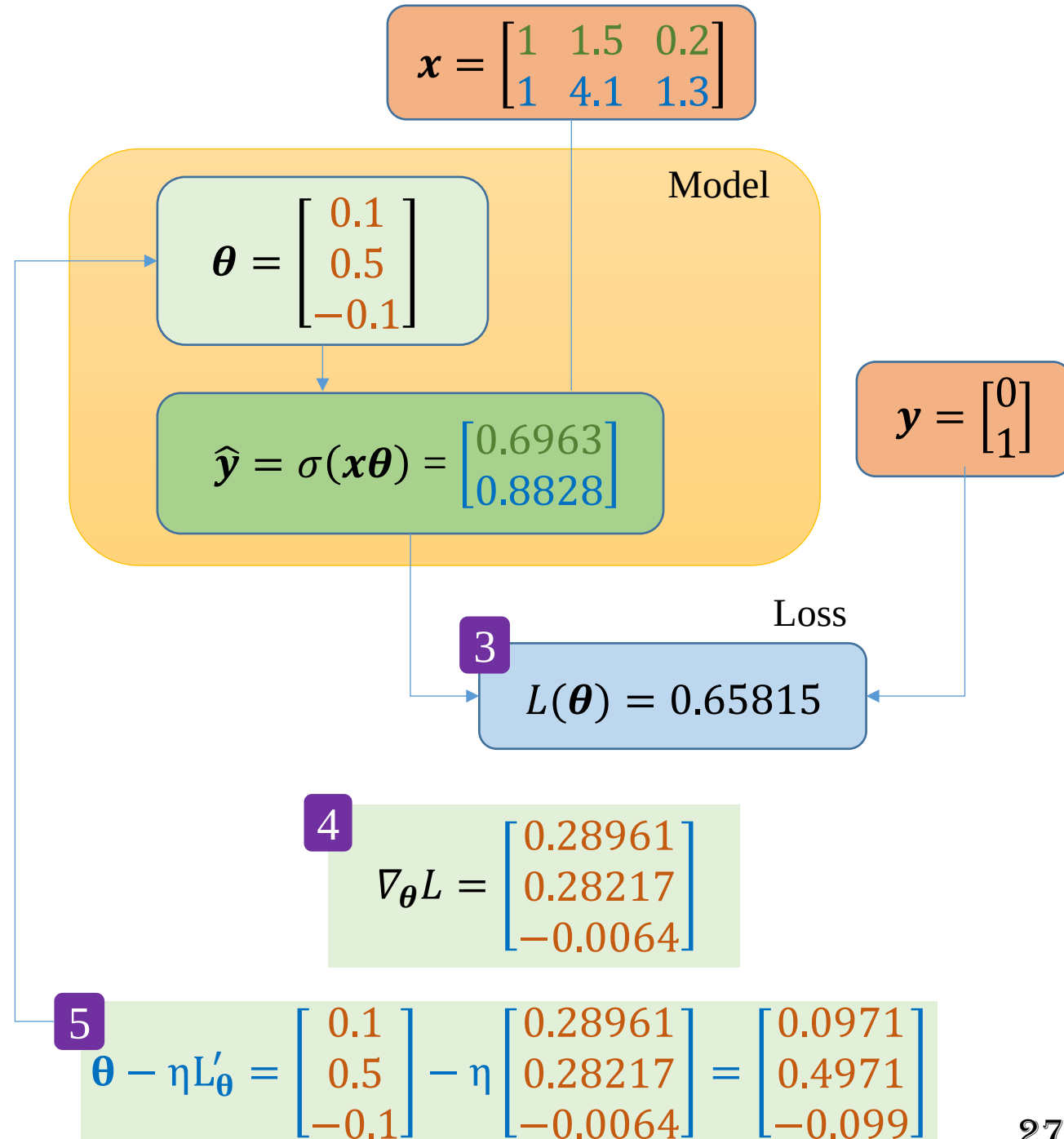
$$L(\hat{y}, y) = \frac{1}{m} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

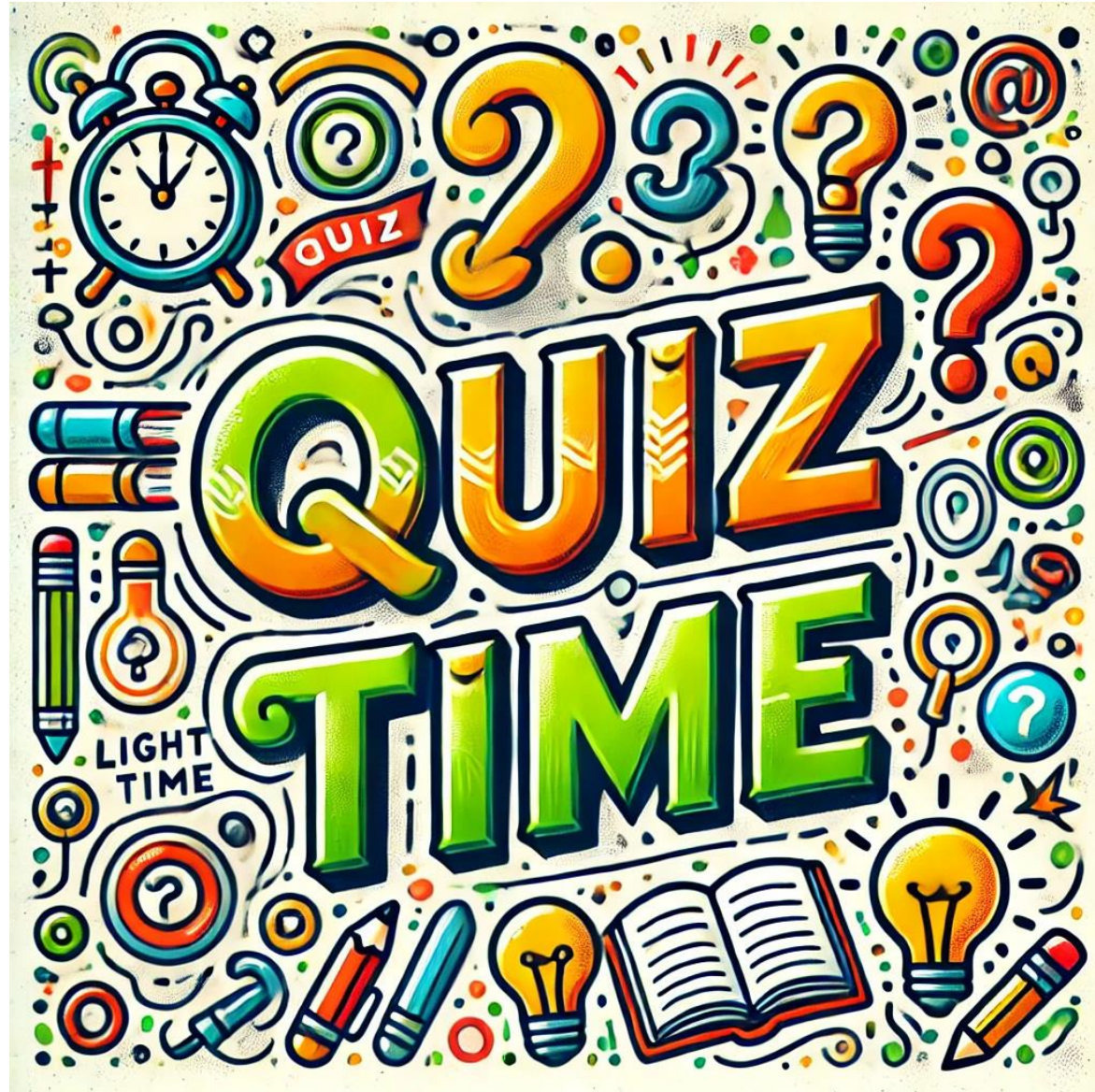
4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{m} x^T (\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta \nabla_{\theta} L$$





Question 1

❖ Output của đoạn code sau là gì?

```
import numpy as np

data = np.array([0.1, 0.6, 0.4])
print(data.round())
```

a) [0.0 0.0 0.0]

b) [1.0 1.0 1.0]

c) [0.0 1.0 0.0]

d) [1.0 0.0 1.0]

Question 2

❖ Trong vectorization cho m sample, $\mathbf{z}$ được tính như thế nào?

$$\mathbf{x} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix} \quad \boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

a) $\mathbf{z} = \mathbf{w}\mathbf{x} + b$

b) $\mathbf{z} = \mathbf{x}^T \boldsymbol{\theta}$

c) $\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$

d) Cả B và C đều đúng

Question 3

❖ Trong vectorization cho m sample, hàm loss BCE có công thức là gì?

$$\mathbf{x} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix} \quad \boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

a) $L(\hat{\mathbf{y}}, \mathbf{y}) = -\mathbf{y} \log \hat{\mathbf{y}} - (\mathbf{1} - \mathbf{y}) \log(\mathbf{1} - \hat{\mathbf{y}})$

b) $L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (\mathbf{1} - \mathbf{y})^T \log(\mathbf{1} - \hat{\mathbf{y}}))$

c) $L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y} \log \hat{\mathbf{y}}^T - (\mathbf{1} - \mathbf{y}) \log(\mathbf{1} - \hat{\mathbf{y}})^T)$

d) $L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y} \log \hat{\mathbf{y}} - (\mathbf{1} - \mathbf{y}) \log(\mathbf{1} - \hat{\mathbf{y}}))$

Question 4

❖ Cài đặt nào (L1 và L2) đúng cho hàm loss sau (cả 2 hàm chạy đều không có lỗi)?

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{m} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (\mathbf{1} - \mathbf{y})^T \log(\mathbf{1} - \hat{\mathbf{y}}))$$

```
L1 = (-y*np.log(y_hat)
      - (1 - y)*np.log(1 - y_hat)).mean()

L2 = (-y.T.dot(np.log(y_hat))
      - (1 - y).T.dot(np.log(1 - y_hat))) / m
```

a) Chỉ L1 đúng

b) Chỉ L2 đúng

c) Cả L1 và L2 đều đúng

d) Cả L1 và L2 đều sai



Question 5

❖ Trong vectorization cho 1 sample, z được tính như thế nào (có thể chọn nhiều đáp án)?

a) $z = wx + b$

b) $z = x^T \theta$

c) $z = \theta^T x$

d) Tất cả đều sai

Question 6

❖ Tìm công thức để tính được các giá trị z (có thể chọn nhiều đáp án)?

$$\mathbf{x} = \begin{bmatrix} 1 & 1 \\ x_1^{(1)} & x_1^{(2)} \\ x_2^{(1)} & x_2^{(2)} \end{bmatrix} \quad \boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

a) $\mathbf{z} = \boldsymbol{\theta}^T \mathbf{x}$

b) $\mathbf{z} = \mathbf{x}^T \boldsymbol{\theta}$

c) $\mathbf{z} = \boldsymbol{\theta} \mathbf{x}$

d) $\mathbf{z} = \mathbf{x} \boldsymbol{\theta}$

Outline

SECTION 1

Vectorization for 1-sample

SECTION 2

Vectorization for m-sample

SECTION 3

Vectorization for N-sample

SECTION 3

Sigmoid & Tanh Functions

$$\mathbf{x} = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix} \quad \hat{\mathbf{y}} = \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix}$$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{y} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1-\mathbf{y})^T \log(1-\hat{\mathbf{y}}))$$

Logistic Regression - Batch

1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$z = \mathbf{x}\boldsymbol{\theta}$$
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1 - \mathbf{y})^T \log(1 - \hat{\mathbf{y}}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{N} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

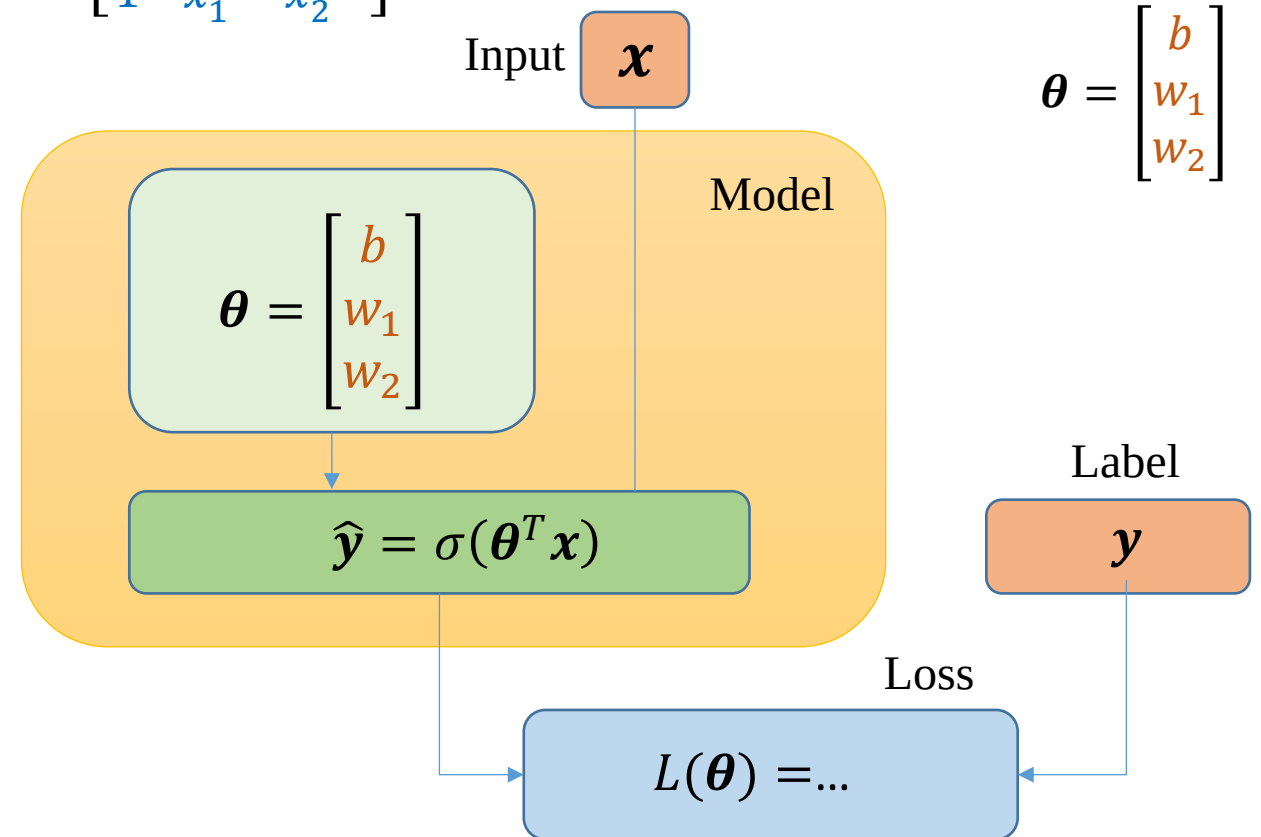
5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathbf{x} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \\ 1 & x_1^{(3)} & x_2^{(3)} \\ 1 & x_1^{(4)} & x_2^{(4)} \end{bmatrix}$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$z = x\theta$$
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = \frac{1}{N} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{N} x^T (\hat{y} - y)$$

5) Update parameters

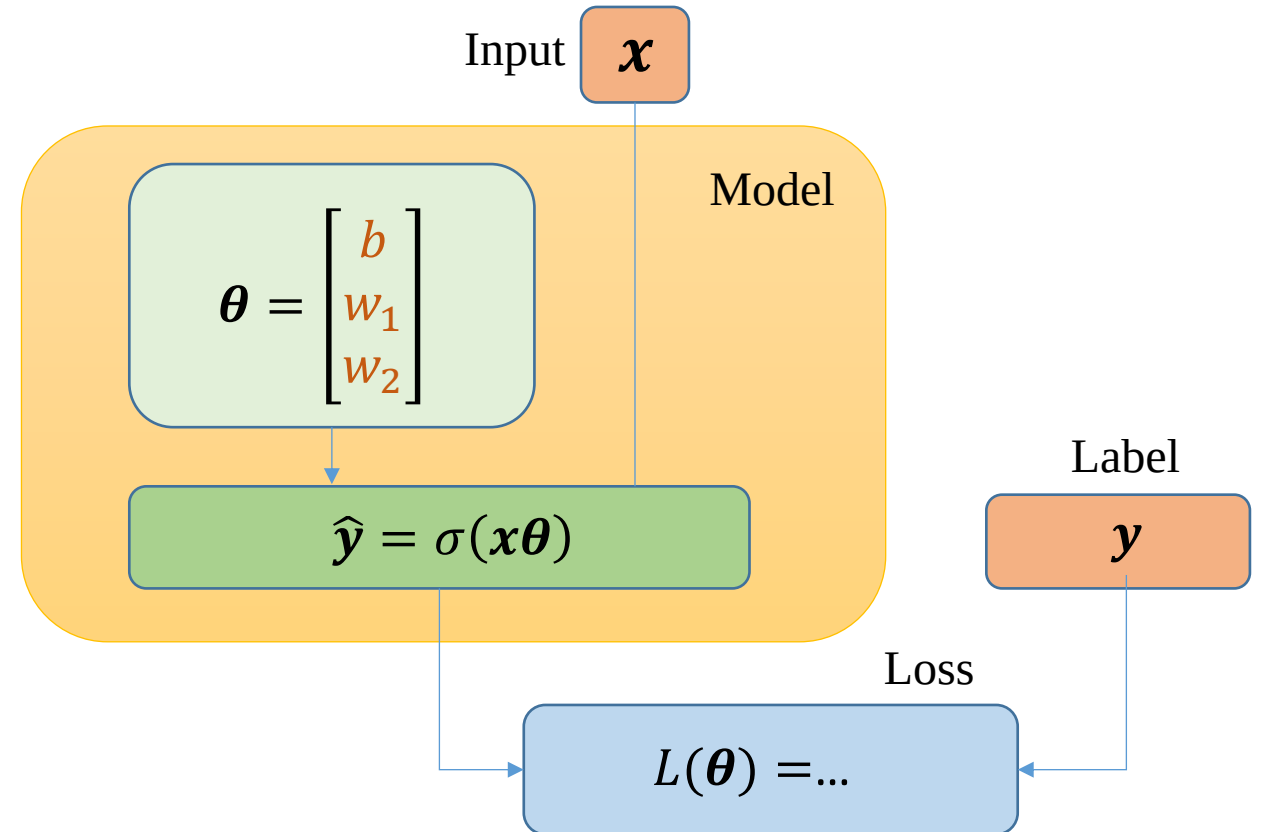
$$\theta = \theta - \eta \nabla_{\theta} L$$

Logistic Regression - Batch

$$x = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathbf{x} = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

1) Pick all the samples from training data

2) Compute output $\hat{\mathbf{y}}$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{\mathbf{y}} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1 - \mathbf{y})^T \log(1 - \hat{\mathbf{y}}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{N} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

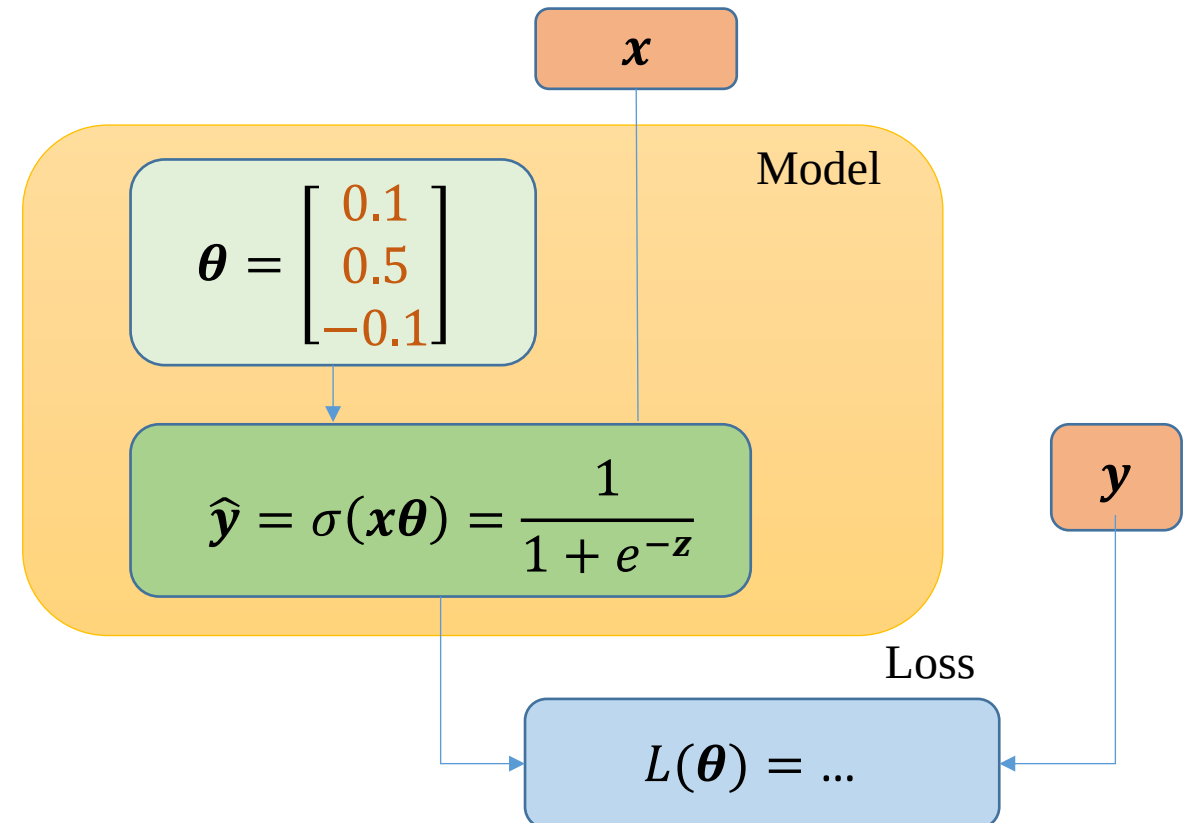
5) Update parameters

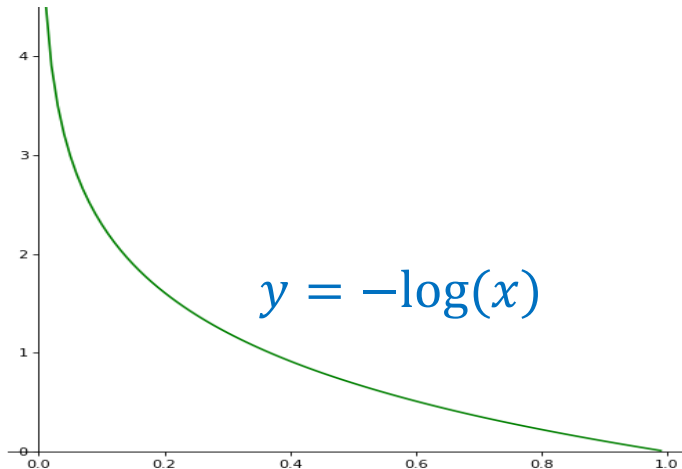
$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

2

$$\hat{\mathbf{y}} = \sigma(\mathbf{x}\boldsymbol{\theta}) = \sigma \left(\begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix} \right)$$

$$= \sigma \left(\begin{bmatrix} 0.78 \\ 0.83 \\ 1.49 \\ 2.02 \end{bmatrix} \right) = \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix}$$





- 1) Pick all the samples from training data
- 2) Compute output $\hat{y}$

$$z = x\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

- 3) Compute loss

$$L(\hat{y}, y) = \frac{1}{N} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

- 4) Compute derivative

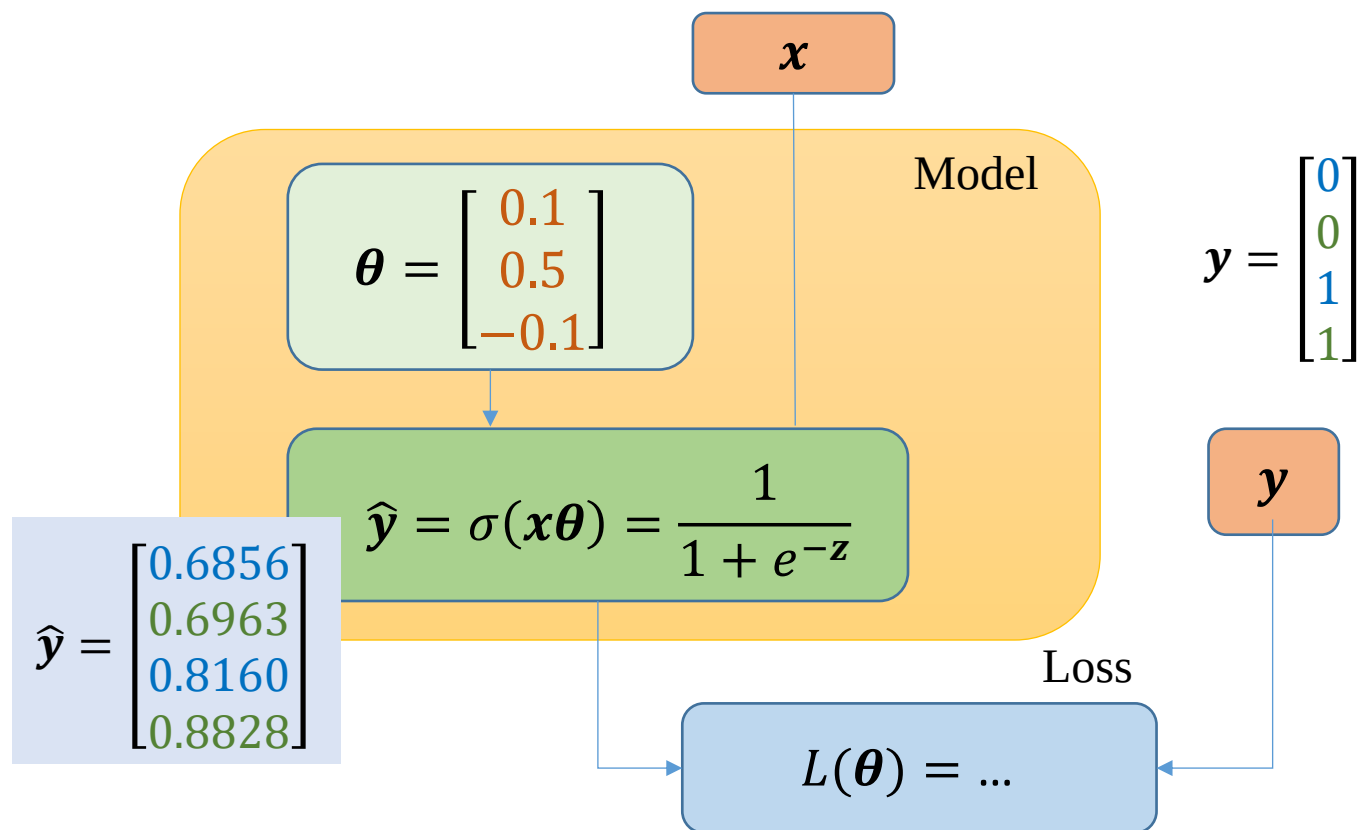
$$\nabla_{\theta} L = \frac{1}{N} x^T (\hat{y} - y)$$

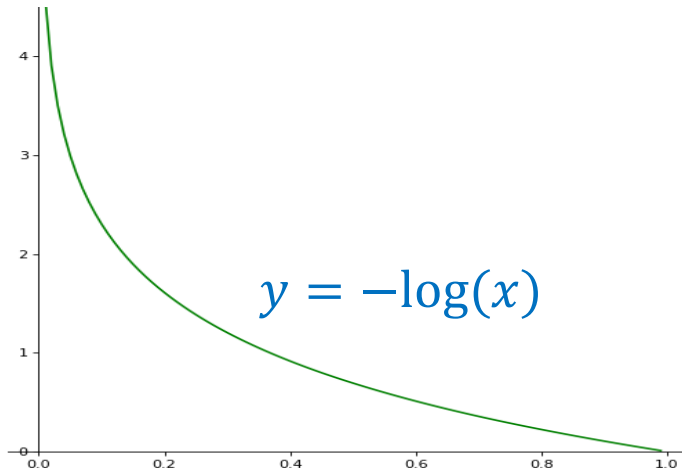
- 5) Update parameters

$$\theta = \theta - \eta \nabla_{\theta} L$$

3

$$L(\theta) = \frac{1}{N} \left\{ - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}^T \log \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix} - \left(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right)^T \log \left(1 - \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix} \right) \right\}$$





1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$z = x\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = \frac{1}{N} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{N} x^T (\hat{y} - y)$$

5) Update parameters

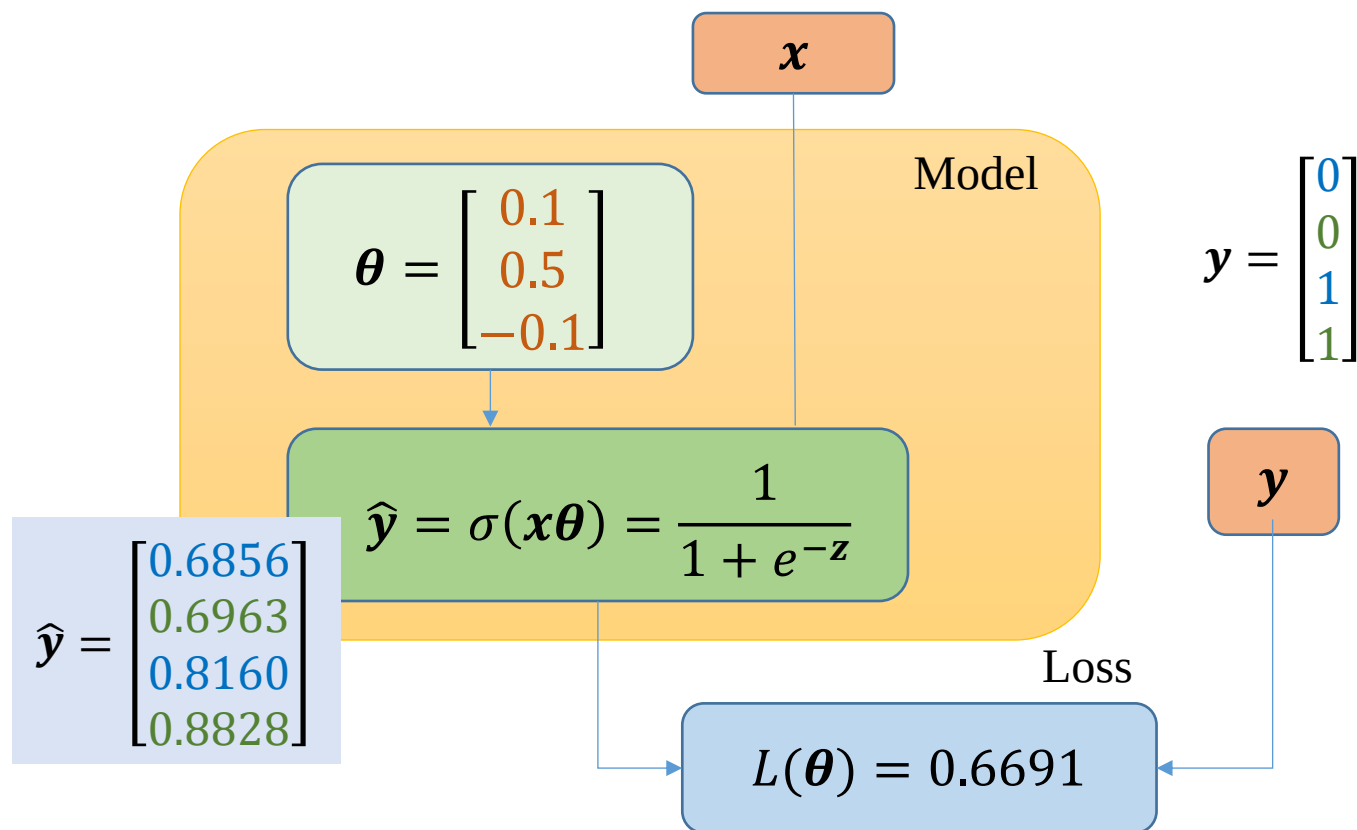
$$\theta = \theta - \eta \nabla_{\theta} L$$

3

$$L(\theta) = \frac{1}{N} \left\{ - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}^T \log \left(\begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix} \right) - \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}^T \log \left(\begin{bmatrix} 0.3144 \\ 0.3037 \\ 0.1840 \\ 0.1172 \end{bmatrix} \right) \right\}$$

$$= \frac{1}{N} (-\log 0.8160 - \log 0.8828 - \log 0.3144 - \log 0.3037)$$

$$= 0.6691$$



$$\mathbf{x} = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix} \quad \hat{\mathbf{y}} = \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix}$$

- 1) Pick all the samples from training data
- 2) Compute output $\hat{\mathbf{y}}$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{\mathbf{y}} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

- 3) Compute loss

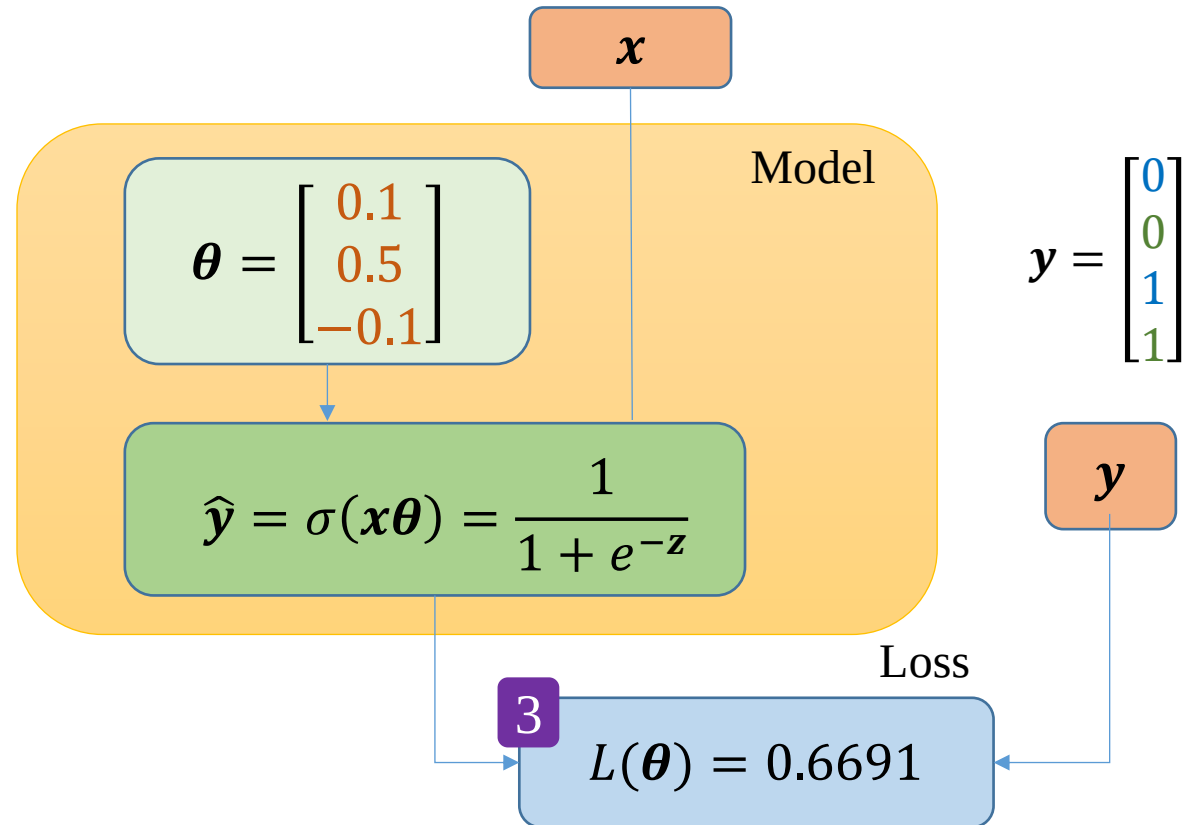
$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1 - \mathbf{y})^T \log(1 - \hat{\mathbf{y}}))$$

- 4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{N} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

- 5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$



$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1.4 & 1.5 & 3.0 & 4.1 \\ 0.2 & 0.2 & 1.1 & 1.3 \end{bmatrix} \left(\begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right)$$

$$= \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1.4 & 1.5 & 3.0 & 4.1 \\ 0.2 & 0.2 & 1.1 & 1.3 \end{bmatrix} \begin{bmatrix} 0.6856 \\ 0.6963 \\ -0.184 \\ -0.117 \end{bmatrix} = \begin{bmatrix} 0.2702 \\ 0.2431 \\ -0.019 \end{bmatrix}$$

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$z = x\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

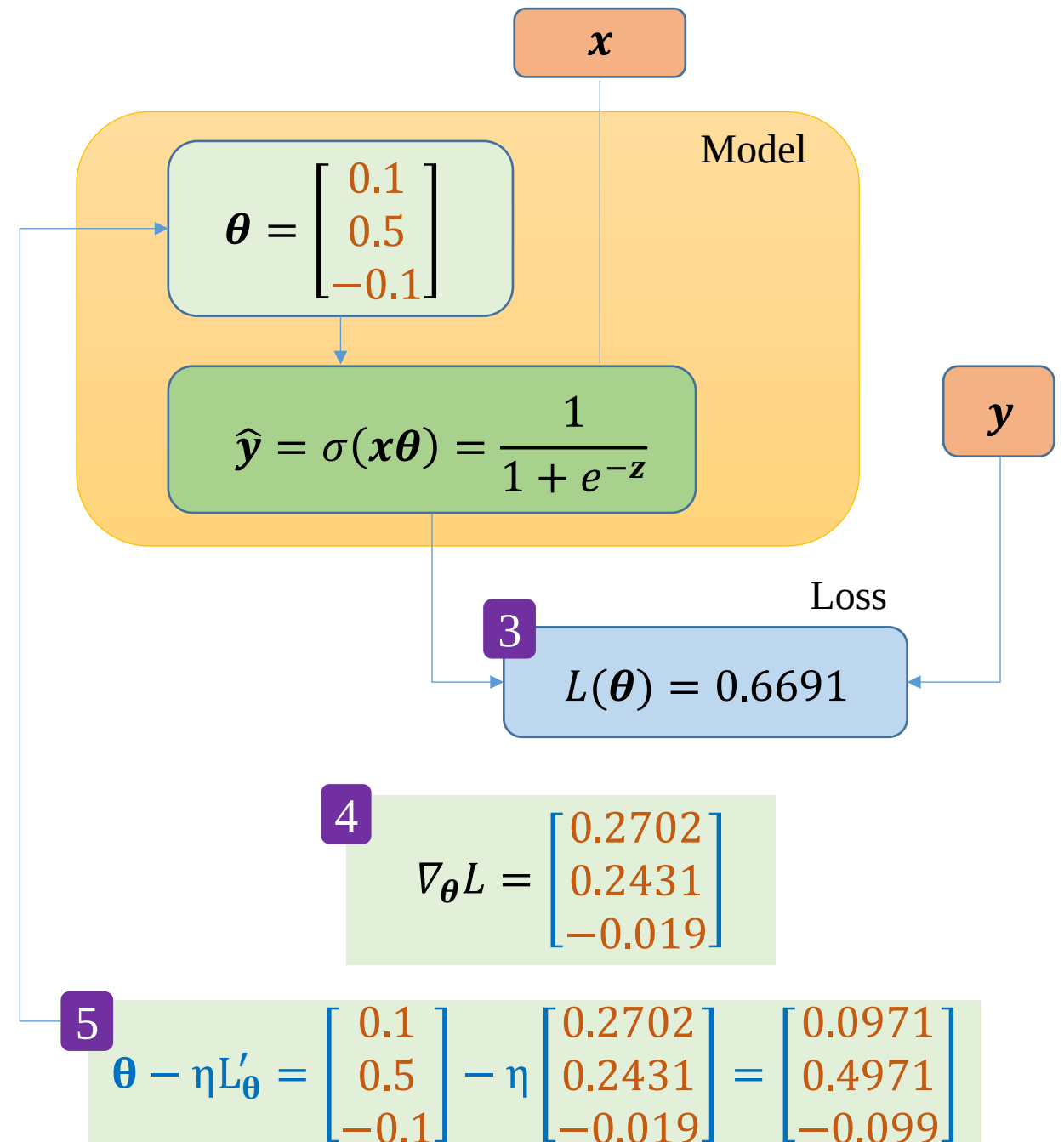
$$L(\hat{y}, y) = \frac{1}{N} (-y^T \log \hat{y} - (1-y)^T \log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{N} x^T (\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta \nabla_{\theta} L$$



Outline

SECTION 1

Vectorization for 1-sample

SECTION 2

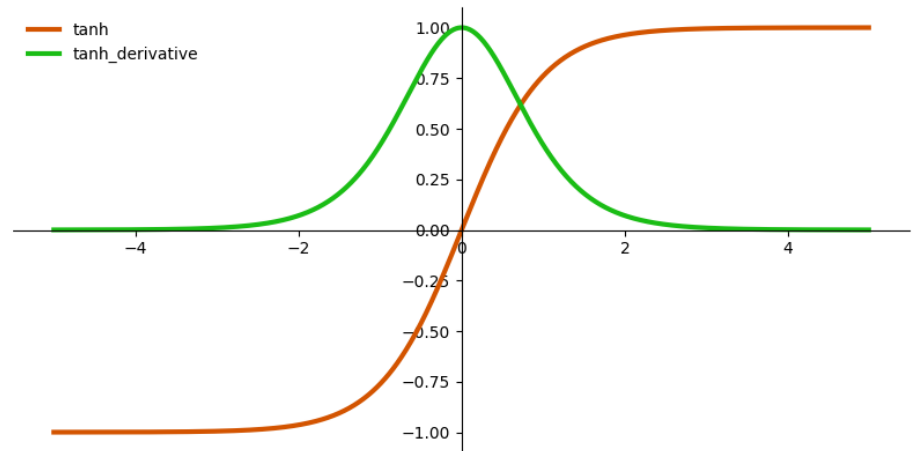
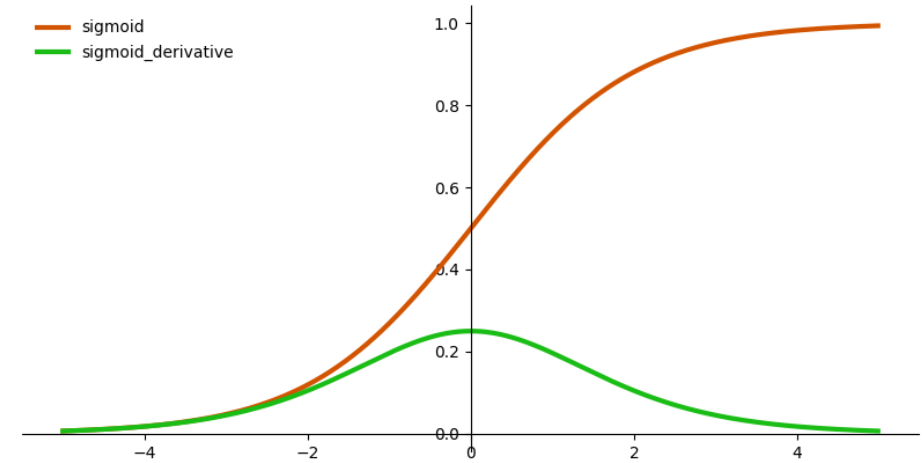
Vectorization for m-sample

SECTION 3

Vectorization for N-sample

SECTION 3

Sigmoid & Tanh Functions

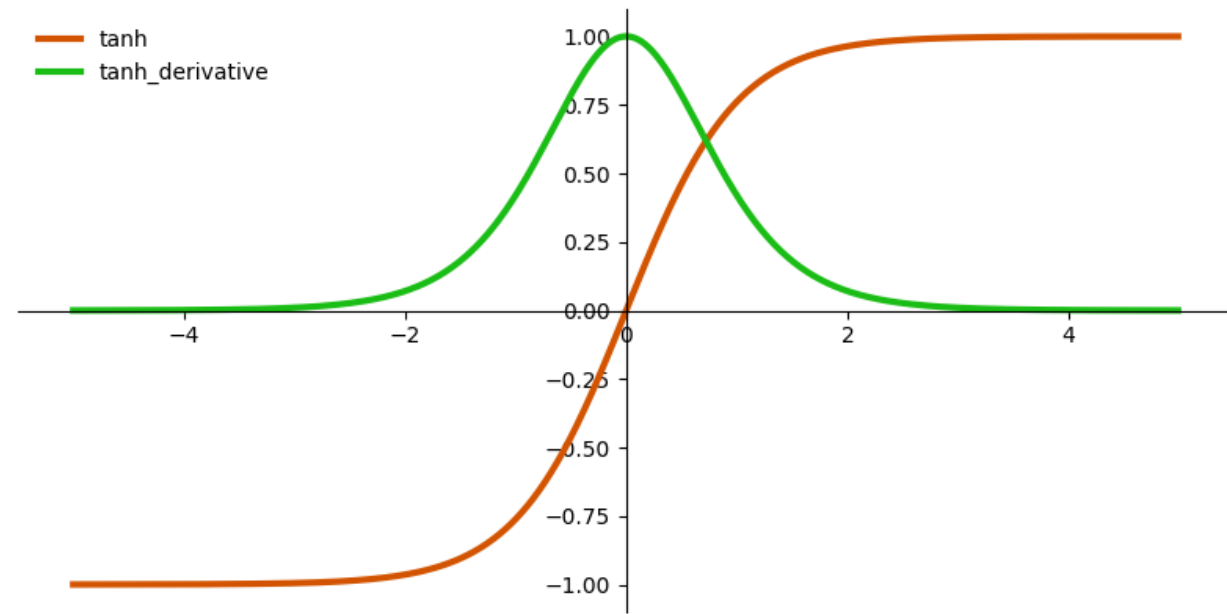
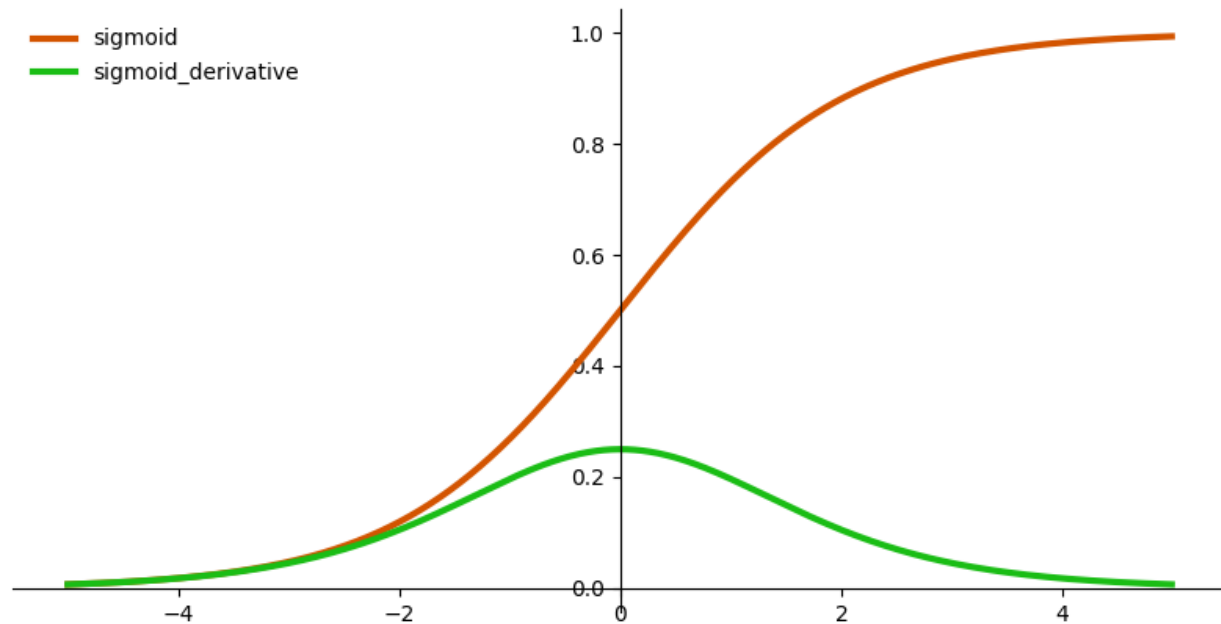


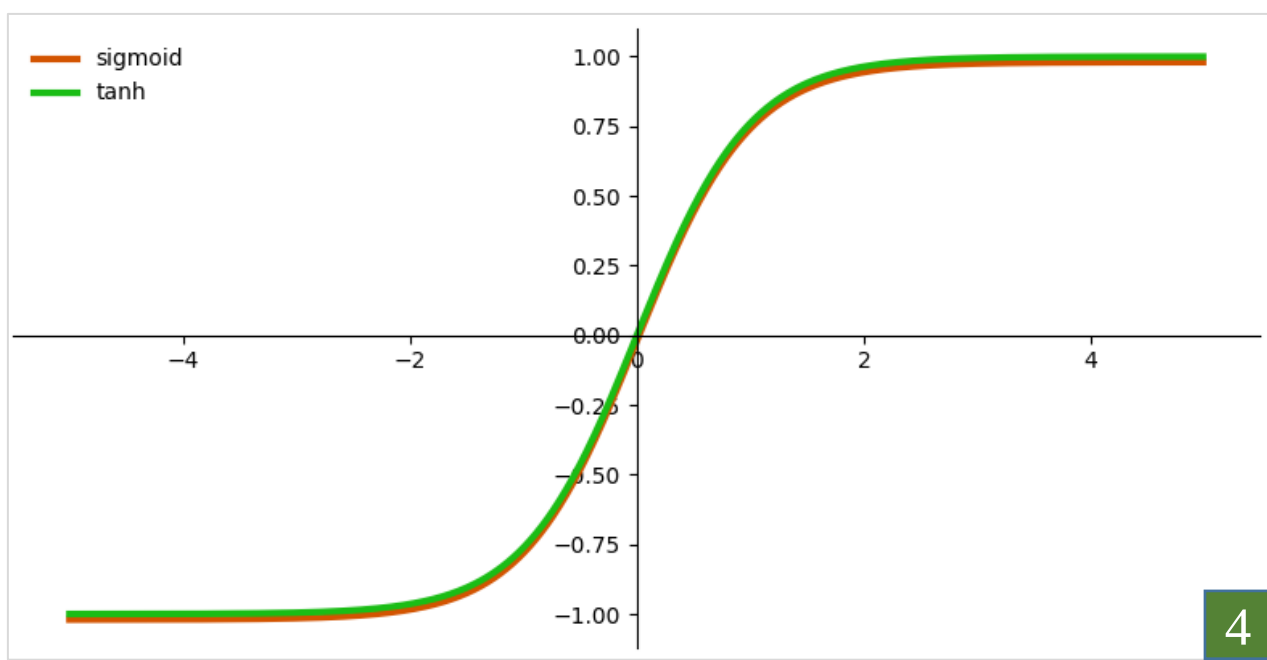
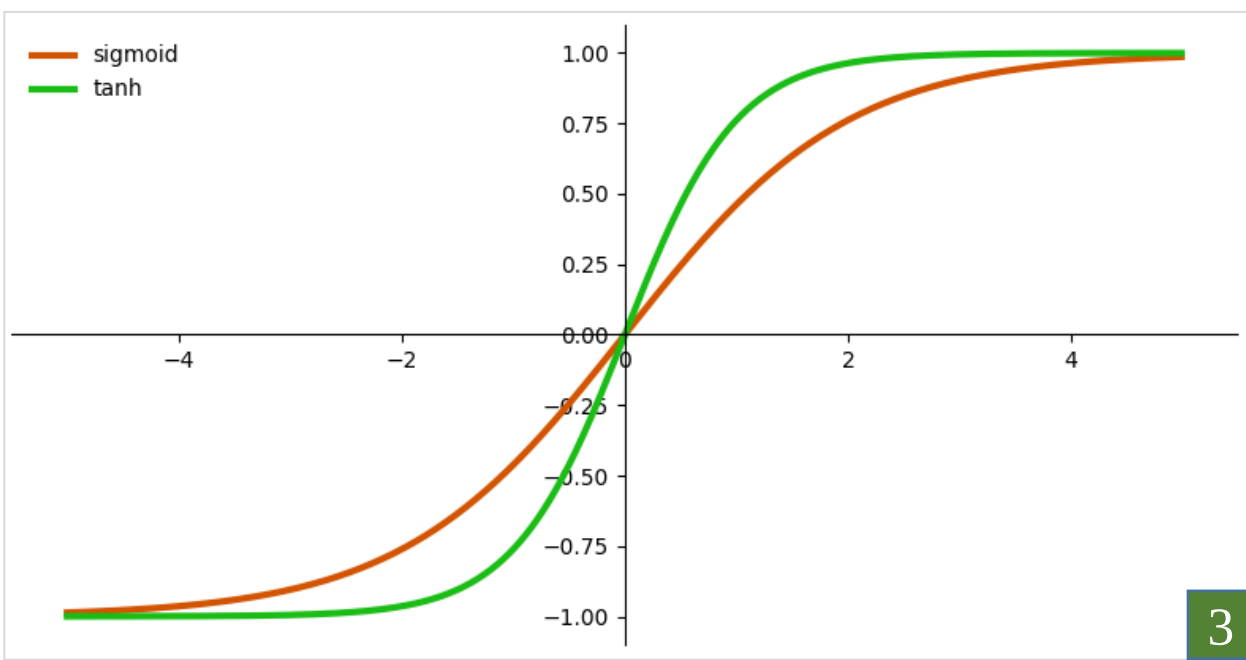
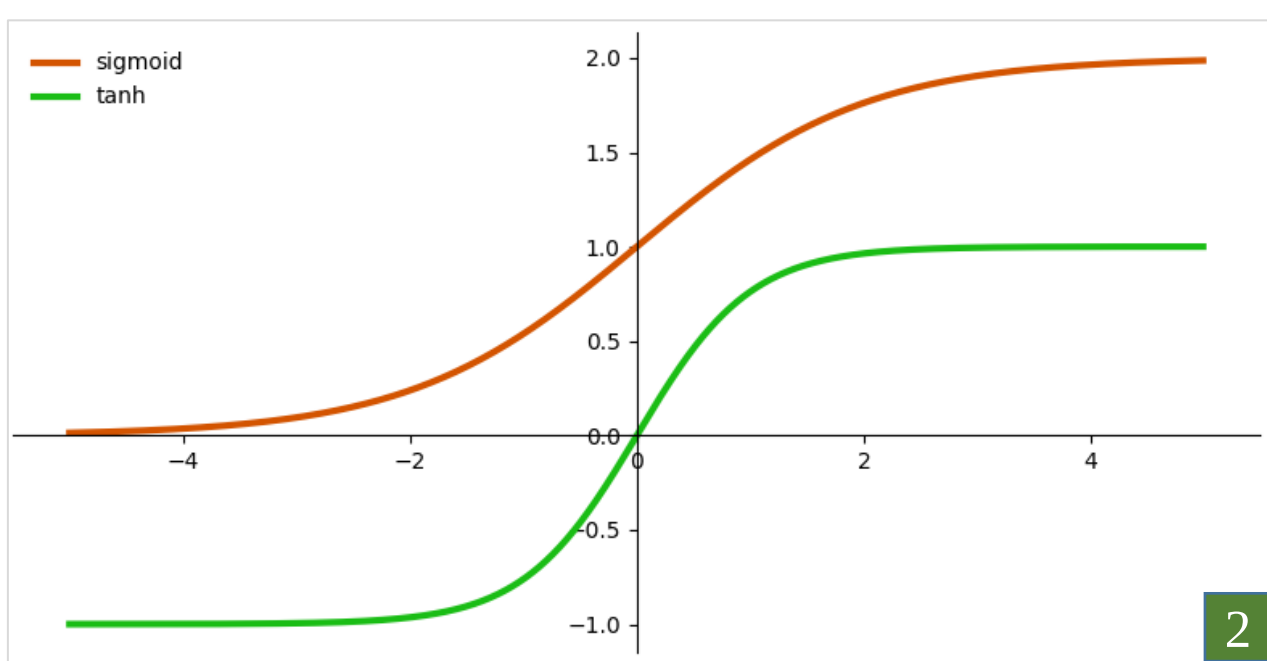
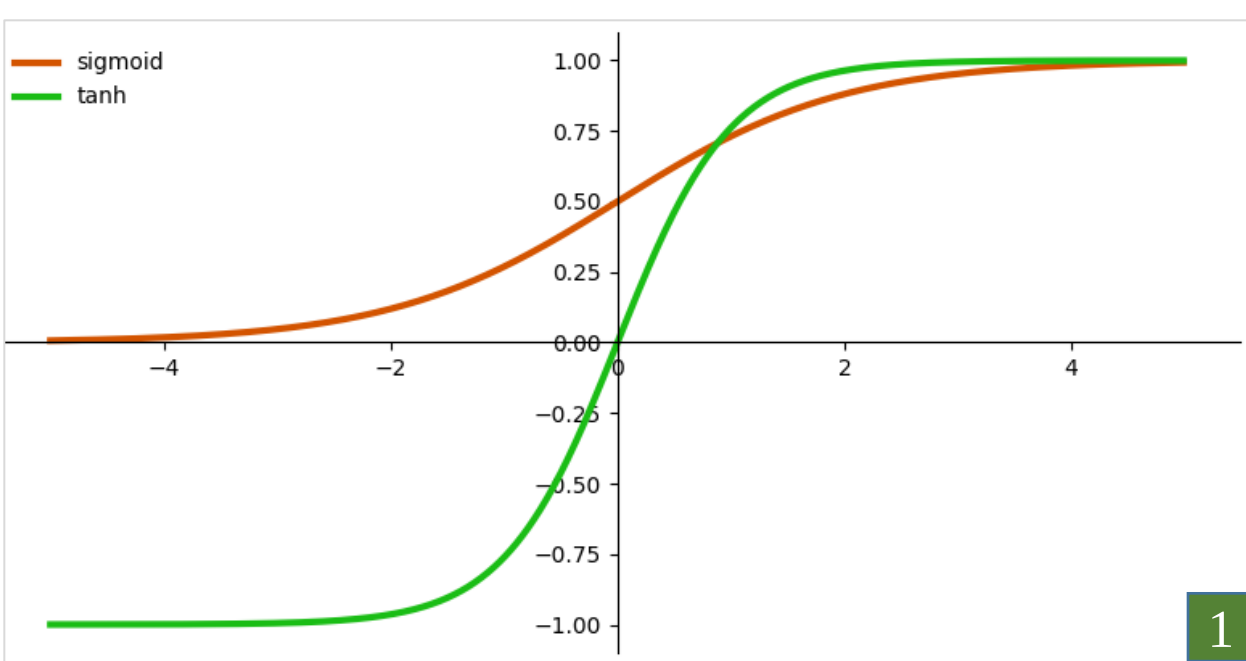
Sigmoid and Tanh Functions

37

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$





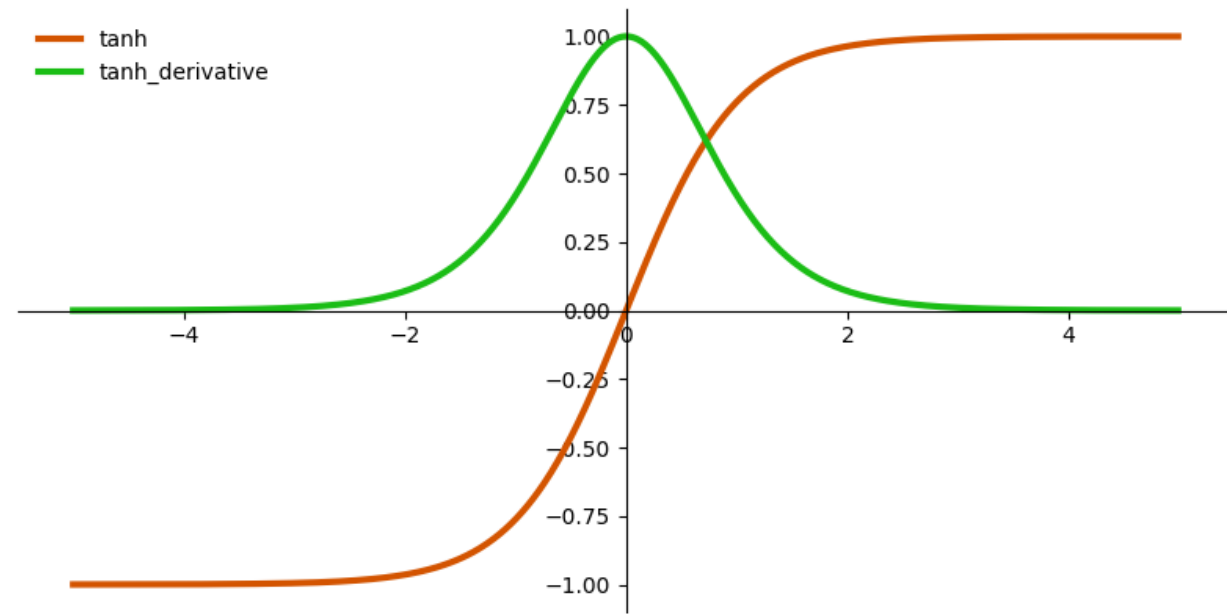
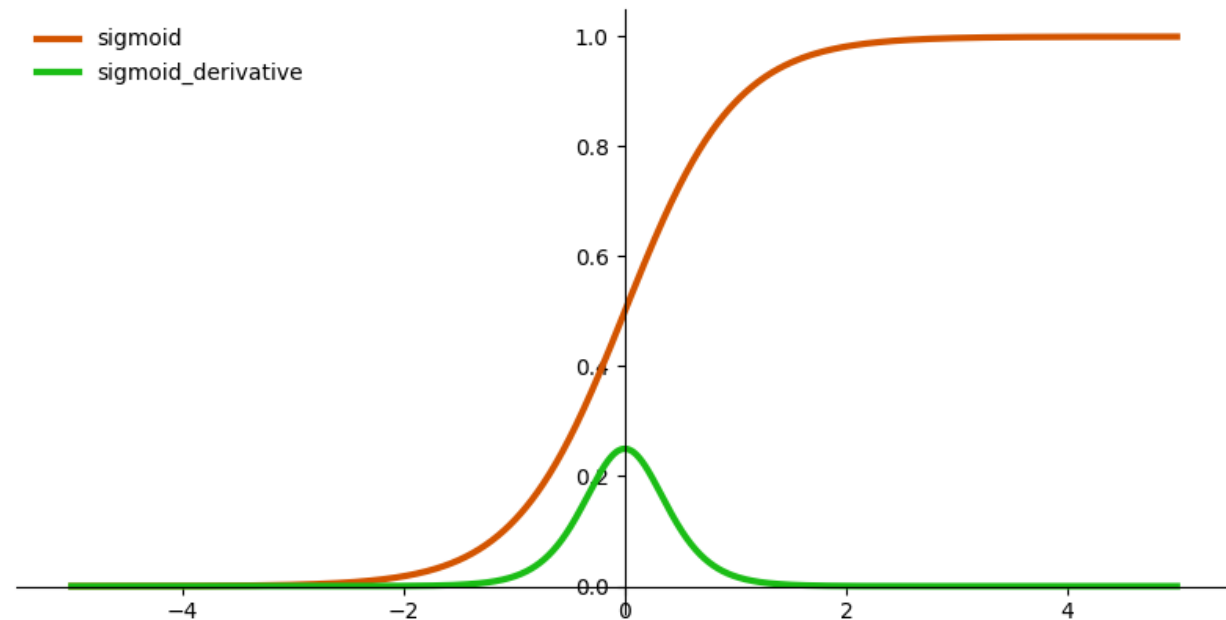
Sigmoid and Tanh Functions

39

$$\text{sigmoid}(2x) = \frac{1}{1 + e^{-2x}}$$

$$\tanh(x) = 2 \times \frac{1}{1 + e^{-2x}} - 1$$

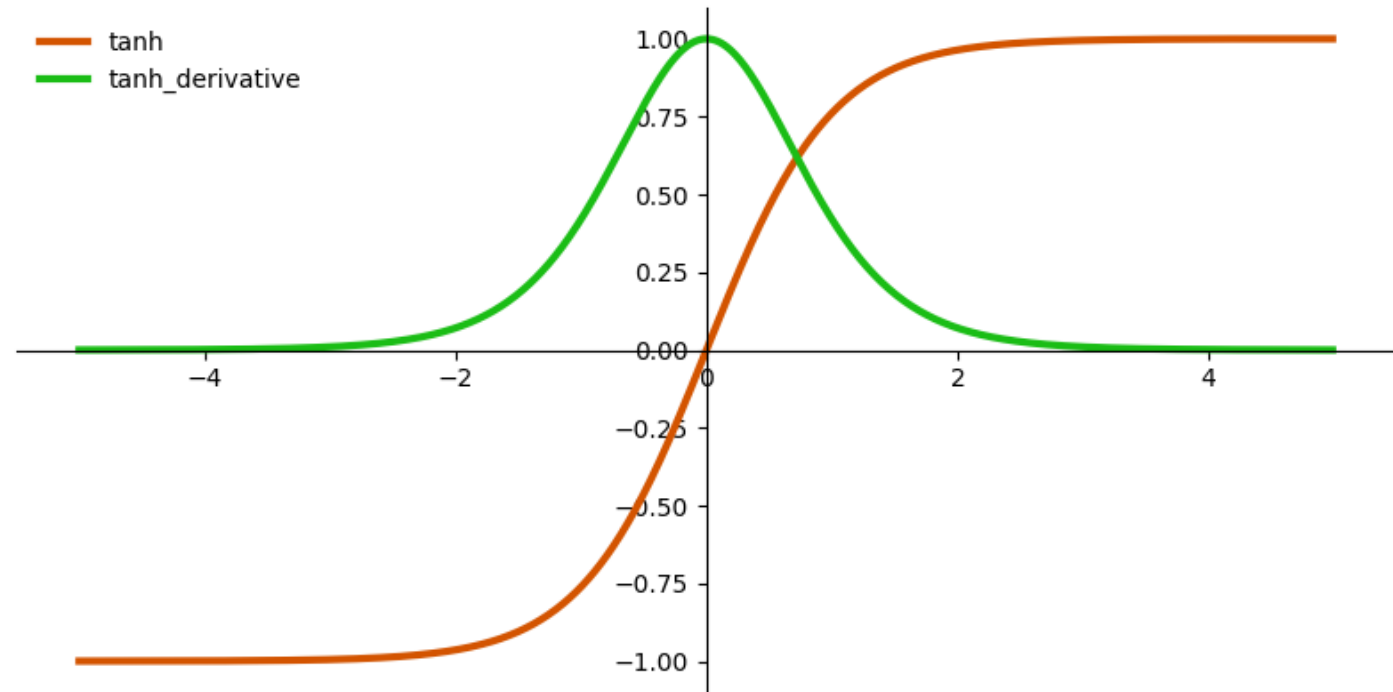
$$\tanh(x) = 2 \times \text{sigmoid}(2x) - 1$$



Tanh function

$$\begin{aligned}\tanh(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} \\ &= 1 - \frac{2}{e^{2x} + 1}\end{aligned}$$

$$\begin{aligned}\tanh(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} \\ &= -\frac{e^{-2x} - 1}{e^{-2x} + 1} = \frac{2}{e^{-2x} + 1} - 1\end{aligned}$$



$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} = -\frac{e^{-2x} - 1}{e^{-2x} + 1} = \frac{2}{e^{-2x} + 1} - 1$$

$$\begin{aligned} \tanh'(x) &= \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)' = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2} \\ &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\ &= 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2 = 1 - \tanh^2(x) \end{aligned}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}$$

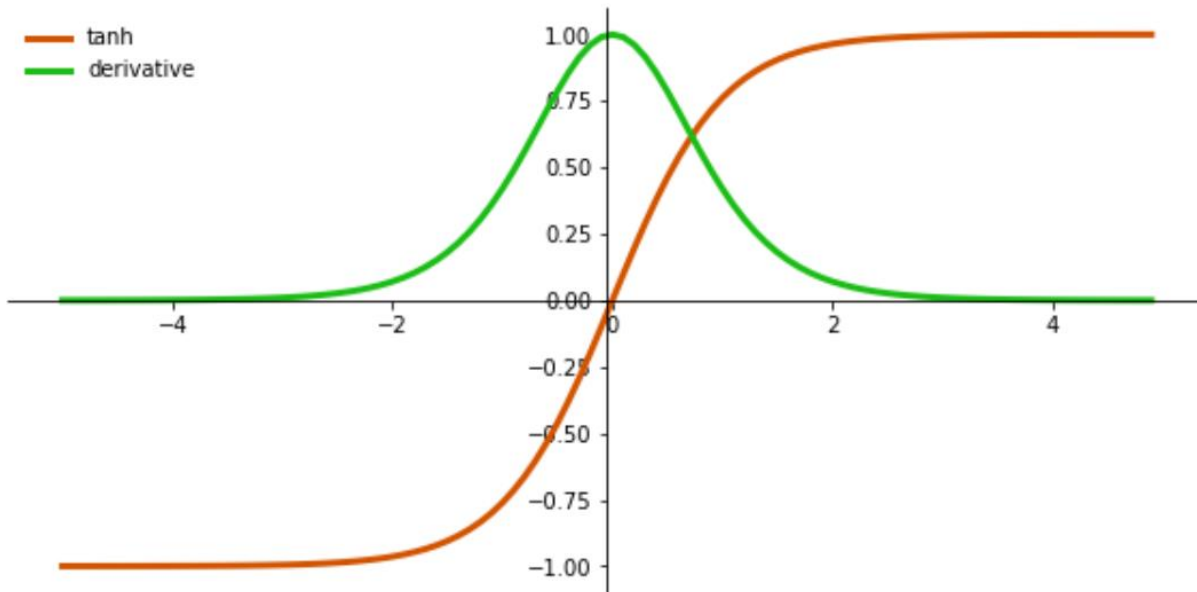
$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} = -\frac{e^{-2x} - 1}{e^{-2x} + 1} = \frac{2}{e^{-2x} + 1} - 1$$

$$\begin{aligned} \tanh'(x) &= \left(\frac{2}{e^{-2x} + 1} - 1 \right)' = \frac{4e^{-2x}}{(e^{-2x} + 1)^2} = 4 \left(\frac{e^{-2x} + 1 - 1}{(e^{-2x} + 1)^2} \right) \\ &= 4 \left(\frac{1}{e^{-2x} + 1} - \frac{1}{(e^{-2x} + 1)^2} \right) = - \left(\frac{4}{(e^{-2x} + 1)^2} - \frac{4}{e^{-2x} + 1} \right) \\ &= - \left(\frac{4}{(e^{-2x} + 1)^2} - \frac{4}{e^{-2x} + 1} + 1 - 1 \right) = 1 - \left(\frac{2}{e^{-2x} + 1} - 1 \right)^2 = 1 - \tanh^2(x) \end{aligned}$$

Logistic Regression-Tanh

❖ Construct loss

$$\tanh(x) = \frac{2}{1 + e^{-2x}} - 1$$



Model and Loss

$$z = \theta^T x = x^T \theta$$

$$\hat{y} = \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\hat{y}_s = \frac{\hat{y} + 1}{2}$$

$$L = -y \log(\hat{y}_s) - (1 - y) \log(1 - \hat{y}_s)$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}_s} \frac{\partial \hat{y}_s}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial L}{\partial \hat{y}_s} = -\frac{y}{\hat{y}_s} + \frac{1 - y}{1 - \hat{y}_s} = \frac{\hat{y}_s - y}{\hat{y}_s(1 - \hat{y}_s)}$$

$$\frac{\partial \hat{y}_s}{\partial \hat{y}} = \frac{1}{2}$$

$$\frac{\partial \hat{y}}{\partial z} = 1 - \hat{y}^2$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\hat{y}_s - y)(1 - \hat{y}^2)}{2\hat{y}_s(1 - \hat{y}_s)}$$

Logistic Regression-Tanh

❖ Construct loss

$$\tanh(x) = \frac{2}{1 + e^{-2x}} - 1$$

$$z = \theta^T x = x^T \theta$$

Model and Loss

$$\hat{y} = \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\hat{y}_s = \frac{\hat{y} + 1}{2}$$

$$L = -y \log(\hat{y}_s) - (1 - y) \log(1 - \hat{y}_s)$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}_s} \frac{\partial \hat{y}_s}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial L}{\partial \hat{y}_s} = -\frac{y}{\hat{y}_s} + \frac{1 - y}{1 - \hat{y}_s} = \frac{\hat{y}_s - y}{\hat{y}_s(1 - \hat{y}_s)}$$

$$\frac{\partial \hat{y}_s}{\partial \hat{y}} = \frac{1}{2}$$

$$\frac{\partial \hat{y}}{\partial z} = 1 - \hat{y}^2$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

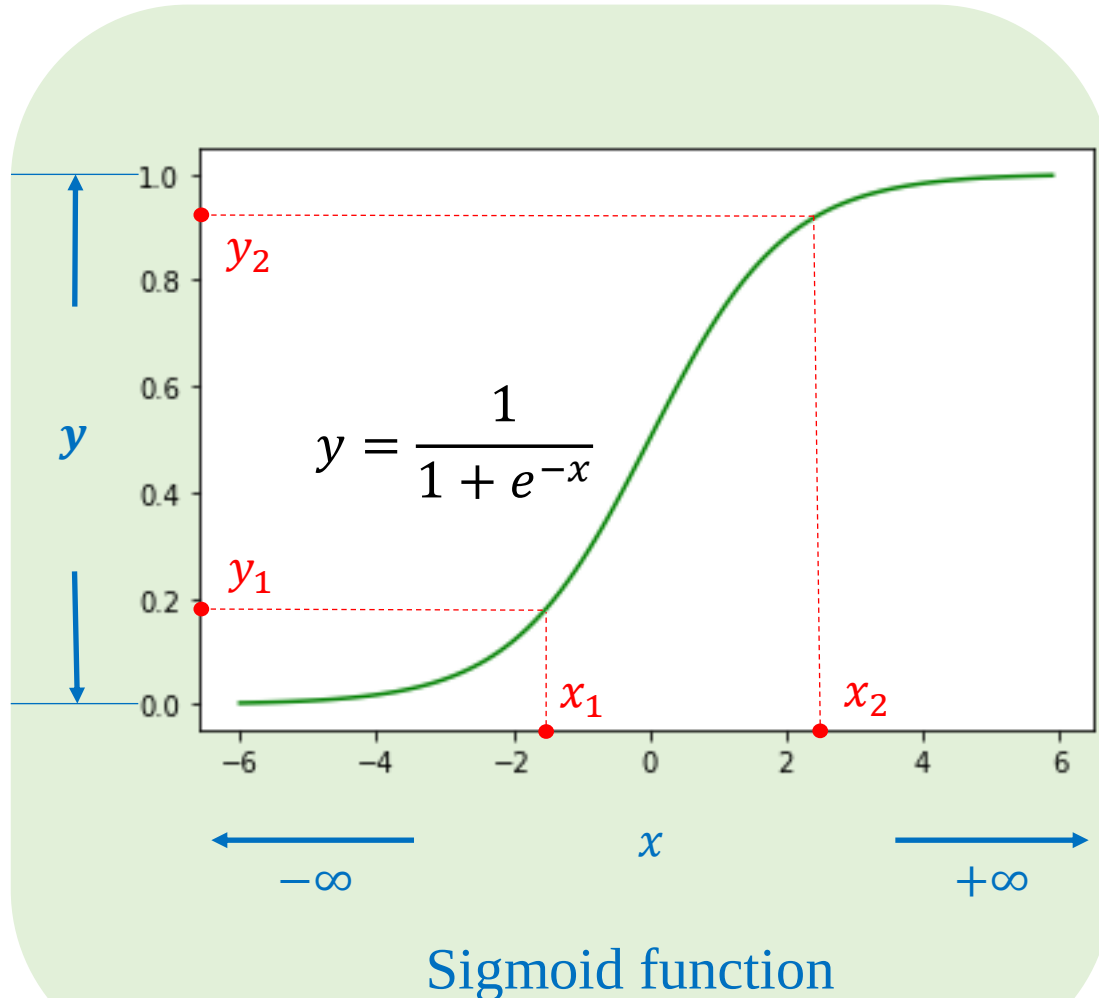
$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\hat{y}_s - y)(1 - \hat{y}^2)}{2\hat{y}_s(1 - \hat{y}_s)}$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\frac{\hat{y} + 1}{2} - y)(1 - \hat{y}^2)}{2 \frac{\hat{y} + 1}{2} (1 - \frac{\hat{y} + 1}{2})}$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\hat{y} + 1 - 2y)(1 - \hat{y}^2)}{(\hat{y} + 1)(1 - \hat{y})}$$

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} + 1 - 2y)$$

Summary



1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$\mathbf{z} = \mathbf{x}\boldsymbol{\theta}$$

$$\hat{y} = \sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

3) Compute loss (binary cross-entropy)

$$L(\boldsymbol{\theta}) = \frac{1}{N} (-\mathbf{y}^T \log \hat{\mathbf{y}} - (1 - \mathbf{y})^T \log(1 - \hat{\mathbf{y}}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{N} \mathbf{x}^T (\hat{\mathbf{y}} - \mathbf{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta L'_{\boldsymbol{\theta}}$$

η is learning rate

